Online Supplement to "Exact Finite-Sample Distribution and Moments of the Least Squares Estimator in First-Order Autoregressions: $O(1)$ Algorithms"

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This supplement is organized as follows. Section S1 derives the limiting characteristic functions (44)–(46) of Proposition 4 of the paper from known representations and reduces the demeaned limit to the single parameter $\varkappa$. It also analyzes the branches of the limiting characteristic functions and shows that the limits have moments of all orders. Section S2 gives the χ^2 representation of $A + cB$. Section S3 gives the recursions of Taylor-mode arithmetic used by the programs. Section S4 contains additional tables and figures, with comparisons against published tables. Section S5 treats the finite-sample distribution in the explosive case with an intercept. Section S6 proves Theorem 1 and Propositions 1 and 3 of the paper. Section S7 extends the exact and limiting analysis to the regression with a linear trend. The MATLAB programs are described in the user manual distributed with them. The scaled initial value of the local-to-unity limits is written γ_0, as in the paper, to distinguish it from the finite-sample parameter γ of case DU. Equation and theorem numbers without the prefix S refer to the paper.

We use the model and case definitions of Section 2 of the paper: n is the number of regression observations, ϕ the autoregressive coefficient, α the intercept, σ the innovation standard deviation, and (u, v) the standardized numerator and denominator of $\hat{\phi} - \phi = u/v$. The labels DK and DU mean deterministic (fixed) starts with known and estimated intercepts; SK and SU mean the corresponding stationary starts. In cases DK and DU the standardized start and drift parameters are $z_0 = y_0/\sigma$ and $\gamma = [\alpha - (1 - \phi)y_0]/\sigma$,

respectively; SK and SU require $|\phi| < 1$. We retain $\mathbb{E}$, $\mathbb{P}$, Var, Cov and Corr for expectation, probability, variance, covariance and correlation. The standard normal distribution function and density are $\Phi_{\mathcal{N}}$ and $\varphi_{\mathcal{N}}$, and $\chi^2_k(\delta)$ is the noncentral chi-square distribution with k degrees of freedom and noncentrality δ, χ^2_k the central one. Matrix absolute bars denote determinants, scalar absolute bars denote moduli, a prime denotes transpose and an overbar denotes complex conjugation; $\mathrm{i} = \sqrt{-1}$. The threshold r and the real transform arguments (t_1, t_2) are those of (6). Additional symbols are defined locally below.

S1 The limiting characteristic functions

S1.1 Reduction of the demeaned limit to the parameter $\varkappa$

Here $\phi = 1 + c/n$ with fixed $c \in \mathbb{R}$, $y_0 = \sqrt{n}\,\gamma_0\sigma$ with fixed γ_0, $\alpha = b\sigma/\sqrt{n}$ with fixed real b, and $\varkappa = b + c\gamma_0$ is the demeaned local-to-unity drift parameter. Also, W is standard Brownian motion and $J_c(r) = \int_0^r e^{(r-s)c}\mathrm{d}W(s)$ for $0 \le r \le 1$; the limiting numerator and denominator $(\hat{A}, \hat{B})$ are those of (42) of the paper.

Let $H_c(r) = (e^{cr} - 1)/c$, so that $e^{cr} = 1 + cH_c(r)$ ($H_c(r) = r$ at $c = 0$). The limiting process of the standardized data $y_{\lfloor nr \rfloor}/(\sqrt{n}\sigma)$ is

$$X_0(r) = \gamma_0 e^{cr} + bH_c(r) + J_c(r) = \gamma_0 + \varkappa H_c(r) + J_c(r),$$

and the demeaned functionals

$$\int_0^1 X_0(r)\,\mathrm{d}W(r) - W(1)\int_0^1 X_0(r)\,\mathrm{d}r \qquad \text{and} \qquad \int_0^1 X_0(r)^2\,\mathrm{d}r - \Big(\int_0^1 X_0(r)\,\mathrm{d}r\Big)^2$$

are unchanged when the constant γ_0 is dropped from X_0. This gives (42) of the paper with $X = J_c + \varkappa H_c$ and shows that the limit depends on (b, γ_0) only through $\varkappa$. For $c \neq 0$, with $\gamma_c = \varkappa/c = b/c + \gamma_0$, $X(r) = J_c(r) + \gamma_c(e^{cr} - 1)$, and dropping the constant $-\gamma_c$ and expanding the squares gives the equivalent form

$$\hat{A} = \int_0^1 J_c(r)\,\mathrm{d}W(r) + \gamma_c\int_0^1 e^{cr}\mathrm{d}W(r) - W(1)\Big[\int_0^1 J_c(r)\mathrm{d}r + \gamma_c\,\frac{e^c - 1}{c}\Big],$$
$$\hat{B} = \gamma_c^2\,\frac{e^{2c} - 1}{2c} + 2\gamma_c\int_0^1 e^{cr}J_c(r)\mathrm{d}r + \int_0^1 J_c^2(r)\mathrm{d}r - \Big[\int_0^1 J_c(r)\mathrm{d}r + \gamma_c\,\frac{e^c - 1}{c}\Big]^2,$$

which involves γ_c and is therefore not defined at $c = 0$ when $b \neq 0$; the representation (42) is continuous in c.

The same limit is Theorem 2.1 of Zhou and Yu (2015), the in-fill limit of the demeaned estimator in an Ornstein–Uhlenbeck model; its proof is in the working paper of Zhou and Yu (2010). It is written there in (b, γ_0) with the constants

$$c_2 = \frac{e^c - c - 1}{c^2}, \qquad c_3 = \frac{e^{2c} - 4e^c + 2c + 3}{2c^3}, \qquad c_4 = \frac{e^c - 1}{c}$$

(with their continuous values $1/2$, $1/3$ and 1 at $c = 0$); for $c \neq 0$ their expression reduces to ours by the identities

$$\frac{b}{c}(e^{cr} - 1) + \gamma_0 e^{cr} - c_2 b - c_4 \gamma_0 = \gamma_c \, (e^{cr} - c_4),$$
$$c_3 b^2 + c_4^2 b \gamma_0 + \gamma_0^2 \, \frac{e^{2c} - 1}{2c} - (c_2 b + c_4 \gamma_0)^2 = \gamma_c^2 \Big(\frac{e^{2c} - 1}{2c} - c_4^2 \Big).$$

S1.2 Derivation of (44)

The joint characteristic function of $(\hat{A}, \hat{B})$ is expressed in terms of

$$\omega = c + \mathrm{i}t_1, \qquad \theta^2 = 2\mathrm{i}t_2 - 2\mathrm{i}ct_1 - c^2, \qquad \tau_0 = t_1^2 - 2\mathrm{i}t_2,$$
$$\tau_1 = 2(t_1^2 + \mathrm{i}c^2 t_1 - 2\mathrm{i}ct_2), \qquad \tau_2 = c^3 + \mathrm{i}(c - 2)ct_1 - \tau_0.$$

It is

$$e^{-\omega/2} \Big(\frac{\tilde{\tau}_3}{\theta^4} \Big)^{-1/2} \exp \Big(- \frac{\varkappa^2 \tau_0 (2 - 2\cos\theta - \theta \sin\theta)}{2\tilde{\tau}_3} \Big),$$
$$\tilde{\tau}_3 = \tau_1 - (\tau_1 + c^2\theta^2)\cos\theta + \tau_2 \theta \sin\theta,$$

which follows from the Fredholm-determinant representation of Nabeya and Tanaka (1990) for the demeaned Ornstein–Uhlenbeck functionals (see also Tanaka, 2017, Chapter 5); we use that representation without rederiving it, and Table S1 checks it against the finite-sample characteristic function. The stationary-start forms (45) and (46) are derived in the proof of Lemma S1.2. The half-angle identities $1 - \cos\theta = 2\sin^2(\theta/2)$ and $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$ give $\tilde{\tau}_3 = \sin^2(\theta/2)\, \tau_3$ with

$$\tau_3 = 2\tau_1 - c^2\theta^2 \cos\theta \, \csc^2(\theta/2) + 2\tau_2 \theta \cot(\theta/2),$$

and

$$2-2\cos\theta-\theta\sin\theta=4\sin^2(\theta/2)\big[1-(\theta/2)\cot(\theta/2)\big],$$

so that

$$\Big(\frac{\tilde\tau_3}{\theta^4}\Big)^{-1/2}=\frac{\theta}{\sin(\theta/2)}\Big(\frac{\tau_3}{\theta^2}\Big)^{-1/2},$$
$$\frac{\tau_0(2-2\cos\theta-\theta\sin\theta)}{2\tilde\tau_3}=\frac{2\tau_0[1-(\theta/2)\cot(\theta/2)]}{\tau_3},$$

and hence the characteristic function is

$$e^{-\omega/2}\,\frac{\theta}{\sin(\theta/2)}\Big(\frac{\tau_3}{\theta^2}\Big)^{-1/2}\exp\Big(-\frac{2\varkappa^2\tau_0[1-(\theta/2)\cot(\theta/2)]}{\tau_3}\Big),\tag{S.1}$$

which Lemma S1.4 below turns into (44). The first identity of the display before (S.1) holds with the branches on both sides continuous and equal to their positive values at the origin. The reason is that $\theta/\sin(\theta/2)$ is an even function of θ, hence a single-valued function of θ^2, without zeros or poles on the parameter domain (its poles are at $\theta^2=4\pi^2k^2$, $k=1,2,\ldots$); it is also positive at the origin, where $\theta=\mathrm{i}c$ and $\theta/\sin(\theta/2)=c/\sinh(c/2)$. Put $\mathcal{C}=\cos\theta$ and $\mathcal{S}=\sin\theta/\theta$. In terms of the even functions $\mathcal{C}_h=\cos(\theta/2)$ and $\mathcal{S}_h=\sin(\theta/2)/(\theta/2)$,

$$\csc^2(\theta/2)=\frac{4}{\theta^2\mathcal{S}_h^2},\qquad \theta\cot(\theta/2)=\frac{2\mathcal{C}_h}{\mathcal{S}_h},\qquad \mathcal{C}_h\mathcal{S}_h=\mathcal{S},$$

which gives

$$\tau_3=\frac{2\tau_1\mathcal{S}_h^2-4c^2\mathcal{C}+4\tau_2\mathcal{S}}{\mathcal{S}_h^2},$$

the form in which the programs evaluate the value of (S.1); its phase is taken from the factorization of Lemma S1.4.

For small $|\theta^2|$ the programs use a power series. The numerator of τ_3 is

$$N=2\tau_1\mathcal{S}_h^2-4c^2\mathcal{C}+4\tau_2\mathcal{S}.$$

We regard it as a function of $z=\theta^2$ at fixed (t_1,c), so that $t_2=-\mathrm{i}(z+2\mathrm{i}ct_1+c^2)/2$ and

$$\tau_1=\tau_1^0-2cz,\qquad \tau_2=\tau_2^0+z,\qquad \tau_1^0=2(t_1^2-\mathrm{i}c^2t_1-c^3),\qquad \tau_2^0=c^3+c^2+\mathrm{i}c^2t_1-t_1^2.$$

The functions $\mathcal{S}_h^2$, $\mathcal{C}$ and $\mathcal{S}$ have the Taylor coefficients $a_k=2(-1)^k/(2k+2)!$, $b_k=$

$(-1)^k/(2k)!$ and $s_k = (-1)^k/(2k+1)!$ in z, so $N = \sum_{k=0}^{\infty} N_k z^k$ with

$$N_k = 2\tau_1^0 a_k - 4c a_{k-1} - 4c^2 b_k + 4\tau_2^0 s_k + 4 s_{k-1} \qquad (a_{-1} = s_{-1} = 0).$$

Since

$$N_0 = N(0) = 2\tau_1^0 - 4c^2 + 4\tau_2^0 = 0,$$

the ratio

$$\frac{N}{z} = \sum_{k=0}^{\infty} N_{k+1} z^k$$

is entire. The programs use this series for $|z| < 4$ (26 terms) and the closed forms otherwise.

S1.3 Branch analysis

Lemma S1.1. *For all real c, t_1, t_2, the factor $\mathcal{G} = 1 - \omega\mathcal{S}/\mathcal{C} = 1 - (\omega/\theta)\tan\theta$ of $\mathcal{C} - \omega\mathcal{S} = \mathcal{C}\,\mathcal{G}$ in (43) (Section 6.1 of the paper) is not zero and not a negative real number.*

Proof. Since $\theta^2 = -c^2 + 2\mathrm{i}(t_2 - c t_1)$ is never a positive real number and the zeros of $\mathcal{S} = \sin\theta/\theta$ (with its continuous value 1 at $\theta = 0$) have $\theta^2 > 0$, $\mathcal{S} \neq 0$ on the parameter domain. So

$$u(r) = \frac{\sin(\theta r)}{\sin\theta}, \qquad r \in [0,1]$$

(with $u(r) = r$ when $\theta = 0$), an auxiliary function distinct from the numerator u of the paper, is well defined, with $u(0) = 0$, $u(1) = 1$ and $u'' = -\theta^2 u$. Put $a = u'(1) = \theta\cot\theta$. Integrating by parts and using the boundary values,

$$a = \int_0^1 \left\{\bar{u}(r)u''(r) + |u'(r)|^2\right\} \mathrm{d}r = \int_0^1 |u'(r)|^2\,\mathrm{d}r - \theta^2 \int_0^1 |u(r)|^2\,\mathrm{d}r.$$

Because $\mathrm{Re}(-\theta^2) = c^2$,

$$\mathrm{Re}\ a = \int_0^1 \left\{|u'(r)|^2 + c^2|u(r)|^2\right\} \mathrm{d}r > 0.$$

Moreover

$$2\int_0^1 \mathrm{Re}\{\bar{u}(r)u'(r)\}\,\mathrm{d}r = |u(1)|^2 - |u(0)|^2 = 1,$$

so

$$\operatorname{Re} a - c = \int_0^1 \left\{ |u'(r)|^2 - 2c \operatorname{Re}\{\bar{u}(r)u'(r)\} + c^2|u(r)|^2 \right\} \mathrm{d}r = \int_0^1 |u'(r) - cu(r)|^2 \, \mathrm{d}r > 0.$$

The last integral is strictly positive because $u' = cu$ with $u(0) = 0$ would force $u \equiv 0$, contradicting $u(1) = 1$. Hence a and $a - c - \mathrm{i}t_1 = a - \omega$ both lie in the open right half-plane, and

$$\mathcal{G} = 1 - \frac{\omega}{\theta} \tan\theta = \frac{\theta \cot\theta - \omega}{\theta \cot\theta} = \frac{a - \omega}{a}$$

is nonzero with argument $\operatorname{Arg}(a-\omega) - \operatorname{Arg} a \in (-\pi, \pi)$, so it is not on the closed negative real axis. The argument covers $c = 0$ and $t_2 = ct_1$ (where $\theta = \mathrm{i}c$) without special treatment. □

Lemma S1.2. *For $c < 0$ and all real t_1, t_2, the mixing factors*

$$1 - \frac{\tau_0 \mathcal{S}}{2c(\mathcal{C} - \omega\mathcal{S})} \quad \textit{of (45)} \qquad \textit{and} \qquad 1 - \frac{c\,\tau_0 \bar{d}_1}{2\rho_\infty \mathcal{S}_\infty} \quad \textit{of (46)}$$

have real part at least one.

Proof. Write (43) as $\varphi_0(t_1, t_2) \exp(\gamma_0^2 \Lambda)$ with

$$\Lambda = -\frac{\tau_0 \mathcal{S}}{2(\mathcal{C} - \omega\mathcal{S})}, \qquad \varphi_0 = e^{-\omega/2}(\mathcal{C} - \omega\mathcal{S})^{-1/2},$$

where φ_0 does not depend on γ_0 and does not vanish. For every real γ_0 this is the characteristic function of (A, B), so $|\varphi_0| \, e^{\gamma_0^2 \operatorname{Re}\Lambda} \le 1$ for all $\gamma_0 \in \mathbb{R}$, which is possible only if $\operatorname{Re}\Lambda \le 0$. Now let $\gamma_0 \sim \mathcal{N}(0, V_0)$ with $V_0 = -1/(2c) > 0$. The Gaussian integral

$$\mathbb{E}[e^{\gamma_0^2 \Lambda}] = (1 - 2\Lambda V_0)^{-1/2}$$

then converges, and $1 - 2\Lambda V_0 = 1 + \Lambda/c$ has real part $1 + \operatorname{Re}\Lambda/c \ge 1$. The same argument applies to (44) when $b = 0$, so that $\varkappa = c\gamma_0$. Write (44) as $\hat{\varphi}_0 \exp(\varkappa^2 \hat{\Lambda})$ with

$$\hat{\Lambda} = -\frac{\tau_0 \bar{d}_1}{2\rho_\infty \mathcal{S}_\infty}.$$

Then, as before, $\operatorname{Re}\hat{\Lambda} \le 0$, and $1 - 2c^2 \hat{\Lambda} V_0 = 1 + c\hat{\Lambda}$ has real part at least one. □

Lemma S1.3. *Write* Arg_c *for the continuous argument equal to zero at* $(t_1, t_2) = (0, 0)$. *On the parameter domain* $\operatorname{Re} \theta^2 = -c^2 \le 0$, *with* $\vartheta = (-\theta^2)^{1/2}$ *the principal root, the continuous argument of* $\cos\theta$ *along any path from the origin is*

$$\operatorname{Arg}_c \cos\theta = \operatorname{Im}\vartheta + \operatorname{Arg}(1 + e^{-2\vartheta}), \tag{S.2}$$

the form given in the paper. Equivalently,

$$\operatorname{Arg}_c \cos\theta = \operatorname{Arg}\cos\theta + 2\pi \left\lfloor \frac{-\operatorname{sgn}(\operatorname{Im}\theta)\operatorname{Re}\theta - \operatorname{Arg}\cos\theta}{2\pi} + \frac{1}{2} \right\rfloor. \tag{S.3}$$

Proof. On the half-plane $\operatorname{Re}\theta^2 \le 0$ the principal root ϑ is a continuous function of θ^2 with $\operatorname{Re}\vartheta \ge 0$, and $\cos\theta = \cosh\vartheta$ because cos is even. Write

$$\cosh\vartheta = \tfrac{1}{2}e^{\vartheta}(1 + e^{-2\vartheta}), \qquad |e^{-2\vartheta}| = e^{-2\operatorname{Re}\vartheta} \le 1.$$

The factor e^{ϑ} has the continuous argument $\operatorname{Im}\vartheta$, a real number without jumps; $|\operatorname{Im}\vartheta|$ grows like $|t_2 - ct_1|^{1/2}$ along every path $t_2 = -rt_1$ with $r \ne -c$, and each change by 2π is one circuit of $\cos\theta$ around the origin. The factor $1 + e^{-2\vartheta}$ lies in the closed disk of radius one about 1, hence in the closed right half-plane, and it does not vanish, because $\operatorname{Re}\vartheta > 0$ except at $\theta^2 = 0$, where it equals 2; on the closed right half-plane without the origin the principal argument is continuous. Both arguments vanish at the origin, where $\vartheta = |c|$ is real, which gives (S.2).

For the equivalent form (S.3), write $\theta = \theta_r + \mathrm{i}\theta_i$. Then

$$\cos\theta = \cosh\theta_i \cos\theta_r - \mathrm{i}\sinh\theta_i \sin\theta_r,$$
$$\operatorname{Re}\left[\cos\theta \cdot e^{\mathrm{i}\operatorname{sgn}(\theta_i)\theta_r}\right] = \cosh\theta_i \cos^2\theta_r + |\sinh\theta_i| \sin^2\theta_r > 0$$

whenever $\theta_i \ne 0$, so the continuous argument of $\cos\theta$ stays within $\pi/2$ of $-\operatorname{sgn}(\theta_i)\theta_r$, which is itself continuous along the paths (θ_r vanishes where θ^2 crosses the negative real axis, and θ^2 never crosses the positive one; $\theta_i = 0$ only at $\theta = 0$). The multiple of 2π in (S.3) is the one that brings the principal argument within $\pi/2$ of $-\operatorname{sgn}(\theta_i)\theta_r$; with $\theta = \mathrm{i}\vartheta$, $-\operatorname{sgn}(\theta_i)\theta_r = \operatorname{Im}\vartheta$. Either form replaces the sign-switching rule of Evans and Savin (1981) and Perron (1989). □

S1.4 The factor τ_3/θ^2

Lemma S1.4 turns (S.1) into the factored form (44) of the paper and gives its phase, by carrying the finite-sample factorization $|\boldsymbol{K}| = |\boldsymbol{D}|\rho\,\delta_3$ of case DU of Proposition 2 to the limit factor by factor.

Lemma S1.4 (The demeaned limit, factored)**.** *Let*

$$\mathcal{W} = \frac{\theta}{2}\cot\frac{\theta}{2}, \qquad \bar{d}_2 = \frac{\tan(\theta/2)}{\theta}, \qquad \bar{d}_1 = \frac{2\bar{d}_2 - 1}{\theta^2},$$

and

$$\rho_\infty = 1 + (\theta^2 + c^2)\bar{d}_1, \qquad \mathcal{S}_\infty = \theta\cot\theta - \omega + \frac{(\theta^2 + c^2)\bar{d}_2^2 + 2\mathrm{i}t_1\bar{d}_2 + t_1^2\bar{d}_1}{\rho_\infty}, \tag{S.4}$$

for fixed (c, t_1) all meromorphic functions of θ^2, analytic on the parameter domain with removable singularities at $\theta^2 = 0$ (note $\theta^2 + c^2 = 2\mathrm{i}(t_2 - ct_1)$). Then $\tau_3/\theta^2 = 4\,\mathcal{W}\rho_\infty\mathcal{S}_\infty$, so that (S.1) equals

$$\mathbb{E}\big[e^{\mathrm{i}t_1\hat{A} + \mathrm{i}t_2\hat{B}}\big] = e^{-\omega/2}(\mathcal{S}\rho_\infty\mathcal{S}_\infty)^{-1/2}\exp\Big(-\frac{\varkappa^2\tau_0\bar{d}_1}{2\rho_\infty\mathcal{S}_\infty}\Big),$$

which is (44), because $\theta/\sin(\theta/2) = 2/\mathcal{S}_h$, $\mathcal{S} = \mathcal{W}\mathcal{S}_h^2$ and $(1 - \mathcal{W})/(\theta^2\mathcal{W}) = \bar{d}_1$, with $\mathcal{W} = \mathcal{C}_h/\mathcal{S}_h$. Moreover, for every real (t_1, t_2) and every c, $\operatorname{Re}\mathcal{W} \geq (c/2)\coth(c/2) \geq 1$, ρ_∞ is not on the closed negative real axis, and $\operatorname{Re}\mathcal{S}_\infty \geq 2c/(e^{2c} - 1) > 0$ (the bound is 1 at $c = 0$). Consequently $\operatorname{Arg}\mathcal{W} + \operatorname{Arg}\rho_\infty + \operatorname{Arg}\mathcal{S}_\infty$ is the continuous phase of τ_3/θ^2, and $\mathcal{S}_h^{-1}(\mathcal{W}\rho_\infty\mathcal{S}_\infty)^{-1/2}$, with these principal arguments, is the continuous branch of $(\mathcal{S}\rho_\infty\mathcal{S}_\infty)^{-1/2}$.

The proof carries the finite-sample factors $n^{-1}|\boldsymbol{D}|$, ρ_n and $n\delta_3$ to their limits $e^\omega\mathcal{S}$, ρ_∞ and $\mathcal{S}_\infty$ under $(t_1, t_2) \mapsto (t_1/n, t_2/n^2)$ and $\phi = 1 + c/n$, where ρ_n is the finite-sample rank-one factor after this substitution.

Proof of Lemma S1.4.

The limits of the primitives. Put $t_1 \mapsto t_1/n$, $t_2 \mapsto t_2/n^2$ and $\phi = 1 + c/n$ in (9) of the paper. Then

$$q = -(1 + \omega/n), \qquad p + 2q = -\theta^2/n^2, \qquad x = \frac{p}{2q} = -1 + \frac{\theta^2}{2n^2} + O(n^{-3}),$$

the first two identities being exact. Write $x = \cos(\pi - \varepsilon)$ with $\varepsilon = \theta/n + O(n^{-2})$, so that $n\varepsilon = \theta + O(n^{-1})$. The Chebyshev form

$$U_m(\cos(\pi - \varepsilon)) = (-1)^m \frac{\sin((m+1)\varepsilon)}{\sin \varepsilon},$$

with $m + 1 = n$, gives the exact identities

$$U_m(x) = (-1)^m \frac{\sin(n\varepsilon)}{\sin \varepsilon}, \qquad \frac{U_{m-1}(x)}{U_m(x)} = -\frac{\sin((n-1)\varepsilon)}{\sin(n\varepsilon)} = -\cos\varepsilon + \sin\varepsilon \cot(n\varepsilon),$$

and hence the asymptotic ones

$$U_m(x) = (-1)^m n\,\mathcal{S}\left(1 + O(n^{-1})\right), \qquad \frac{U_{m-1}(x)}{U_m(x)} = -1 + \frac{\theta \cot\theta}{n} + O(n^{-2}).$$

Hence, from Lemma 2 and Section 4.1 of the paper,

$$\begin{aligned} &|\boldsymbol{D}| = q^m U_m(x) = e^{\omega} n\,\mathcal{S}\,(1 + O(n^{-1})), \\ &q^2 d_3 = 1 + \frac{\omega - \theta\cot\theta}{n} + O(n^{-2}), \qquad qd_4 = -\frac{1}{n\mathcal{S}}(1 + O(n^{-1})). \end{aligned}$$

In case DK this gives $n\delta_3 = n(1 - q^2 d_3) \to \theta\cot\theta - \omega$. Since $|\boldsymbol{K}| = |\boldsymbol{D}|\delta_3$ in that case and $\mathcal{S}\,\theta\cot\theta = \mathcal{C}$,

$$|\boldsymbol{K}| = \left(n^{-1}|\boldsymbol{D}|\right)\left(n\delta_3\right) \to e^{\omega}\mathcal{S}(\theta\cot\theta - \omega) = e^{\omega}(\mathcal{C} - \omega\mathcal{S}),$$

so that $|\boldsymbol{K}|^{-1/2}$ tends to the factor $e^{-\omega/2}(\mathcal{C} - \omega\mathcal{S})^{-1/2}$ of (43). In case DU,

$$1 + q(d_3 + d_4) = -\frac{\theta\tan(\theta/2)}{n} + O(n^{-2}),$$

whence

$$d_2 = \frac{1 + q(d_3 + d_4)}{p + 2q} = n\bar{d}_2\big(1 + O(n^{-1})\big), \qquad d_1 = \frac{m + 2qd_2}{p + 2q} = n^3\bar{d}_1\big(1 + O(n^{-1})\big),$$

with $\bar{d}_2$ and $\bar{d}_1$ as in the lemma.

The rank-one factor. Since

$$\kappa = \frac{2\mathrm{i}[(1-\phi)t_1/n + t_2/n^2]}{n} = \frac{\theta^2 + c^2}{n^3},$$

the rank-one factor converges:

$$\rho_n = 1 + \kappa d_1 \to \rho_\infty = 1 + (\theta^2 + c^2)\bar{d}_1.$$

The border factor. The border Schur complement of case DU is $\delta_3 = 1 - \boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b}$, with $b_1 = \mathrm{i}t_1/n^2$ and, from (20),

$$w_1 = d_1/\rho_n, \qquad w_2 = d_2/\rho_n, \qquad w_3 = d_3 - \kappa d_2^2/\rho_n.$$

Inserting the limits gives, up to $O(n^{-2})$,

$$b_1^2 w_1 = -\frac{t_1^2 \bar{d}_1}{n\rho_\infty}, \qquad 2b_1 q w_2 = -\frac{2\mathrm{i}t_1 \bar{d}_2}{n\rho_\infty}, \qquad q^2 w_3 = q^2 d_3 - \frac{(\theta^2 + c^2)\bar{d}_2^2}{n\rho_\infty},$$

so $n\delta_3 \to \mathcal{S}_\infty$ as defined in (S.4) above. Therefore

$$|\boldsymbol{K}| = |\boldsymbol{D}|\rho_n\delta_3 = e^\omega \mathcal{S}\rho_\infty\mathcal{S}_\infty\big(1 + O(n^{-1})\big).$$

The factorization. The identity $\tau_3/\theta^2 = 4\mathcal{W}\rho_\infty\mathcal{S}_\infty$ is algebraic. Put $z = \theta^2$ and $D = \bar{d}_2 = \tan(\theta/2)/\theta$. The half-angle formulas express every function of θ involved as a rational function of D and z:

$$\mathcal{W} = \frac{1}{2D}, \qquad \theta\cot\theta = \frac{1}{2D} - \frac{zD}{2}, \qquad \bar{d}_1 = \frac{2D-1}{z}, \qquad \frac{\mathcal{C}}{\mathcal{S}_h^2} = \frac{1 - zD^2}{4D^2}, \qquad \frac{\mathcal{S}}{\mathcal{S}_h^2} = \frac{1}{2D},$$

and, with t_2 eliminated through $z = 2\mathrm{i}t_2 - 2\mathrm{i}ct_1 - c^2$,

$$\tau_1 = 2(t_1^2 - \mathrm{i}c^2 t_1 - cz - c^3), \qquad \tau_2 = c^3 + c^2 + \mathrm{i}c^2 t_1 - t_1^2 + z, \qquad \frac{\tau_3}{\theta^2} = \frac{1}{z}\Big[2\tau_1 - 4c^2\,\frac{\mathcal{C}}{\mathcal{S}_h^2} + 4\tau_2\,\frac{\mathcal{S}}{\mathcal{S}_h^2}\Big].$$

Substituting these and (S.4) into the two sides gives the same rational function of D, z, c and t_1 (a routine computer-algebra check), so the identity holds wherever both sides are defined, and at the removable values by continuity. It can also be read off the convergence of the finite-sample characteristic function: with $\varkappa = 0$, $|\boldsymbol{K}|^{-1/2}$ at $(t_1/n, t_2/n^2)$ converges

to (S.1), so

$$\frac{e^{\omega}\mathcal{S}_h^2\tau_3}{4\theta^2} = e^{\omega}\mathcal{S}\rho_{\infty}\mathcal{S}_{\infty},$$

which is the same statement since $\mathcal{S}/\mathcal{S}_h^2 = \mathcal{W}$.

Re $\mathcal{W} > 0$. This is the energy argument of Lemma S1.1 with $\theta/2$ in place of θ: $(\theta/2)^2 = \theta^2/4$ has real part $-c^2/4 \le 0$, so $\sin(\theta/2)/(\theta/2) \neq 0$, the auxiliary function $u(r) = \sin(\theta r/2)/\sin(\theta/2)$ on $[0,1]$ satisfies $u(0) = 0$, $u(1) = 1$, $u'' = -(\theta^2/4)u$, and

$$\mathcal{W} = u'(1) = \int_0^1 \left\{|u'(r)|^2 - \tfrac{1}{4}\theta^2|u(r)|^2\right\} \mathrm{d}r, \qquad \text{Re } \mathcal{W} = \int_0^1 \left\{|u'(r)|^2 + \tfrac{1}{4}c^2|u(r)|^2\right\} \mathrm{d}r > 0.$$

The minimum of the last integral over functions with $u(0) = 0$ and $u(1) = 1$ is attained at $\sinh(cr/2)/\sinh(c/2)$, so that Re $\mathcal{W} \ge (c/2)\coth(c/2) \ge 1$.

ρ_∞ off the negative axis. Let

$$L = -\frac{\mathrm{d}^2}{\mathrm{d}r^2} - \theta^2 \quad \text{on } [0,1], \qquad u(0) = u(1) = 0,$$

the limit of $n^2\boldsymbol{D}$ acting on functions sampled at $r = k/n$: the core $\boldsymbol{D}$ has eigenvectors $\sin(jk\pi/n)$ and eigenvalues

$$p + 2q\cos(j\pi/n) = \frac{(j\pi)^2 - \theta^2}{n^2} + O(n^{-3}).$$

Since θ^2 is never a positive real number, $\cos(\theta/2) \neq 0$, and the solution u of $Lu = 1$ (again an auxiliary function) and its integral are

$$u(r) = \frac{1}{\theta^2}\Big[\frac{\cos(\theta(r - \frac{1}{2}))}{\cos(\theta/2)} - 1\Big], \qquad \bar{d}_1 = \int_0^1 u(r)\,\mathrm{d}r = \frac{1}{\theta^2}\Big[\frac{2\tan(\theta/2)}{\theta} - 1\Big],$$

the continuum analog of $d_1 = \mathbf{1}_m'\boldsymbol{D}^{-1}\mathbf{1}_m$. Multiplying $Lu = 1$ by $\bar{u}$ and integrating by parts, with the zero boundary values,

$$\int_0^1 \bar{u}(r)\,\mathrm{d}r = \int_0^1 \left\{|u'(r)|^2 - \theta^2|u(r)|^2\right\} \mathrm{d}r,$$

so that

$$\text{Re } \bar{d}_1 = \int_0^1 \left\{|u'(r)|^2 + c^2|u(r)|^2\right\} \mathrm{d}r > 0,$$

because $\text{Re}(-\theta^2) = c^2$ and $u \neq 0$. As $\theta^2 + c^2 = 2\mathrm{i}(t_2 - ct_1)$ is purely imaginary, Im $\rho_\infty =$

$2(t_2 - ct_1)\,\mathrm{Re}\,\bar{d}_1$ vanishes only when $\theta^2 + c^2 = 0$, where $\rho_\infty = 1$: so ρ_∞ is never a nonpositive real number. (This is Lemma 3(b) of the paper with the matrix $\boldsymbol{D}$ replaced by the operator L; it needs no homotopy.)

$\mathcal{S}_\infty$. For every n, δ_3 is the Schur complement of the accretive matrix $\boldsymbol{\Delta}$ in the accretive matrix $\boldsymbol{K}$ of case DU, so $\delta_3 = \boldsymbol{h}^*\boldsymbol{K}\boldsymbol{h}$ for the vector $\boldsymbol{h}$ with $h_n = 1$ and $(h_1, \ldots, h_{n-1})' = -\boldsymbol{\Delta}^{-1}\boldsymbol{b}$ (proof of Lemma 3(c) of the paper), and, with $\mathrm{Re}\,\boldsymbol{K} = (\boldsymbol{I}_n - \phi\boldsymbol{J}_n)'(\boldsymbol{I}_n - \phi\boldsymbol{J}_n)$ from Theorem 1 and $\phi = 1 + c/n$,

$$\mathrm{Re}\,\delta_3 = \boldsymbol{h}^*(\mathrm{Re}\,\boldsymbol{K})\boldsymbol{h} = \sum_{t=1}^{n} |h_t - \phi h_{t-1}|^2, \qquad h_0 = 0.$$

Since, by telescoping,

$$1 = h_n = \sum_{t=1}^{n} \phi^{n-t}(h_t - \phi h_{t-1}),$$

the Cauchy–Schwarz inequality gives

$$\mathrm{Re}\,\delta_3 \geq \Big(\sum_{j=0}^{n-1} \phi^{2j}\Big)^{-1} = \frac{\phi^2 - 1}{\phi^{2n} - 1},$$

so that

$$\mathrm{Re}\,\mathcal{S}_\infty = \lim_{n\to\infty} n\,\mathrm{Re}\,\delta_3 \geq \lim_{n\to\infty} \frac{n(\phi^2 - 1)}{\phi^{2n} - 1} = \frac{2c}{e^{2c} - 1} > 0,$$

with the value 1 at $c = 0$. Hence $\mathcal{S}_\infty$ lies in the open right half-plane, and the three principal arguments are continuous functions of (t_1, t_2) whose sum is, modulo 2π, the phase of τ_3/θ^2 and vanishes at the origin: it is the continuous phase. □

We have also checked the bounds of Lemmas S1.2 and S1.4 numerically. The grid has the eleven values

$$c \in \{-30, -10, -3, -1, -0.3, -0.05, 0.05, 0.3, 1, 3, 10\},$$

35 values of r in $[-60, 8]$, and 3000 logarithmically spaced values of t_1 in $[10^{-5}, 10^{3.5}]$ along each path $t_2 = -rt_1$. Over this grid:

- $\mathrm{Re}\,\mathcal{W} \geq 1.0002$ (the bound $(c/2)\coth(c/2)$ at $c = \pm 0.05$), $|\mathrm{Arg}\,\rho_\infty| \leq 1.35$, and $\mathrm{Re}\,\mathcal{S}_\infty \geq 2c/(e^{2c} - 1)$, the bound being approached only at the smallest values of t_1, near the origin, where it is attained;

Table S1: Convergence of the finite-sample characteristic function to its limit

Case	Parameters	$n = 500$	$n = 2000$	$n = 8000$
DK	$(0, \gamma_0 = 0)$	2.8×10^{-2}	7.2×10^{-3}	1.8×10^{-3}
DK	$(0, \gamma_0 = 1.2)$	1.4×10^{-2}	3.6×10^{-3}	9.1×10^{-4}
DK	$(-5, \gamma_0 = 0.7)$	3.0×10^{-3}	7.7×10^{-4}	1.9×10^{-4}
DK	$(3, \gamma_0 = 0.5)$	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}
DU	$(0, \varkappa = 0)$	2.5×10^{-2}	6.5×10^{-3}	1.6×10^{-3}
DU	$(0, \varkappa = 1.5)$	8.6×10^{-3}	2.2×10^{-3}	5.6×10^{-4}
DU	$(-8, \varkappa = 0.6)$	1.1×10^{-2}	2.7×10^{-3}	6.8×10^{-4}
DU	$(2, \varkappa = -0.4)$	6.8×10^{-3}	1.8×10^{-3}	4.5×10^{-4}
SK	$c = -4$	5.7×10^{-3}	1.5×10^{-3}	3.7×10^{-4}
SK	$c = -0.5$	5.5×10^{-3}	1.4×10^{-3}	3.6×10^{-4}
SU	$c = -6$	1.1×10^{-2}	2.9×10^{-3}	7.2×10^{-4}
SU	$c = -0.2$	2.5×10^{-2}	6.4×10^{-3}	1.6×10^{-3}

Notes: Maximum of $|\varphi_n(t_1/n, t_2/n^2) - \varphi_\infty(t_1, t_2)|$ over the six points $(t_1, t_2) \in \{(10^{-3}, 0.3), (0.1, -1), (0.5, 4), (2, -4), (5, 1.5), (15, -0.2)\}$, where φ_n is the exact characteristic function of Theorem 1 at $\phi = 1 + c/n$ (with $z_0 = \sqrt{n}\gamma_0$ in case DK and $\gamma = \varkappa/\sqrt{n}$ in case DU) and φ_∞ is (43)–(46). The differences fall by a factor of four as n quadruples, i.e. they are $O(1/n)$.

- the sum of the three arguments stays in $(-1.6, 1.6)$ and agrees with the principal argument of τ_3/θ^2 to 4×10^{-16} everywhere (on this grid τ_3/θ^2 does not cross the negative real axis; the lemma is what guarantees continuity in general);
- the smallest real part of either mixing factor of Lemma S1.2 (at the six negative values of c) equals one to twelve digits.

Independently, Table S1 shows that the limiting characteristic functions, evaluated on these branches, agree with $\varphi_n(t_1/n, t_2/n^2)$ to $O(1/n)$ in all four cases. At larger arguments, where $\cos\theta$ winds around the origin, φ_n/φ_∞ likewise tends to $+1$ (for instance at (t_1, t_2) up to $(100, -10)$ and $(50, -350)$, with $c = -7$ and 1.5 and n up to 3000), which rules out a sign error in the branches.

S1.5 Moments of the limits

The local-to-unity limits of Proposition 4 of the paper have moments of all orders. For fixed admissible parameters, write B_* for the relevant denominator (B or $\hat{B}$). The Laplace

transforms $\mathbb{E}[e^{-sB_*}]$, given by the characteristic functions of Proposition 4 at imaginary arguments, satisfy, for every $0 < a < 1/\sqrt{2}$, $\mathbb{E}[e^{-sB_*}] \le C_a e^{-a\sqrt{s}}$ for all sufficiently large s, with a finite positive constant C_a; polynomial prefactors are absorbed by reducing the exponential constant, and a deterministic Gaussian shift cannot increase this Laplace transform. Markov's inequality with $\sqrt{s} = a/(2\ell)$ gives $\mathbb{P}[B_* \le \ell] \le C_a e^{-a^2/(4\ell)}$ for sufficiently small $\ell > 0$, where $a^2/4$ can be any number below $1/8$. Thus B_* has negative moments of all orders. The numerator is a Gaussian linear-plus-quadratic functional with positive moments of all orders, and Hölder's inequality proves the claim.

S2 The distribution of $A + cB$

Setting $t_2 = ct_1$ in (43) makes $\theta = \mathrm{i}c$ independent of t_1 and gives the characteristic function of $A + cB$ in case DK:

$$\mathbb{E}[e^{\mathrm{i}t_1(A+cB)}] = \frac{e^{-\mathrm{i}t_1(1+\gamma_0^2)/2}}{(1-\mathrm{i}c_0t_1)^{1/2}} \exp\Big(-\frac{\delta}{2} + \frac{\delta}{2(1-\mathrm{i}c_0t_1)}\Big),$$

where

$$c_0 = \frac{e^{2c}-1}{2c}, \qquad \delta = \frac{\gamma_0^2 e^{2c}}{c_0}.$$

Hence

$$A + cB \sim \tfrac{1}{2}\big[c_0\chi_1^2(\delta) - (1+\gamma_0^2)\big],$$

with $c_0 = 1$ and $\delta = \gamma_0^2$ at $c = 0$. In case DU, with $c_4 = (e^c - 1)/c$ as in Section S1, let

$$c_5 = c_0 - c_4^2, \qquad g_5 = \Big(c_5^2 + \frac{4c_5}{c^2}\Big)^{1/2}.$$

Then

$$\hat{A} + c\hat{B} \sim \lambda_1 X_1 - \lambda_2 X_2 - \tfrac{1}{2}, \qquad \lambda_{1,2} = \frac{g_5 \pm c_5}{4},$$

where $X_1 \sim \chi_1^2(\mu_1)$ and $X_2 \sim \chi_1^2(\mu_2)$ are independent, with

$$\mu_1 = \frac{\varkappa^2}{2}\Big[1 - \frac{(c-4)c_5}{cg_5}\Big], \qquad \mu_2 = \varkappa^2 - \mu_1.$$

At $c = 0$, $\lambda_1 = \lambda_2 = 1/(4\sqrt{3})$ and $\mu_1 = \mu_2 = \varkappa^2/2$. These characteristic functions decay like $t_1^{-1/2}$ and t_1^{-1}, which is the slow decay along the ray $t_2 = ct_1$, that is, at $r = -c$,

discussed in Section 6.1 of the paper.

They also give closed forms at $r = -c$. In case DK, with $a = \sqrt{(1+\gamma_0^2)/c_0}$,

$$\mathbb{P}[A/B < -c] = \mathbb{P}\big[\chi_1^2(\delta) < (1+\gamma_0^2)/c_0\big] = \Phi_{\mathcal{N}}(a - \sqrt{\delta}) - \Phi_{\mathcal{N}}(-a - \sqrt{\delta}).$$

In case DU, writing $X_2 = (Z + \sqrt{\mu_2})^2$ with Z standard normal,

$$\begin{aligned}\mathbb{P}[\hat{A}/\hat{B} < -c] &= \mathbb{P}\big[\lambda_1 X_1 - \lambda_2 X_2 < \tfrac{1}{2}\big] \\ &= \int_{-\infty}^{\infty} F_1\Big(\frac{\frac{1}{2} + \lambda_2 (z + \sqrt{\mu_2})^2}{\lambda_1}\Big)\varphi_{\mathcal{N}}(z)\mathrm{d}z,\end{aligned}$$

where

$$F_1(x) = \Phi_{\mathcal{N}}(\sqrt{x} - \sqrt{\mu_1}) - \Phi_{\mathcal{N}}(-\sqrt{x} - \sqrt{\mu_1})$$

is the noncentral χ_1^2 distribution function. The programs integrate over $z \in [-12, 12]$ (the omitted normal mass is below 4×10^{-33}), split at those of the following points that lie in $[-12, 12]$: $z = -\sqrt{\mu_2}$, where the argument of F_1 is smallest, and the points where that argument equals μ_1. They form λ_2 as $(c_5/c^2)/(g_5 + c_5)$, which avoids the cancellation in $(g_5 - c_5)/4$. In cases DK and DU at $c = 0$ and $\gamma_0 = \varkappa = 0$ these probabilities are

$$2\Phi_{\mathcal{N}}(1) - 1 = 0.68268949 \qquad \text{and} \qquad \mathbb{P}[X_1 - X_2 < 2\sqrt{3}] = 0.95756559,$$

the values quoted in the paper. Near $c = 0$, $c_5 = c_0 - c_4^2$ is formed from its series

$$c_5 = c^2\Big(\frac{1}{12} + \frac{c}{12} + \frac{17c^2}{360} + \cdots\Big),$$

because the difference cancels to $O(c^2)$. The validation suite checks this integral against 60-digit references at $\varkappa = 200$ and 1000, together with the continuity of the distribution function across $r = -c$. The stationary cases are the mixtures over $\gamma_0 \sim \mathcal{N}(0, -1/(2c))$ (in case SU, with $b = 0$, over $\varkappa = c\gamma_0$): case SU by an outer quadrature over the standardized γ_0 on $[-12, 12]$, and case SK in the closed form below.

In case SK ($c < 0$) the mixture over γ_0 is no longer a shifted noncentral χ_1^2 variable, because both the shift and the noncentrality of the case-DK representation depend on γ_0; it has a representation of its own. Here $X(r) = \gamma_0 e^{cr} + J_c(r)$, with $\gamma_0 \sim \mathcal{N}(0, -1/(2c))$ independent of W, is the stationary Ornstein–Uhlenbeck process. Itô's formula gives $A + cB =$

$(X(1)^2 - X(0)^2 - 1)/2$, where $\operatorname{Var} X(0) = \operatorname{Var} X(1) = -1/(2c)$ and $\operatorname{Corr}(X(0), X(1)) = e^c$. Consequently

$$A + cB \overset{d}{=} \lambda_c(Z_1^2 - Z_2^2) - \tfrac{1}{2}, \qquad \lambda_c = \frac{\sqrt{1 - e^{2c}}}{-4c}, \qquad Z_1, Z_2 \text{ independent } \mathcal{N}(0,1),$$

with characteristic function $e^{-\mathrm{i}t_1/2}/\sqrt{1 + 4\lambda_c^2 t_1^2}$, and

$$\mathbb{P}[A/B < -c] = \mathbb{P}\Big[Z_1^2 - Z_2^2 < \frac{1}{2\lambda_c}\Big] = \int_{-\infty}^{\infty} \operatorname{erf}\Big(\Big(\frac{a_c + z^2}{2}\Big)^{1/2}\Big)\varphi_{\mathcal{N}}(z)\mathrm{d}z, \qquad a_c = \frac{1}{2\lambda_c},$$

since $\mathbb{P}[\chi_1^2 < y] = \operatorname{erf}(\sqrt{y/2})$; this is what the programs evaluate in case SK (as $c \uparrow 0$, $a_c \to 0$ and the probability tends to $1/2$).

S3 Taylor-mode arithmetic and error estimates

A jet of order K in a scalar variable s about s_0 is the vector $(z^{[0]}, \ldots, z^{[K]})$ of Taylor coefficients of a function

$$z(s) = \sum_{k=0}^{K} z^{[k]}(s - s_0)^k + O\big((s - s_0)^{K+1}\big).$$

For jets a and b, with coefficients written $a_k = a^{[k]}$ and $b_k = b^{[k]}$ $(0 \le k \le K)$ for brevity, and a constant λ, the programs use the following recurrences (Griewank and Walther, 2008, Chapter 13):

$$(a + \lambda)_k = a_k + \lambda\,\mathbf{1}\{k = 0\}, \qquad (\lambda a)_k = \lambda a_k, \qquad (ab)_k = \sum_{j=0}^{k} a_j b_{k-j},$$

$$(a/b)_k = \frac{1}{b_0}\Big(a_k - \sum_{j=1}^{k} b_j (a/b)_{k-j}\Big), \qquad (\sqrt{a})_k = \frac{1}{2\sqrt{a_0}}\Big(a_k - \sum_{j=1}^{k-1} (\sqrt{a})_j (\sqrt{a})_{k-j}\Big),$$

$$(\log a)_k = \frac{1}{a_0}\Big(a_k - \sum_{j=1}^{k-1} \frac{j}{k} (\log a)_j a_{k-j}\Big), \qquad (e^a)_k = \sum_{j=1}^{k} \frac{j}{k} a_j (e^a)_{k-j},$$

$$(\cos a)_k = -\sum_{j=1}^{k} \frac{j}{k} a_j (\sin a)_{k-j}, \qquad (\sin a)_k = \sum_{j=1}^{k} \frac{j}{k} a_j (\cos a)_{k-j},$$

where the recurrences for the square root, logarithm, exponential, cosine and sine hold for $k \geq 1$, with $(\sqrt{a})_0 = \sqrt{a_0}$, $(\log a)_0 = \log a_0$, $(e^a)_0 = e^{a_0}$, $(\cos a)_0 = \cos a_0$ and $(\sin a)_0 = \sin a_0$ on the chosen analytic branches, and $\mathbf{1}\{k = 0\}$ equals one if $k = 0$ and zero otherwise. Division requires $b_0 \neq 0$, and the logarithm and square-root recurrences require $a_0 \neq 0$. For a nonnegative integer m, a^m is computed by repeated squaring when $m \leq 12$. Otherwise, with the increment $h = s - s_0$, the leading power is first factored out, $a = h^\ell b$ with ℓ the number of leading coefficients that are exactly zero and $b_0 \neq 0$, so that $a^m = h^{\ell m} b_0^m (b/b_0)^m$, and the power of the normalized jet $u = b/b_0$, $u_0 = 1$, comes from the recurrence for powers of a power series,

$$k\,(u^m)_k = \sum_{j=1}^{k} \big[(m+1)j - k\big]\, u_j\, (u^m)_{k-j}, \qquad (u^m)_0 = 1.$$

The recurrence is applied to the rescaled coefficients $\tilde{u}_j = \varsigma^j u_j$, with ς a power of two chosen so that $\max_{1 \leq j \leq K} |\tilde{u}_j| \leq 1$; the coefficient of order k of the result is then multiplied by ς^{-k}. The factor b_0^m is formed in a fixed number of operations as $2^{\ell - E} e^{\mathrm{i} m \arg b_0} \cdot 2^E$, with $\ell = m \log_2 |b_0|$ and $E = \lfloor \ell \rfloor$, so that its binary exponent is carried separately (for a real negative b_0 the sign $(-1)^m$ is applied exactly), and the exponents of b_0^m and of ς^{-k} are applied at the end. Its relative error is about $\epsilon_{\mathrm{M}} |m \log b_0|$, with ϵ_{M} the machine epsilon of Section 4.3, the same order as the error of order $m\epsilon_{\mathrm{M}}$ that the rounding of b_0 itself propagates into b_0^m. This limits, but does not exclude, overflow: the intermediate coefficients $(\tilde{u}^m)_k$ can still grow with m. A column whose result is not finite is therefore recomputed by repeated squaring of the original jet, a safeguard of $O(\log m)$ jet multiplications, and the recomputed column is used when all its coefficients are finite. In exact arithmetic the result does not depend on the rescaling, also when the constant term is tiny relative to the other coefficients: for instance $(1 + 10^{15} h)^{13} = 1 + 1.3 \times 10^{16} h + O(h^2)$. A vanishing constant term occurs at $\phi = t_1 = 0$, where $q = -(\phi + \mathrm{i} t_1)$ has constant term zero. The jet logarithm returns only the derivative coefficients, which carry no branch. The constant term of each logarithm is then set from the branch results. That of $\log |\boldsymbol{D}|$ is set by Theorem 2; those of $\log \rho$, $\log \delta_4$, $\log \delta_{\mathsf{DK}}$, $\log S$ and of the logarithms of the mixing factors by their principal values (Theorem 2(ii), Proposition 3 and Lemma S1.2), and those of the limiting factors as in Section S1. With $K = 0$ the arithmetic is ordinary complex arithmetic. The same code evaluates the characteristic function ($K = 0$), its t_2-derivative ($K = 1$) and the derivatives in s_1 of the moment generating function ($K = k$). Apart from

that safeguard, the cost is independent of n and polynomial in K. Each of the recurrences above costs $O(K^2)$. The composition of the Chebyshev expansions with a jet, described next, costs $O(K^3)$, because it is carried out by Horner's rule with K jet multiplications.

Two details are specific to the primitives. In the roots regime the formulas are written in $w = q/r_+$, so that $r_-/r_+ = w^2$. The only n-th powers that occur are then w^{2n}, w^{2n-2}, w^{n-2} and, in the explosive forms of Proposition 3, $(-w)^n$, all of modulus at most one, while r_+^n enters only through $n \log r_+$, so that no power overflows. The derivative coefficients can still be large, and their size is part of the conditioning of the derivative. In the confluent regime the Taylor expansions of the Chebyshev polynomials U_m and U_{m-1}, with $m = n-1$, at $x = \pm 1$ have their coefficients carried against powers of $\tau = n^2(x \mp 1)$, so that none of them overflows at any n. For a truncated expansion $\sum_{k=0}^{J} A_k \tau^k$ of degree $J \geq K$, its coefficients are first shifted to the evaluation point τ_*,

$$\sum_{k=0}^{J} A_k \tau^k = \sum_{k=0}^{J} \tilde{A}_k (\tau - \tau_*)^k, \qquad \tilde{A}_k = \sum_{j=0}^{J-k} \binom{k+j}{k} A_{k+j} \tau_*^j,$$

with the binomial coefficients and the powers of τ_* carried by their own recursions, and are then composed with the jet of $\tau - \tau_*$ by Horner's rule.

Error estimates

With each value of the distribution function, the density or a moment, the programs return an estimate of its error. The estimate is the sum of three parts.

The first is the quadrature error, estimated by the adaptive Gauss–Kronrod rule.

The second is the rounding error of the characteristic function. Away from the unit root it is a few units of ϵ_{M}. Near the unit root the rank-one and border factors are formed from quantities of order n or n^2, and the rounding error is of order $n\epsilon_{\mathrm{M}}$, about 10^{-10} at $n = 10^6$. With a drift in case DU it can be larger, of order $n^2\gamma^2\epsilon_{\mathrm{M}}$: the noncentral exponent of Theorem 1 of the paper is the difference of two terms of order $n\gamma^2$, each with a relative rounding error of order $n\epsilon_{\mathrm{M}}$. To flag this loss, the programs add the computed value of $|\log \varphi(0,0)|$, which is zero in exact arithmetic, to the estimate; at $n = 10^6$ and $\gamma = 100$ it is 0.14.

The third, for the limits only, is the truncation error of the asymptotic expansion of the tail of the inversion integral (Section 6.1 of the paper).

The programs warn when the estimate exceeds the tolerance. The estimates are diagnostics, not certified bounds. The user manual gives further details.

S4 Additional tables and figures

Additional critical values

Tables S2 and S3 extend Table 4 of the paper to further sample sizes and probabilities. Each entry is the quantile x_p defined by $\mathbb{P}[n(\hat{\phi}-1) < x_p] = p$, at $\phi = 1$, $z_0 = 0$ in DK and $\gamma = 0$ in DU, to three decimals in Table S2 and four in Table S3.

Comparisons with published tables. The exact quantiles of Table 4 of the paper can be set against Table 10.A.1 of Fuller (1996). Fuller's table, constructed by simulation (50,000 replications at each n, smoothed), carries errors of about ± 0.1 in the tails; the exact values confirm it to its stated precision. In the no-intercept case the correspondence is immediate: at $n = 25$ the exact quantiles -11.920, -9.363, -7.371, -5.350, 1.012, 1.411, 1.789, 2.294 agree with Fuller's -11.9, -9.3, -7.3, -5.3, 1.01, 1.40, 1.79, 2.28 to within 0.08. In the intercept case there is a point of definition that the exact computation brings to light. With $y_0 = 0$ the first lagged value contributes nothing to the sums without an intercept but does enter the sample mean of the regressor. Fuller's $\hat{\rho}_\mu$ panel, where $\hat{\rho}_\mu$ denotes his intercept-estimated autoregressive coefficient, matches the exact distribution only if the regression behind it runs over $t = 2, \ldots, n$, with $n - 1$ observations, while the statistic is scaled by n (at $\phi = 1$ the distribution in case DU does not depend on y_0). His entry at sample size n then estimates $n/(n-1)$ times the exact quantile of $(n-1)(\hat{\phi}-1)$ from $n-1$ regression observations. At $n = 25$ this is -12.486 at 5% against Fuller's -12.5, and at $n = 100$ it is -13.663 against -13.7. The exact quantiles of $n(\hat{\phi}-1)$ from n regression observations, as in Table 4 of the paper, are instead -12.060 and -13.532. A two-million-replication simulation confirms both conventions to ± 0.01, and the limiting values (-8.039 and -14.094 at 5%) agree with the response surfaces of MacKinnon (1996) to the three digits he reports. The exact computations of Evans and Savin (1984) for case DU are reproduced as well. The 24 probabilities of their Table I ($n = 25$, 50, 100, $\phi \in \{0.9, 1\}$, $\gamma \in \{0, 1\}$) agree with ours to all five digits shown. All 216 percentage points of their Tables II and III (γ up to 10) agree to within one unit in the third decimal, 149 of them exactly, once their normalization is taken into account: they

scale $\hat{\phi} - \phi$ by their $S_T^{1/2}$ (their T is our n), where

$$S_T = \sum_{t=1}^{n} \mathbb{E}[w_{t-1}^2] - \frac{1}{n}\Big(\sum_{t=1}^{n} \mathbb{E}[w_{t-1}]\Big)^2$$

is computed from the moments of the undemeaned data $w_t = (y_t - y_0)/\sigma$, rather than by $\mathbb{E}[v]^{1/2}$. The two differ by a variance term:

$$\mathbb{E}[v] = S_T - \frac{1}{n}\,\mathrm{Var}\Big(\sum_{t=1}^{n} w_{t-1}\Big).$$

Table S2: Exact quantiles of $n(\hat{\phi} - 1)$ at $\phi = 1$, no intercept (case DK, $z_0 = 0$)

n	0.005	0.01	0.025	0.05	0.10	0.25	0.50	0.75	0.90	0.95	0.975	0.99	0.995
10	−11.201	−9.946	−8.099	−6.554	−4.887	−2.547	−0.755	0.399	1.165	1.645	2.129	2.815	3.381
15	−12.497	−10.950	−8.753	−6.982	−5.132	−2.632	−0.788	0.352	1.075	1.507	1.927	2.503	2.963
20	−13.272	−11.538	−9.125	−7.220	−5.266	−2.677	−0.804	0.331	1.035	1.445	1.839	2.368	2.786
25	−13.782	−11.920	−9.363	−7.371	−5.350	−2.704	−0.814	0.318	1.012	1.411	1.789	2.294	2.688
30	−14.142	−12.188	−9.529	−7.475	−5.407	−2.723	−0.820	0.310	0.997	1.388	1.757	2.246	2.626
40	−14.615	−12.538	−9.743	−7.609	−5.481	−2.747	−0.829	0.300	0.979	1.361	1.719	2.189	2.552
50	−14.912	−12.756	−9.876	−7.692	−5.526	−2.761	−0.833	0.294	0.968	1.345	1.696	2.156	2.509
75	−15.323	−13.057	−10.058	−7.804	−5.587	−2.781	−0.840	0.286	0.955	1.325	1.667	2.113	2.454
100	−15.536	−13.212	−10.151	−7.862	−5.618	−2.791	−0.843	0.282	0.948	1.315	1.653	2.093	2.428
150	−15.753	−13.370	−10.246	−7.920	−5.650	−2.801	−0.846	0.279	0.941	1.305	1.639	2.072	2.401
200	−15.864	−13.450	−10.294	−7.950	−5.666	−2.806	−0.848	0.277	0.938	1.300	1.632	2.062	2.388
250	−15.931	−13.499	−10.323	−7.967	−5.675	−2.809	−0.849	0.276	0.936	1.297	1.628	2.056	2.381
300	−15.975	−13.531	−10.342	−7.979	−5.682	−2.811	−0.850	0.275	0.935	1.295	1.626	2.052	2.376
400	−16.032	−13.572	−10.367	−7.994	−5.690	−2.813	−0.850	0.274	0.933	1.293	1.622	2.047	2.369
500	−16.066	−13.596	−10.381	−8.003	−5.694	−2.815	−0.851	0.273	0.932	1.291	1.620	2.044	2.365
750	−16.111	−13.629	−10.401	−8.015	−5.701	−2.817	−0.852	0.273	0.931	1.289	1.618	2.040	2.360
1000	−16.134	−13.646	−10.411	−8.021	−5.704	−2.818	−0.852	0.272	0.930	1.288	1.616	2.038	2.358
2000	−16.168	−13.671	−10.425	−8.030	−5.709	−2.819	−0.852	0.272	0.929	1.287	1.614	2.035	2.354
5000	−16.189	−13.685	−10.434	−8.036	−5.712	−2.820	−0.853	0.271	0.928	1.286	1.613	2.034	2.352
∞	−16.203	−13.695	−10.440	−8.039	−5.714	−2.821	−0.853	0.271	0.928	1.285	1.612	2.033	2.350

Notes: Column headings are probabilities p; entries are the quantiles x_p satisfying $\mathbb{P}[n(\hat{\phi}-1) < x_p] = p$, for case DK at $\phi = 1$, $z_0 = 0$.

Confluence diagnostics

Table S4 compares relative errors of the two evaluations of the Toeplitz primitive $d_1 = \mathbf{1}_m' \boldsymbol{D}^{-1} \mathbf{1}_m$ at the transform origin. Figure S1 makes the same comparison along the path $t_2 = -r t_1$. Both support the confluence discussion in Sections 4.1 and 7.1 of the paper.

Table S3: Exact quantiles of $n(\hat{\phi}-1)$ at $\phi=1$, intercept estimated (case DU, $\gamma=0$)

n	0.005	0.01	0.025	0.05	0.10	0.25	0.50	0.75	0.90	0.95	0.975	0.99	0.995
10	−13.7762	−12.7483	−11.2044	−9.8644	−8.3415	−5.9630	−3.7382	−1.9245	−0.5273	0.2561	0.9597	1.8595	2.5526
15	−16.0203	−14.6530	−12.6547	−10.9778	−9.1350	−6.3809	−3.9215	−2.0329	−0.6417	0.1062	0.7507	1.5380	2.1190
20	−17.4483	−15.8350	−13.5266	−11.6320	−9.5908	−6.6130	−4.0213	−2.0859	−0.6955	0.0383	0.6584	1.3997	1.9356
25	−18.4218	−16.6311	−14.1047	−12.0604	−9.8856	−6.7606	−4.0840	−2.1177	−0.7267	−0.0005	0.6065	1.3229	1.8345
30	−19.1252	−17.2020	−14.5150	−12.3622	−10.0916	−6.8626	−4.1269	−2.1389	−0.7472	−0.0256	0.5732	1.2740	1.7706
40	−20.0710	−17.9645	−15.0582	−12.7590	−10.3606	−6.9944	−4.1820	−2.1655	−0.7724	−0.0561	0.5329	1.2153	1.6942
50	−20.6766	−18.4499	−15.4012	−13.0080	−10.5283	−7.0759	−4.2159	−2.1816	−0.7873	−0.0740	0.5094	1.1814	1.6502
75	−21.5321	−19.1318	−15.8797	−13.3534	−10.7597	−7.1874	−4.2618	−2.2031	−0.8069	−0.0975	0.4790	1.1375	1.5935
100	−21.9816	−19.4885	−16.1285	−13.5321	−10.8788	−7.2444	−4.2852	−2.2139	−0.8167	−0.1090	0.4641	1.1162	1.5659
150	−22.4464	−19.8561	−16.3838	−13.7150	−11.0003	−7.3023	−4.3088	−2.2247	−0.8263	−0.1204	0.4494	1.0952	1.5390
200	−22.6847	−20.0442	−16.5140	−13.8080	−11.0619	−7.3316	−4.3208	−2.2301	−0.8311	−0.1260	0.4421	1.0849	1.5257
250	−22.8296	−20.1584	−16.5929	−13.8643	−11.0992	−7.3492	−4.3280	−2.2334	−0.8340	−0.1294	0.4378	1.0787	1.5177
300	−22.9271	−20.2351	−16.6459	−13.9021	−11.1242	−7.3611	−4.3328	−2.2356	−0.8359	−0.1316	0.4349	1.0746	1.5125
400	−23.0498	−20.3317	−16.7125	−13.9496	−11.1556	−7.3759	−4.3388	−2.2383	−0.8383	−0.1344	0.4313	1.0695	1.5059
500	−23.1239	−20.3900	−16.7527	−13.9782	−11.1745	−7.3848	−4.3424	−2.2399	−0.8397	−0.1361	0.4291	1.0664	1.5020
750	−23.2234	−20.4682	−16.8065	−14.0165	−11.1997	−7.3968	−4.3472	−2.2421	−0.8416	−0.1383	0.4263	1.0624	1.4968
1000	−23.2733	−20.5074	−16.8335	−14.0357	−11.2124	−7.4027	−4.3497	−2.2432	−0.8426	−0.1394	0.4249	1.0604	1.4942
2000	−23.3486	−20.5665	−16.8742	−14.0646	−11.2315	−7.4117	−4.3533	−2.2449	−0.8440	−0.1411	0.4227	1.0573	1.4904
5000	−23.3939	−20.6021	−16.8987	−14.0820	−11.2429	−7.4171	−4.3555	−2.2458	−0.8449	−0.1421	0.4214	1.0555	1.4880
∞	−23.4243	−20.6259	−16.9150	−14.0936	−11.2506	−7.4207	−4.3569	−2.2465	−0.8454	−0.1428	0.4206	1.0543	1.4865

Notes: Column headings are probabilities p; entries are the quantiles x_p satisfying $\mathbb{P}[n(\hat{\phi}-1)<x_p]=p$, for case DU at $\phi=1$, $\gamma=0$.

Table S4: Confluence: the two regimes for $d_1=\mathbf{1}_m'\boldsymbol{D}^{-1}\mathbf{1}_m$ at $t_1=t_2=0$, $n=120$

ϕ	$\lvert\tau\rvert$	d_1 (reference)	roots (18)	Chebyshev (28)	regime used
0.9	8.0×10^{1}	10100.012271	5.6×10^{-16}	2.1×10^{-11}	roots
0.98	2.9×10^{0}	92761.572975	1.6×10^{-14}	1.8×10^{-15}	roots
0.99	7.3×10^{-1}	127001.645853	1.2×10^{-13}	1.8×10^{-15}	Chebyshev
0.999	7.2×10^{-3}	143926.661429	4.8×10^{-9}	1.8×10^{-15}	Chebyshev
0.9999	7.2×10^{-5}	144002.326455	1.4×10^{-4}	1.8×10^{-15}	Chebyshev
0.99999	7.2×10^{-7}	143991.419178	2.4×10^{1}	1.8×10^{-15}	Chebyshev
0.999999	7.2×10^{-9}	143990.143783	7.6×10^{5}	1.6×10^{-15}	Chebyshev
1	0	143990.000000	–	2.0×10^{-15}	Chebyshev

Notes: Relative errors in double precision of the closed form in the roots and of the Chebyshev expansion, against a reference value computed from the spectral representation $d_1=\sum_{j=1}^{m}(\mathbf{1}_m'\boldsymbol{u}_j)^2/\lambda_j$, $m=n-1$, in its stabilized form: $(\mathbf{1}_m'\boldsymbol{u}_j)^2=(2/n)\cot^2(j\pi/(2n))$ for odd j and 0 for even j, and $\lambda_j=(p+2q)-4q\sin^2(j\pi/(2n))$ with $p+2q=(1-\phi)^2$ supplied in closed form, so that no eigenvalue is a difference of nearly equal numbers. At $\phi=1$ the reference reproduces the exact value $d_1=n(n^2-1)/12$ to 10^{-15}. The reference is a double-precision evaluation, not an exact value. $\tau=n^2(x+1)=-n^2(1-\phi)^2/(2\phi)$; the algorithm uses the roots when $\lvert\tau\rvert\geq5/2$ and the expansion otherwise.

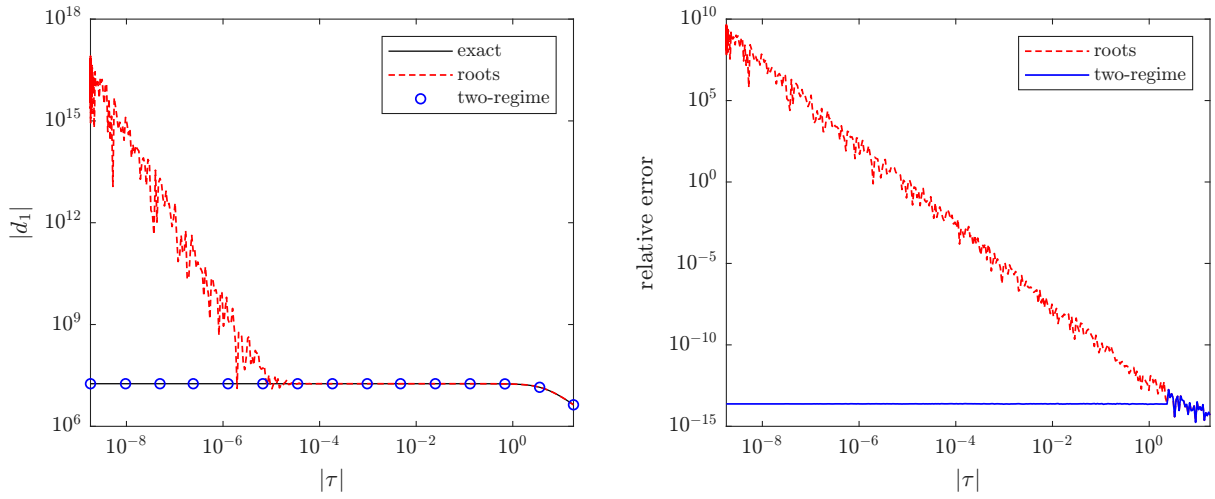


Figure S1: Two-regime evaluation of d_1 along an inversion path, $n = 600$, $\phi = 1 - 10^{-7}$, $r = 0.1/n$. Left: $|d_1|$ by the stabilized spectral reference (solid), by the closed form in the roots (dashed) and by the two-regime algorithm (circles) against $|\tau|$. Right: relative errors against the reference.

Moments and drift

Table S5 gives the bias, standard deviation, skewness and excess kurtosis of $\hat{\phi}$ in the four cases for $\phi \in \{0, 0.5, 0.9, 0.99, 1\}$ and $n \in \{25, 100, 400\}$, with $z_0 = 0$ and $\gamma = 0$ in the fixed-start cases; Section 7.3 of the paper discusses the bias and the comparison with Vinod and Shenton (1996). The stationary and fixed starts differ visibly only when ϕ is close to one. At $\phi = 0.99$ and $n = 25$, the stationary initial observation has variance 50 and dominates the regressor. In case SK it roughly halves the bias and reduces the standard deviation by a third (-0.035 and 0.077 against -0.065 and 0.117), but it leaves a much heavier left tail (skewness -2.57 and excess kurtosis 10.7 against -1.76 and 4.7). Table S6 shows the convergence of the normalized finite-sample moments to the Dickey–Fuller moments, which is slow: the skewness and excess kurtosis of $n(\hat{\phi} - 1)$ have the limits -2.270 and 8.37 in case DK and -1.551 and 4.01 in case DU, and in case DK the excess kurtosis at $n = 100$ is still 13% below its limit and at $n = 400$ still 3.5%. Table S7 gives moments and quantiles with drift, as discussed in Section 7.4. In the explosive region the moments of $|\phi|^n(\hat{\phi} - \phi)/(\phi^2 - 1)$ are dominated by the far tails. As a check of the exact moments there, in case DK with $z_0 = 0$, at $\phi = 1.3$ and $n = 12$, its mean, standard deviation, skewness and excess kurtosis are -2.198, 6.002, -3.177 and 12.79, and four million simulated samples give -2.200, 6.007, -3.176 and 12.78, with standard

errors 0.003, 0.005, 0.005 and 0.05.

Table S5: Exact bias, standard deviation, skewness and excess kurtosis of $\hat{\phi}$, four cases

Panel A: fixed start									
		DK ($z_0 = 0$)				DU ($\gamma = 0$)			
ϕ	n	bias	sd	skew	kurt	bias	sd	skew	kurt
0	25	0.0000	0.1969	0.000	−0.192	−0.0417	0.1963	0.028	−0.188
0	100	0.0000	0.0995	0.000	−0.057	−0.0101	0.0995	0.004	−0.057
0	400	0.0000	0.0499	0.000	−0.015	−0.0025	0.0499	0.000	−0.015
0.5	25	−0.0368	0.1790	−0.540	0.260	−0.1037	0.1888	−0.457	0.119
0.5	100	−0.0098	0.0874	−0.326	0.115	−0.0253	0.0890	−0.315	0.102
0.5	400	−0.0025	0.0434	−0.171	0.033	−0.0063	0.0436	−0.169	0.032
0.9	25	−0.0640	0.1319	−1.393	2.831	−0.1781	0.1624	−0.946	1.222
0.9	100	−0.0176	0.0508	−1.068	1.779	−0.0413	0.0578	−0.953	1.377
0.9	400	−0.0045	0.0228	−0.598	0.575	−0.0096	0.0237	−0.584	0.544
0.99	25	−0.0648	0.1170	−1.763	4.664	−0.1936	0.1522	−1.077	1.739
0.99	100	−0.0182	0.0332	−1.944	6.010	−0.0531	0.0449	−1.341	2.902
0.99	400	−0.0048	0.0103	−1.608	4.182	−0.0125	0.0131	−1.289	2.657
1	25	−0.0642	0.1154	−1.815	4.935	−0.1920	0.1514	−1.104	1.817
1	100	−0.0174	0.0310	−2.142	7.281	−0.0522	0.0430	−1.425	3.291
1	400	−0.0044	0.0079	−2.237	8.078	−0.0133	0.0111	−1.519	3.819
Panel B: stationary start									
		SK				SU			
ϕ	n	bias	sd	skew	kurt	bias	sd	skew	kurt
0	25	0.0000	0.1931	0.000	−0.186	−0.0400	0.1928	0.027	−0.182
0	100	0.0000	0.0990	0.000	−0.056	−0.0100	0.0990	0.004	−0.056
0	400	0.0000	0.0499	0.000	−0.015	−0.0025	0.0499	0.000	−0.015
0.5	25	−0.0351	0.1742	−0.534	0.260	−0.0988	0.1842	−0.457	0.132
0.5	100	−0.0097	0.0868	−0.324	0.114	−0.0250	0.0884	−0.314	0.102
0.5	400	−0.0025	0.0433	−0.170	0.033	−0.0063	0.0435	−0.169	0.032
0.9	25	−0.0555	0.1183	−1.445	3.173	−0.1624	0.1573	−0.980	1.373
0.9	100	−0.0168	0.0493	−1.057	1.759	−0.0395	0.0563	−0.953	1.394
0.9	400	−0.0044	0.0226	−0.595	0.570	−0.0095	0.0236	−0.582	0.541
0.99	25	−0.0350	0.0769	−2.571	10.659	−0.1883	0.1518	−1.089	1.776
0.99	100	−0.0138	0.0270	−2.175	7.876	−0.0494	0.0440	−1.373	3.055
0.99	400	−0.0044	0.0096	−1.612	4.286	−0.0116	0.0126	−1.314	2.803

Notes: bias = $\mathbb{E}[\hat{\phi}] - \phi$, sd the standard deviation, skew the skewness and kurt the excess kurtosis (the kurtosis minus 3) of $\hat{\phi}$, computed from (39) centered at $\mathbb{E}[\hat{\phi}]$. The stationary start requires $|\phi| < 1$.

Table S6: Moments of $n(\hat{\phi}-1)$ at $\phi=1$ and of its limit

Case	n	mean	sd	skewness	excess kurtosis
DK	25	−1.6047	2.8852	−1.8150	4.9349
DK	100	−1.7354	3.0968	−2.1423	7.2809
DK	400	−1.7698	3.1585	−2.2369	8.0778
DK	1600	−1.7785	3.1746	−2.2615	8.2947
DK	6400	−1.7807	3.1786	−2.2678	8.3501
DK	∞	−1.7814	3.1800	−2.2699	8.3686
DU	25	−4.8001	3.7841	−1.1038	1.8173
DU	100	−5.2223	4.3013	−1.4254	3.2914
DU	400	−5.3391	4.4564	−1.5187	3.8193
DU	1600	−5.3691	4.4971	−1.5431	3.9649
DU	6400	−5.3766	4.5074	−1.5492	4.0022
DU	∞	−5.3791	4.5108	−1.5513	4.0148

Notes: Fixed start, $z_0=0$ (DK) and $\gamma=0$ (DU). The rows $n=\infty$ are the moments of the Dickey–Fuller limits A/B and $\hat{A}/\hat{B}$ of Section 6.1; the limiting means −1.7814 and −5.3791 agree with Nabeya (1999) and Gonzalo and Pitarakis (1998).

Table S7: A random walk with drift: case DU, $\phi=1$, $\gamma=\alpha/\sigma$

		moments of $n(\hat{\phi}-1)$				exact quantiles			limit at $\varkappa=\sqrt{n}\gamma$		
n	γ	mean	sd	skew	kurt	0.05	0.50	0.95	0.05	0.50	0.95
50	0	−5.0739	4.1118	−1.310	2.703	−13.008	−4.216	−0.074	−14.094	−4.357	−0.143
50	0.1	−4.4746	4.0141	−1.391	3.001	−12.271	−3.586	0.273	−13.245	−3.703	0.204
50	0.5	−0.5955	1.2496	−1.934	13.994	−2.619	−0.470	1.071	−2.706	−0.489	1.031
50	1	−0.1198	0.5047	−0.063	0.456	−0.951	−0.116	0.698	−0.964	−0.121	0.687
50	2	−0.0286	0.2466	0.006	0.089	−0.433	−0.029	0.377	−0.436	−0.030	0.374
200	0	−5.2996	4.4034	−1.487	3.634	−13.808	−4.321	−0.126	−14.094	−4.357	−0.143
200	0.1	−3.2724	3.7384	−1.916	5.920	−10.636	−2.278	0.697	−10.816	−2.297	0.680
200	0.5	−0.1255	0.5052	−0.093	0.466	−0.961	−0.119	0.690	−0.964	−0.121	0.687
200	1	−0.0300	0.2466	−0.006	0.089	−0.436	−0.030	0.375	−0.436	−0.030	0.374
200	2	−0.0074	0.1227	0.001	0.021	−0.209	−0.007	0.194	−0.209	−0.008	0.194

Notes: sd denotes standard deviation, skew skewness and kurt excess kurtosis. The probability columns are quantiles of $n(\hat{\phi}-1)$ in case DU. The final block uses the local-to-unity limit with $c=0$ and $\varkappa=\sqrt{n}\gamma$.

Density comparisons

Figure S2 compares finite-sample coefficient densities across the four cases; it accompanies Section 7.4 of the paper.

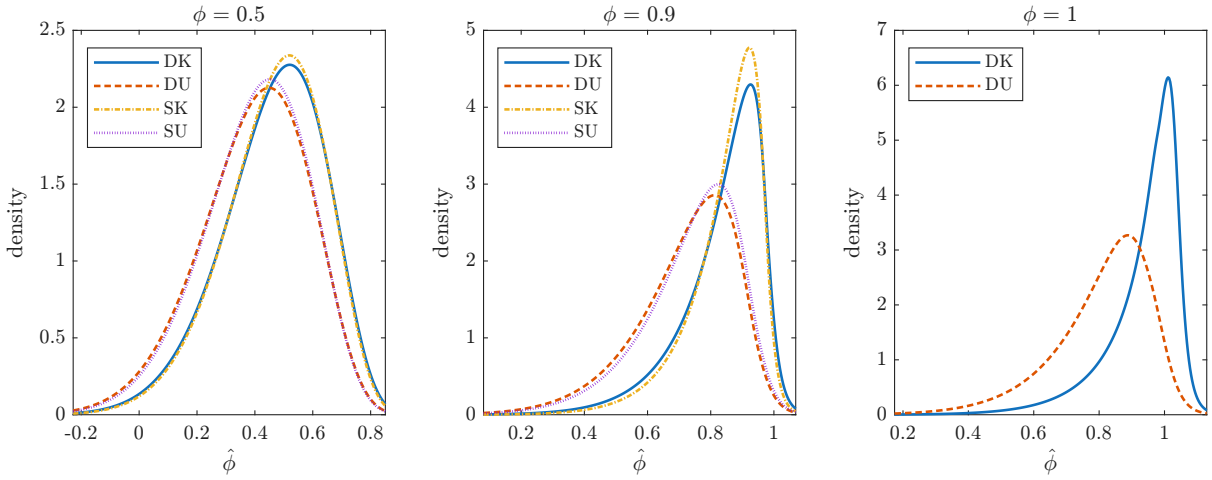


Figure S2: Exact densities of $\hat{\phi}$ at $n = 25$: all four cases for $\phi = 0.5$ and 0.9, and only DK and DU for $\phi = 1$. The fixed starts have $z_0 = 0$ or $\gamma = 0$; stationary starts require $|\phi| < 1$.

Quantiles near the unit root and in the explosive region

Table S8 compares the exact quantiles of $n(\hat{\phi} - \phi)$ at $\phi = 1 + c/n$ with their local-to-unity limits, and Table S9 those of $|\phi|^n(\hat{\phi} - \phi)/(\phi^2 - 1)$ in the explosive region with the limits (47) and (48); both are discussed in Section 7.4 of the paper.

Table S8: Local to unity: exact quantiles of $n(\hat{\phi} - \phi)$ at $\phi = 1 + c/n$ and their limits

		$n = 50$			$n = 200$			$n = 1000$			limit		
Case	c	0.05	0.50	0.95	0.05	0.50	0.95	0.05	0.50	0.95	0.05	0.50	0.95
DK	−10	−10.499	−0.773	4.850	−11.780	−0.930	4.815	−12.137	−0.973	4.806	−12.228	−0.983	4.803
DK	−2	−8.521	−0.887	1.971	−8.978	−0.927	1.931	−9.106	−0.939	1.921	−9.138	−0.941	1.918
DK	2	−6.501	−0.324	0.824	−6.544	−0.315	0.774	−6.556	−0.312	0.762	−6.559	−0.311	0.758
DK	5	−1.512	−0.016	0.279	−1.174	−0.012	0.235	−1.092	−0.011	0.224	−1.072	−0.011	0.221
DU	−10	−13.640	−2.960	3.610	−15.647	−3.353	3.542	−16.223	−3.462	3.524	−16.370	−3.489	3.519
DU	−2	−13.448	−4.087	0.565	−14.514	−4.251	0.526	−14.819	−4.297	0.516	−14.896	−4.308	0.513
DU	2	−11.030	−1.756	0.159	−11.334	−1.705	0.135	−11.418	−1.691	0.129	−11.439	−1.687	0.127
DU	5	−2.160	−0.083	0.239	−1.656	−0.064	0.205	−1.534	−0.059	0.197	−1.505	−0.058	0.194
SK	−10	−10.144	−0.735	4.759	−11.395	−0.887	4.736	−11.745	−0.929	4.730	−11.834	−0.939	4.729
SK	−2	−7.400	−0.729	1.770	−7.792	−0.764	1.746	−7.901	−0.773	1.740	−7.929	−0.775	1.738
SU	−10	−13.199	−2.812	3.595	−15.164	−3.200	3.533	−15.728	−3.307	3.516	−15.871	−3.334	3.512
SU	−2	−12.781	−3.682	0.797	−13.784	−3.840	0.747	−14.070	−3.883	0.734	−14.143	−3.894	0.731

Notes: Fixed start with $z_0 = 0$, $\gamma = 0$ (DK, DU); stationary start (SK, SU), for which the limit is that of a random scaled initial condition $\gamma_0 \sim \mathcal{N}(0, -1/(2c))$.

Table S9: The explosive case: exact quantiles of $|\phi|^n(\hat{\phi}-\phi)/(\phi^2-1)$ and the limit

Case	ϕ	n	0.05	0.25	0.50	0.75	0.95
DK, $z_0=0$	1.05	10	−10.3567	−3.9192	−1.0868	0.5743	2.4032
DK, $z_0=0$	1.05	25	−9.0533	−2.9885	−0.7042	0.3237	1.4214
DK, $z_0=0$	1.05	50	−13.6171	−2.6371	−0.4827	0.3755	1.5916
DK, $z_0=0$	1.05	100	−16.4481	−1.4511	−0.1747	0.6954	3.2006
DK, $z_0=0$	1.05	400	−6.3138	−1.0000	0.0000	1.0000	6.3137
DK, $z_0=0$	1.05	∞	−6.3138	−1.0000	0.0000	1.0000	6.3138
DK, $z_0=0$	1.2	10	−8.8249	−2.5690	−0.5085	0.3858	1.6169
DK, $z_0=0$	1.2	25	−17.5410	−1.5280	−0.1982	0.6619	2.9916
DK, $z_0=0$	1.2	50	−6.7339	−1.0233	−0.0113	0.9772	5.9378
DK, $z_0=0$	1.2	100	−6.3140	−1.0000	0.0000	1.0000	6.3135
DK, $z_0=0$	1.2	400	−6.3138	−1.0000	0.0000	1.0000	6.3138
DK, $z_0=0$	1.2	∞	−6.3138	−1.0000	0.0000	1.0000	6.3138
DK, $z_0=1$	1.05	10	−9.3470	−3.5383	−0.9953	0.4815	2.0180
DK, $z_0=1$	1.05	25	−8.6527	−2.8331	−0.6599	0.3043	1.3332
DK, $z_0=1$	1.05	50	−13.2277	−2.4830	−0.4523	0.3616	1.5291
DK, $z_0=1$	1.05	100	−15.3292	−1.3661	−0.1620	0.6676	3.0822
DK, $z_0=1$	1.05	400	−5.9987	−0.9509	0.0000	0.9509	5.9985
DK, $z_0=1$	1.05	∞	−5.9986	−0.9509	0.0000	0.9509	5.9986
DK, $z_0=1$	1.2	10	−7.7649	−2.0875	−0.4102	0.3114	1.2690
DK, $z_0=1$	1.2	25	−13.2485	−1.1931	−0.1484	0.5633	2.5387
DK, $z_0=1$	1.2	50	−5.3588	−0.8320	−0.0079	0.8003	4.8117
DK, $z_0=1$	1.2	100	−5.0724	−0.8160	0.0000	0.8160	5.0721
DK, $z_0=1$	1.2	400	−5.0722	−0.8160	0.0000	0.8160	5.0722
DK, $z_0=1$	1.2	∞	−5.0722	−0.8160	0.0000	0.8160	5.0722
DU, $\gamma=0$	1.05	10	−15.7433	−9.3268	−5.6754	−2.7825	0.3689
DU, $\gamma=0$	1.05	25	−15.0678	−7.6290	−3.9423	−1.8032	0.0897
DU, $\gamma=0$	1.05	50	−22.6841	−6.6166	−2.4787	−1.0300	0.5309
DU, $\gamma=0$	1.05	100	−23.3035	−2.4722	−0.8932	0.1084	2.7734
DU, $\gamma=0$	1.05	400	−6.7397	−1.1636	−0.1142	0.9352	6.5111
DU, $\gamma=0$	1.05	∞	−6.3138	−1.0000	0.0000	1.0000	6.3138
DU, $\gamma=0$	1.2	10	−13.5740	−6.3191	−2.7859	−1.2066	0.4956
DU, $\gamma=0$	1.2	25	−25.3437	−2.6799	−0.9935	0.0077	2.4882
DU, $\gamma=0$	1.2	50	−7.6914	−1.4056	−0.2946	0.7893	6.2155
DU, $\gamma=0$	1.2	100	−6.7708	−1.1764	−0.1236	0.9292	6.5231
DU, $\gamma=0$	1.2	400	−6.4282	−1.0419	−0.0283	0.9854	6.3716
DU, $\gamma=0$	1.2	∞	−6.3138	−1.0000	0.0000	1.0000	6.3138

Notes: The rows $n=\infty$ are the quantiles of Z_1/Z_2 in (47): standard Cauchy for case DK with $z_0=0$ and for case DU with $\gamma=0$ (its limit (48) is (47) with z_0 replaced by $\gamma/(\phi-1)$).

S5 The explosive limit with an intercept

In case DU with $|\phi| > 1$ the limit of $T_n = \phi^n(\hat{\phi} - \phi)/(\phi^2 - 1)$ is the ratio of normals (48) of the paper, which is (47) with z_0 replaced by $\gamma/(\phi - 1)$ (Wang and Yu, 2015, Theorem 2.3(b)), but the approach to it is slow: the error is of order c_{E}/n with

$$c_{\mathrm{E}} = \frac{\phi + 1}{\phi - 1},$$

which is large when ϕ is close to one ($c_{\mathrm{E}} \approx 12.4$ at $\phi = 1.175$) and small for negative ϕ ($c_{\mathrm{E}} \approx 0.09$ at $\phi = -1.2$). This section shows that, for fixed ϕ, the finite-sample distribution is approximated, with an error $O(n|\phi|^{-n/5})$, by an explicit Gaussian-ratio modification of the limit, which for $\gamma = 0$ is a location–scale change; the leading corrections and the exact order of the error of the limit follow. The power ϕ^n in T_n is signed, unlike the $|\phi|^n$ of (47) and (48); the limit is the same, but with $|\phi|^n$ the location correction below changes sign for negative ϕ and odd n.

Setting. Take $\sigma = 1$ and $w_t = y_t - y_0$, so that $w_t = \gamma + \phi w_{t-1} + e_t$ with $w_0 = 0$. Put

$$b = \frac{\gamma}{\phi - 1}, \qquad x_t = w_t + b,$$

so that $x_t = \phi x_{t-1} + e_t$ with $x_0 = b$. The estimator with an intercept is invariant to the shift, so $\hat{\phi}$ is the slope of the regression of x_t on a constant and x_{t-1}. Let

$$q_{\mathrm{E}} = |\phi|, \qquad \nu = \frac{1}{\phi^2 - 1}, \qquad \nu_n = \nu(1 - \phi^{-2n}), \qquad d = \frac{c_{\mathrm{E}}}{n},$$

and

$$X_n = \sum_{j=1}^{n} \phi^{-j} e_j, \qquad Y_n = \sum_{j=1}^{n} \phi^{-(n-j+1)} e_j, \qquad S_n = \sum_{j=1}^{n} e_j,$$
$$U_n = Y_n - \frac{S_n}{n(\phi - 1)}, \qquad L_n = b + X_n.$$

Finally, for n large enough that $0 < d < (\sqrt{5} - 1)/2$, let

$$m_d = -\frac{d}{1 - d}, \qquad s_d = \frac{\sqrt{1 - d - d^2}}{1 - d}, \qquad a = b\sqrt{\phi^2 - 1} = \frac{\gamma(\phi + 1)}{\sqrt{\phi^2 - 1}},$$

and let G_n be the distribution function of

$$m_d + \frac{s_d Z_1 - m_d a}{Z_2 + a}, \qquad Z_1, Z_2 \text{ independent } \mathcal{N}(0,1). \tag{S.5}$$

At $d = 0$ this is the limit $Z_1/(Z_2 + a)$. When $\gamma = 0$ it is a Cauchy location–scale family,

$$G_n(x) = F_C\Big(\frac{(1-d)x + d}{\sqrt{1 - d - d^2}}\Big), \qquad F_C(x) = \frac{1}{2} + \frac{1}{\pi}\arctan x. \tag{S.6}$$

Proposition S5.1. *Let $|\phi| > 1$ and γ be fixed. As $n \to \infty$,*

$$\sup_{x \in \mathbb{R}} \big|\mathbb{P}[T_n \le x] - G_n(x)\big| = O\big(n|\phi|^{-n/5}\big),$$

which is $o(n^{-K})$ for every $K > 0$. The error of the limit is exactly of order n^{-1}: it is $O(n^{-1})$ uniformly in x, and at $x = 0$

$$\mathbb{P}[T_n \le 0] = \frac{1}{2} + \frac{c_{\mathrm{E}}}{\pi n}\, e^{-a^2/2} + O(n^{-2}), \tag{S.7}$$

whereas the limit distribution function equals $1/2$ there. When $\gamma = 0$, uniformly in x,

$$\mathbb{P}[T_n \le x] = F_C\Big(\frac{x + d}{1 + d/2}\Big) + O(n^{-2}), \tag{S.8}$$

$$\mathbb{P}[T_n \le x] = F_C\Big(\frac{x + d + d^2}{1 + d/2 - d^2/8}\Big) + O(n^{-3}), \tag{S.9}$$

$$\sup_{x \in \mathbb{R}} \Big|\mathbb{P}[T_n \le x] - F_C\Big(\frac{x + d}{1 + d/2}\Big)\Big| = \frac{c_{\mathrm{E}}^2}{n^2}\,\frac{8 + \sqrt{65}}{16\pi} + O(n^{-3}). \tag{S.10}$$

When $\gamma = 0$, to first order T_n is thus the limit shifted by $-c_{\mathrm{E}}/n$ and dilated by $1 + c_{\mathrm{E}}/(2n)$. When $\gamma \neq 0$ the location correction also enters the numerator, through the term $-m_d a$ in (S.5), and a shift and dilation of the limit alone are not enough. At $\phi = 1.175$, $\gamma = -1$ and $n = 200$, the location–scale change $(x - m_d)/s_d$ of the argument of the limit leaves a maximal error of 0.09, larger than that of the limit itself. The constants depend on ϕ and are not uniform as $|\phi| \downarrow 1$ (and $c_{\mathrm{E}} \to \infty$ as $\phi \downarrow 1$); the statement is for fixed ϕ as $n \to \infty$ and does not cover local-to-unity sequences. The exponent $n/5$ in the rate is a convenient choice, not the best possible one.

Proof. Step 1: reduction to a Gaussian ratio. Put

$$r_{t,n} = \sum_{j=t}^{n} \phi^{t-1-j} e_j.$$

Then, exactly,

$$x_{t-1} = \phi^{t-1} L_n - r_{t,n}, \qquad \sum_{t=1}^{n} \phi^{t-1} e_t = \phi^n Y_n,$$

and summing $x_t - x_{t-1} = (\phi - 1)x_{t-1} + e_t$ over t gives

$$\sum_{t=1}^{n} x_{t-1} = \frac{\phi^n L_n - b - S_n}{\phi - 1}.$$

With $\bar{x}_- = n^{-1} \sum_{t=1}^{n} x_{t-1}$ and $1/(\phi-1)^2 = c_{\mathrm{E}}\nu$, these identities give, exactly,

$$\begin{aligned} N_n &:= \phi^{-n} \sum_{t=1}^{n} (x_{t-1} - \bar{x}_-) e_t = L_n U_n + R_{N,n}, \\ R_{N,n} &= \phi^{-n} \Big\{ \frac{(b + S_n) S_n}{n(\phi - 1)} - \sum_{t=1}^{n} r_{t,n} e_t \Big\}, \\ D_n &:= \phi^{-2n} \sum_{t=1}^{n} (x_{t-1} - \bar{x}_-)^2 = \nu(1-d) L_n^2 + R_{D,n}, \end{aligned}$$

with

$$\begin{aligned} R_{D,n} = & -\nu \phi^{-2n} L_n^2 - 2\phi^{-2n} L_n \sum_{t=1}^{n} \phi^{t-1} r_{t,n} + \phi^{-2n} \sum_{t=1}^{n} r_{t,n}^2 \\ & + \frac{2\phi^{-n} L_n (b + S_n)}{n(\phi-1)^2} - \frac{\phi^{-2n}(b+S_n)^2}{n(\phi-1)^2}. \end{aligned}$$

Write $\|X\|_p = (\mathbb{E}|X|^p)^{1/p}$ for the L^p norm. For each $p \geq 1$ the Gaussian variables X_n, Y_n and $r_{t,n}$ have L^p norms bounded uniformly in n and t, and, by the triangle inequality for the second,

$$\|S_n\|_p = O(\sqrt{n}), \qquad \Big\| \sum_{t=1}^{n} \phi^{t-1} r_{t,n} \Big\|_p = O(q_{\mathrm{E}}^n).$$

Hölder's inequality then gives

$$\|R_{N,n}\|_p = O(n q_{\mathrm{E}}^{-n}), \qquad \|R_{D,n}\|_p = O(q_{\mathrm{E}}^{-n}). \tag{S.11}$$

Since $T_n = \nu N_n/D_n$, its Gaussian-ratio approximation is

$$Q_n = \frac{U_n}{(1-d)L_n}.$$

Step 2: the exact law of the ratio. The pair (X_n, U_n) is centered Gaussian with

$$\operatorname{Var} X_n = \nu_n, \qquad A_n := \operatorname{Var} U_n = \nu_n - \frac{1-2\phi^{-n}}{n(\phi-1)^2},$$
$$B_n := \operatorname{Cov}(X_n, U_n) = n\phi^{-(n+1)} - \frac{1-\phi^{-n}}{n(\phi-1)^2},$$

as follows from

$$\operatorname{Cov}(X_n, Y_n) = n\phi^{-(n+1)}, \qquad \operatorname{Cov}(X_n, S_n) = \operatorname{Cov}(Y_n, S_n) = \frac{1-\phi^{-n}}{\phi-1}.$$

Regressing U_n on $X_n = L_n - b$ gives

$$U_n = \frac{B_n}{\nu_n} X_n + V_n, \qquad \operatorname{Var} V_n = A_n - \frac{B_n^2}{\nu_n},$$

with V_n independent of X_n, so that, exactly,

$$Q_n \overset{d}{=} \tilde{m}_n + \frac{\tilde{s}_n Z_1 - \tilde{m}_n \tilde{a}_n}{Z_2 + \tilde{a}_n},$$
$$\tilde{m}_n = \frac{B_n/\nu_n}{1-d}, \qquad \tilde{s}_n = \frac{\{A_n/\nu_n - (B_n/\nu_n)^2\}^{1/2}}{1-d}, \qquad \tilde{a}_n = \frac{b}{\sqrt{\nu_n}},$$

with Z_1, Z_2 independent standard normal random variables; no independence approximation is involved. Since

$$\frac{A_n}{\nu_n} = 1 - d + O(q_{\mathrm{E}}^{-n}/n), \qquad \frac{B_n}{\nu_n} = -d + O(nq_{\mathrm{E}}^{-n}),$$

we have

$$\tilde{m}_n = m_d + O(nq_{\mathrm{E}}^{-n}), \qquad \tilde{s}_n = s_d + O(nq_{\mathrm{E}}^{-n}), \qquad \tilde{a}_n = a + O(q_{\mathrm{E}}^{-2n}).$$

To compare the laws of (S.5) at the parameters $(\tilde{m}_n, \tilde{s}_n, \tilde{a}_n)$ and (m_d, s_d, a), write the

variable as a ratio of jointly normal variables,

$$m + \frac{sZ_1 - ma}{Z_2 + a} = \frac{sZ_1 + mZ_2}{Z_2 + a}, \qquad \begin{pmatrix} sZ_1 + mZ_2 \\ Z_2 + a \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ a \end{pmatrix}, \begin{pmatrix} s^2 + m^2 & m \\ m & 1 \end{pmatrix} \right).$$

For (m, a) in compact sets and s bounded away from zero, this bivariate normal family is locally Lipschitz in (m, s, a) in total variation (by Pinsker's inequality and the explicit relative entropy between two Gaussian laws), and the map to the ratio cannot increase the total variation distance. Since $\tilde{m}_n - m_d$, $\tilde{s}_n - s_d$ and $\tilde{a}_n - a$ are $O(nq_{\mathrm{E}}^{-n})$, with $m_d \to 0$, $s_d \to 1$ and a fixed, the parameter errors give

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}[Q_n \le x] - G_n(x) \right| = O(nq_{\mathrm{E}}^{-n}).$$

Step 3: the small denominator. Let

$$\mathcal{E}_n = \left\{ |L_n| \ge q_{\mathrm{E}}^{-n/5}, \quad |U_n| \le n, \quad |R_{N,n}| + |R_{D,n}| \le q_{\mathrm{E}}^{-4n/5} \right\}.$$

The density of L_n is bounded uniformly in n, so $\mathbb{P}[|L_n| < q_{\mathrm{E}}^{-n/5}] = O(q_{\mathrm{E}}^{-n/5})$; the variance of U_n is bounded, so $\mathbb{P}[|U_n| > n] = O(e^{-c_* n^2})$ for some $c_* > 0$; and Markov's inequality with (S.11) gives

$$\mathbb{P}\big[|R_{N,n}| + |R_{D,n}| > q_{\mathrm{E}}^{-4n/5} \big] = O\big(nq_{\mathrm{E}}^{-n} q_{\mathrm{E}}^{4n/5} \big) = O\big(nq_{\mathrm{E}}^{-n/5} \big).$$

Hence

$$\mathbb{P}[\mathcal{E}_n^c] = O(nq_{\mathrm{E}}^{-n/5}).$$

On $\mathcal{E}_n$, $|R_{D,n}| \le q_{\mathrm{E}}^{-4n/5} \le q_{\mathrm{E}}^{-2n/5} L_n^2$, so for large n the denominator $\nu(1-d)L_n^2 + R_{D,n}$ is at least $\nu(1-d)L_n^2/2$, and subtracting the two ratios,

$$\begin{aligned} |T_n - Q_n| &= \left| \frac{\nu(L_n U_n + R_{N,n})}{\nu(1-d)L_n^2 + R_{D,n}} - \frac{U_n}{(1-d)L_n} \right| \\ &\le C \left\{ \frac{|R_{N,n}|}{L_n^2} + \frac{|U_n|\,|R_{D,n}|}{|L_n|^3} \right\} \le C\big(q_{\mathrm{E}}^{-2n/5} + nq_{\mathrm{E}}^{-n/5} \big) =: h_n \end{aligned}$$

for a constant C depending on ϕ. The density f_{Q_n} of Q_n is bounded uniformly in n: the density of a ratio of independent normals with numerator standard deviation $\tilde{s}_n \to 1$ is

Table S10: The explosive limit with an intercept: largest difference between the exact distribution function of T_n and its approximations

ϕ	γ	n	c_{E}	limit	first order (S.8)	G_n
1.175	0	50	12.43	1.08×10^{-1}	3.15×10^{-2}	8.4×10^{-3}
1.175	0	200	12.43	2.18×10^{-2}	1.28×10^{-3}	3.6×10^{-12}
1.175	0	800	12.43	5.28×10^{-3}	7.78×10^{-5}	1.6×10^{-14}
1.5	0	50	5	3.59×10^{-2}	3.39×10^{-3}	3.3×10^{-7}
1.5	0	200	5	8.55×10^{-3}	2.02×10^{-4}	7.4×10^{-15}
-1.2	0	100	0.09091	3.04×10^{-4}	1.90×10^{-6}	2.0×10^{-6}
-1.2	0	201	0.09091	1.52×10^{-4}	6.54×10^{-8}	7.0×10^{-14}
1.175	-1	50	12.43	4.29×10^{-2}	–	7.3×10^{-4}
1.175	-1	200	12.43	9.44×10^{-3}	–	1.0×10^{-13}
1.175	-1	800	12.43	2.27×10^{-3}	–	1.7×10^{-14}
1.5	0.4	100	5	1.23×10^{-2}	–	3.0×10^{-15}
1.2	2	200	11	8.50×10^{-3}	–	2.5×10^{-14}

Notes: $T_n = \phi^n(\hat{\phi} - \phi)/(\phi^2 - 1)$ with the signed power, case DU; $c_{\mathrm{E}} = (\phi+1)/(\phi-1)$. "Limit": $Z_1/(Z_2 + a)$ of Wang and Yu (2015) (standard Cauchy when $\gamma = 0$); G_n: (S.5). Differences below about 10^{-11} are within the accuracy of the exact computation.

at most $\mathbb{E}|Z_2 + \tilde{a}_n|/(\tilde{s}_n\sqrt{2\pi})$. Therefore

$$\sup_{x\in\mathbb{R}} \big|\mathbb{P}[T_n \le x] - \mathbb{P}[Q_n \le x]\big| \le \mathbb{P}[\mathcal{E}_n^c] + 2h_n \sup_{x\in\mathbb{R}} f_{Q_n}(x) = O(nq_{\mathrm{E}}^{-n/5}),$$

which, together with Step 2, proves the first claim.

Step 4: the expansions. The limit $Z_1/(Z_2 + a)$ is (S.5) at $(m, s) = (0, 1)$, and $m_d = O(d)$, $s_d = 1 + O(d)$, so its error is $O(n^{-1})$ uniformly in x, by the smoothness of (S.5) in (m, s). For (S.7), condition on $Z_2 = z$ in (S.5): the variable is at most zero when $Z_1 \le -m_d z/s_d$ if $z + a > 0$ and when $Z_1 \ge -m_d z/s_d$ if $z + a < 0$, so

$$G_n(0) = \mathbb{E}\big[\Phi_{\mathcal{N}}(-m_d Z_2/s_d)\,\mathbf{1}\{Z_2 + a > 0\} + \big(1 - \Phi_{\mathcal{N}}(-m_d Z_2/s_d)\big)\mathbf{1}\{Z_2 + a < 0\}\big].$$

Differentiating in d at $d = 0$, where $m_0 = 0$, $\partial m_d/\partial d = -1$ and $s_0 = 1$, gives

$$\varphi_{\mathcal{N}}(0)\,\mathbb{E}[Z_2 \operatorname{sgn}(Z_2 + a)] = 2\varphi_{\mathcal{N}}(0)\varphi_{\mathcal{N}}(a) = e^{-a^2/2}/\pi,$$

since, for a standard normal Z,

$$\mathbb{E}[Z\,\mathbf{1}\{Z > -a\}] = -\mathbb{E}[Z\,\mathbf{1}\{Z < -a\}] = \varphi_{\mathcal{N}}(a).$$

As $G_n(0) = 1/2$ at $d = 0$ and $G_n(0)$ is smooth in d,

$$G_n(0) = \frac{1}{2} + \frac{d\, e^{-a^2/2}}{\pi} + O(d^2),$$

and the first claim transfers this to $\mathbb{P}[T_n \le 0]$. Numerically, at $\phi = 1.175$ and $\gamma = 0$ the exact $\mathbb{P}[T_n \le 0] - 1/2$ is 0.02044 at $n = 200$ and 0.004984 at $n = 800$, against $c_{\mathrm{E}}/(\pi n) = 0.01978$ and 0.004945; the differences are $26/n^2$ and $25/n^2$, as the $O(n^{-2})$ term requires. For $\gamma = 0$, (S.5) is $m_d + s_d Z_C$ with Z_C standard Cauchy, which is (S.6), and

$$m_d = -d - d^2 + O(d^3), \qquad s_d = 1 + \frac{d}{2} - \frac{d^2}{8} + O(d^3)$$

give (S.8) and (S.9). Expanding the arguments,

$$\frac{(1-d)x + d}{\sqrt{1 - d - d^2}} = x + d\Big(1 - \frac{x}{2}\Big) + d^2\Big(\frac{1}{2} + \frac{3x}{8}\Big) + O(d^3),$$
$$\frac{x + d}{1 + d/2} = x + d\Big(1 - \frac{x}{2}\Big) + d^2\Big(\frac{x}{4} - \frac{1}{2}\Big) + O(d^3),$$

so that, uniformly in x by the smoothness of the Cauchy location–scale family,

$$G_n(x) - F_C\Big(\frac{x + d}{1 + d/2}\Big) = \frac{d^2(1 + x/8)}{\pi(1 + x^2)} + O(d^3).$$

Finally

$$\sup_{x \in \mathbb{R}} \frac{|1 + x/8|}{1 + x^2} = \frac{8 + \sqrt{65}}{16},$$

attained at $x = \sqrt{65} - 8$, which gives (S.10). □

Numerical confirmation. Table S10 compares the exact distribution function of T_n with the limit, with (S.8) when $\gamma = 0$, and with G_n. Each entry is the largest difference over the grid

$$x \in \{\pm 8, \pm 4, \pm 2, \pm 1, \pm 0.5, \pm 0.2, \pm 0.1, 0, 0.03, \sqrt{65} - 8\};$$

the quadrature error of the exact values is below 5×10^{-12}. By (S.10) the error of (S.8) should be $49.36/n^2$ at $\phi = 1.175$ and $7.99/n^2$ at $\phi = 1.5$; the table gives $51.1/n^2$ at $n = 200$ and $49.8/n^2$ at $n = 800$ for the first, and $8.1/n^2$ at $n = 200$ for the second. G_n agrees with the exact distribution to the accuracy of the quadrature once the exponentially small

terms are negligible; the rows with $n = 50$, and the row $n = 100$ at $\phi = -1.2$, show these terms before they become negligible. The statement is asymptotic in n for fixed ϕ, not a finite-sample bound.

S6 Proofs

This section collects the proofs of Theorem 1 and of Propositions 1 and 3 of the paper; the proofs of Theorem 2 and of Lemmas 1 and 2 are in the Appendix of the paper.

Proof of Theorem 1

We use the Gaussian integral: if $\boldsymbol{K}$ is a complex symmetric $N \times N$ matrix with $\operatorname{Re} \boldsymbol{K}$ positive definite and $\boldsymbol{\beta} \in \mathbb{C}^N$, then

$$\int_{\mathbb{R}^N} \exp\big(-\tfrac{1}{2}\boldsymbol{x}'\boldsymbol{K}\boldsymbol{x} + \boldsymbol{\beta}'\boldsymbol{x}\big)\mathrm{d}\boldsymbol{x} = (2\pi)^{N/2}|\boldsymbol{K}|^{-1/2}\exp\big(\tfrac{1}{2}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta}\big), \tag{S.12}$$

where $|\boldsymbol{K}|^{-1/2}$ is the branch obtained by continuity from the positive value at $\boldsymbol{K} = \operatorname{Re}\boldsymbol{K}$ along the segment $\operatorname{Re}\boldsymbol{K} + \lambda\,\mathrm{i}\operatorname{Im}\boldsymbol{K}$, $\lambda \in [0,1]$, on which the matrix remains accretive and hence nonsingular (Lemma 3(a)); both sides are analytic in $\boldsymbol{K}$ on the convex set of accretive matrices and agree for real $\boldsymbol{K}$. Since the matrices $\boldsymbol{K}$ of Theorem 1 are affine in (t_1, t_2) with $\operatorname{Re}\boldsymbol{K}$ fixed, this branch is continuous in (t_1, t_2) and positive at the origin, so it is the branch of the theorem.

Case DU. With $w_t = (y_t - y_0)/\sigma$ and $\boldsymbol{w} = (w_1, \ldots, w_n)'$, the model is $w_t = \gamma + \phi w_{t-1} + e_t$, $w_0 = 0$, and the density of $\boldsymbol{w}$ is

$$(2\pi)^{-n/2}\exp\big\{ -\tfrac{1}{2}\big[(\boldsymbol{I}_n - \phi\boldsymbol{J}_n)\boldsymbol{w} - \gamma\mathbf{1}_n\big]'\big[(\boldsymbol{I}_n - \phi\boldsymbol{J}_n)\boldsymbol{w} - \gamma\mathbf{1}_n\big]\big\}.$$

Let $\boldsymbol{w}_- = \boldsymbol{J}_n\boldsymbol{w} = (0, w_1, \ldots, w_{n-1})'$. Since u and v are standardized and invariant to a common shift of the data, $u = \boldsymbol{w}_-'\boldsymbol{M}_n(\boldsymbol{I}_n - \phi\boldsymbol{J}_n)\boldsymbol{w}$ (the term $\gamma\boldsymbol{w}_-'\boldsymbol{M}_n\mathbf{1}_n$ vanishes) and $v = \boldsymbol{w}_-'\boldsymbol{M}_n\boldsymbol{w}_-$. Hence

$$\mathbb{E}\big[e^{\mathrm{i}t_1u + \mathrm{i}t_2v}\big] = (2\pi)^{-n/2}\int_{\mathbb{R}^n}\exp\big(-\tfrac{1}{2}\boldsymbol{w}'\boldsymbol{K}\boldsymbol{w} + \gamma\mathbf{1}_n'(\boldsymbol{I}_n - \phi\boldsymbol{J}_n)\boldsymbol{w} - \tfrac{1}{2}n\gamma^2\big)\mathrm{d}\boldsymbol{w},$$

with

$$\boldsymbol{K} = (\boldsymbol{I}_n - \phi\boldsymbol{J}_n)'(\boldsymbol{I}_n - \phi\boldsymbol{J}_n) - \mathrm{i}t_1\big[\boldsymbol{J}_n'\boldsymbol{M}_n(\boldsymbol{I}_n - \phi\boldsymbol{J}_n) + (\boldsymbol{I}_n - \phi\boldsymbol{J}_n)'\boldsymbol{M}_n\boldsymbol{J}_n\big] - 2\mathrm{i}t_2\boldsymbol{J}_n'\boldsymbol{M}_n\boldsymbol{J}_n.$$

Now $(\boldsymbol{I}_n - \phi\boldsymbol{J}_n)'(\boldsymbol{I}_n - \phi\boldsymbol{J}_n)$ is tridiagonal with diagonal $(1+\phi^2, \dots, 1+\phi^2, 1)$ and off-diagonal $-\phi$; $\boldsymbol{J}_n'\boldsymbol{J}_n = \mathrm{diag}(1, \dots, 1, 0)$; and

$$\boldsymbol{J}_n'(\boldsymbol{I}_n - \phi\boldsymbol{J}_n) + (\boldsymbol{I}_n - \phi\boldsymbol{J}_n)'\boldsymbol{J}_n = \boldsymbol{J}_n + \boldsymbol{J}_n' - 2\phi\boldsymbol{J}_n'\boldsymbol{J}_n.$$

With $\boldsymbol{\iota} = \boldsymbol{J}_n'\mathbf{1}_n = \mathbf{1}_n - \boldsymbol{e}_{n,n}$ and

$$\boldsymbol{\delta} = (\boldsymbol{I}_n - \phi\boldsymbol{J}_n)'\mathbf{1}_n = (1-\phi)\mathbf{1}_n + \phi\boldsymbol{e}_{n,n},$$

the demeaning matrix $\boldsymbol{M}_n$ contributes

$$\frac{\mathrm{i}t_1}{n}\big(\boldsymbol{\iota}\boldsymbol{\delta}' + \boldsymbol{\delta}\boldsymbol{\iota}'\big) + \frac{2\mathrm{i}t_2}{n}\boldsymbol{\iota}\boldsymbol{\iota}'$$

to $\boldsymbol{K}$. Collecting terms, the leading $m \times m$ block of $\boldsymbol{K}$ is the tridiagonal matrix with diagonal entries $1 + \phi^2 + 2\mathrm{i}\phi t_1 - 2\mathrm{i}t_2 = p$ and off-diagonal entries $-\phi - \mathrm{i}t_1 = q$, that is, $\boldsymbol{D}$, plus the demeaning term $\kappa\mathbf{1}_m\mathbf{1}_m'$: for $s, t \le n-1$, $\iota_s\delta_t + \delta_s\iota_t = 2(1-\phi)$ and $\iota_s\iota_t = 1$, so every entry of the block gains $2\mathrm{i}[(1-\phi)t_1 + t_2]/n = \kappa$. The entries (t, n), $t \le n-1$, are $\mathrm{i}t_1/n$ (from $\iota_t\delta_n$), plus q when $t = n-1$, and the (n,n) entry is 1. This is (11), with $\boldsymbol{\beta} = \gamma\boldsymbol{\delta}$, whose core part is $\gamma(1-\phi)\mathbf{1}_m$ and last entry γ, and $\eta = -\frac{1}{2}n\gamma^2$. Formula (S.12) gives the result, and $\mathrm{Re}\,\boldsymbol{K} = (\boldsymbol{I}_n - \phi\boldsymbol{J}_n)'(\boldsymbol{I}_n - \phi\boldsymbol{J}_n)$, positive definite because $\boldsymbol{I}_n - \phi\boldsymbol{J}_n$ is unit lower triangular.

Case DK. With $x_t = y_t/\sigma$ and $x_0 = z_0$ fixed, $u = \sum_{t=1}^n x_{t-1}(x_t - \phi x_{t-1})$, $v = \sum_{t=1}^n x_{t-1}^2$, and the terms involving x_0 in the exponent are

$$-\tfrac{1}{2}(x_1 - \phi z_0)^2 + \mathrm{i}t_1 z_0(x_1 - \phi z_0) + \mathrm{i}t_2 z_0^2,$$

whose linear part is $(\phi + \mathrm{i}t_1)z_0x_1 = -qz_0x_1$ and whose constant part is

$$-\tfrac{1}{2}z_0^2(\phi^2 + 2\mathrm{i}\phi t_1 - 2\mathrm{i}t_2) = -\tfrac{1}{2}z_0^2(p-1).$$

The remaining terms give the matrix $\boldsymbol{K}$ of case DK, which is that of case DU without the

demeaning.

Cases SK *and* SU. Now, with $x_t = y_t/\sigma$ (after subtracting the mean),

$$\boldsymbol{x} = (x_0, \dots, x_n)' \sim \mathcal{N}\big(\mathbf{0}_{n+1}, \boldsymbol{V}_{n+1}/(1-\phi^2)\big),$$

and $(1-\phi^2)\boldsymbol{V}_{n+1}^{-1}$ is tridiagonal with diagonal $(1, 1+\phi^2, \dots, 1+\phi^2, 1)$ and off-diagonal $-\phi$; its determinant is $1-\phi^2$, which gives the factor $\varrho = (1-\phi^2)^{1/2}$. With $\tilde{\boldsymbol{J}} = [\boldsymbol{I}_n\ \mathbf{0}_n]$ and $\boldsymbol{P} = [\mathbf{0}_n\ \boldsymbol{I}_n]$ selecting $(x_0, \dots, x_{n-1})$ and $(x_1, \dots, x_n)$, $u = (\tilde{\boldsymbol{J}}\boldsymbol{x})'\boldsymbol{M}_n(\boldsymbol{P} - \phi\tilde{\boldsymbol{J}})\boldsymbol{x}$ and $v = (\tilde{\boldsymbol{J}}\boldsymbol{x})'\boldsymbol{M}_n\tilde{\boldsymbol{J}}\boldsymbol{x}$ in case SU ($\boldsymbol{M}_n$ replaced by $\boldsymbol{I}_n$ in case SK), and the same bookkeeping as in case DU, now with $\boldsymbol{\iota} = \tilde{\boldsymbol{J}}'\mathbf{1}_n = (1, \dots, 1, 0)'$ and

$$\tilde{\boldsymbol{\delta}} = (\boldsymbol{P} - \phi\tilde{\boldsymbol{J}})'\mathbf{1}_n = (-\phi, 1-\phi, \dots, 1-\phi, 1)',$$

gives (12)–(14). With the rows and columns of $\boldsymbol{K}$ indexed $0, \dots, n$, the entries of the first row are

$$K_{00} = 1 + 2\mathrm{i}(\phi t_1 - t_2)\Big(1 - \frac{1}{n}\Big), \qquad K_{01} = q + \frac{\mathrm{i}[(1-2\phi)t_1 + 2t_2]}{n},$$
$$K_{0j} = \frac{\mathrm{i}[(1-2\phi)t_1 + 2t_2]}{n} \quad (2 \le j \le n-1), \qquad K_{0n} = \frac{\mathrm{i}t_1}{n},$$

and the rest is as in case DU. The real part $\mathrm{Re}\,\boldsymbol{K} = (1-\phi^2)\boldsymbol{V}_{n+1}^{-1}$ is positive definite. □

The density criterion and its exceptions

This subsection justifies the statement after (7) of the paper: in the four cases with $n \ge 3$, the integral in (7) converges absolutely except at $n = 3$, in case DK with $z_0 = 0$ at $\hat{\phi} = 0$ and in cases DU and SU at $\hat{\phi} = -1/2$, where the density has a logarithmic singularity. Write $x = \phi + r$ for the threshold on $\hat{\phi}$, so that $u - rv = N - xv$, where $N = \sum_{t=1}^n y_{t-1}^* y_t$ is the numerator of $\hat{\phi}$ and y_{t-1}^* is y_{t-1} in cases DK and SK and $y_{t-1} - \bar{y}_-$ in cases DU and SU, with $\bar{y}_- = n^{-1}\sum_{t=1}^n y_{t-1}$.

The rank. The last observation y_n enters $N - xv$ only through the term $y_{n-1}^* y_n$. In coordinates that contain y_n and y_{n-1}^*, the quadratic part of $N - xv$ therefore has a 2×2 block in these two coordinates with a zero entry for y_n^2 and the entry $\frac{1}{2}$ for $y_n y_{n-1}^*$, which is invertible and indefinite. Eliminating these two coordinates shows that the rank of the quadratic part is two plus the rank of its restriction to $y_{n-1}^* = 0$. In cases DU and SU, N

and v are invariant to a common shift of the data, and $N = \sum_{t=1}^{n-1} h_t h_{t+1} + h_n(y_n - \bar{y}_-)$ with $h_t = y_{t-1} - \bar{y}_-$; the restriction is

$$Q_x(\boldsymbol{h}) = \sum_{t=1}^{n-2} h_t h_{t+1} - x \sum_{t=1}^{n-1} h_t^2$$

on the subspace $\sum_{t=1}^{n-1} h_t = 0$, of dimension $n-2$. It vanishes identically only if x equals the ratio $\sum_{t=1}^{n-2} h_t h_{t+1} / \sum_{t=1}^{n-1} h_t^2$ for every $\boldsymbol{h}$ in the subspace. For $n \geq 4$ the vectors $(1, -1, 0, \ldots, 0)$ and $(1, 0, -1, 0, \ldots, 0)$ give the ratios $-1/2$ and 0, so the rank is at least three for every x; for $n = 3$ the subspace is spanned by $(1, -1)$, and the rank is two exactly at $x = -1/2$. In cases SK and DK the free lags, after the same elimination, are $y_0, \ldots, y_{n-2}$ and $y_1, \ldots, y_{n-2}$, of dimensions $n-1$ and $n-2$, with the same adjacent-product quadratic part and no sum constraint; in dimension two or more it cannot vanish identically, and in dimension one (case DK with $n = 3$) it is $-xy_1^2$, which vanishes at $x = 0$.

The linear term. At a point of rank two the integral still converges when the linear term of $N - xv$, in centered and whitened coordinates, has a nonzero projection onto the null space of the quadratic part (Section 4.4). In cases DU and SU, $N - xv$ is a homogeneous quadratic form $Q(\boldsymbol{w})$ in the differences $w_t = y_t - y_0$, and with $\boldsymbol{w} = \boldsymbol{\mu} + \boldsymbol{L}\boldsymbol{z}$, $\boldsymbol{z}$ standard normal, its linear term in $\boldsymbol{z}$ is $2\boldsymbol{L}'\boldsymbol{H}\boldsymbol{\mu}$, where $\boldsymbol{H}$ is the matrix of Q; this is orthogonal to the null vector $\boldsymbol{L}^{-1}\boldsymbol{k}$ of the quadratic part, for every null vector $\boldsymbol{k}$ of $\boldsymbol{H}$. So the projection vanishes, whatever ϕ and γ. In case DK with $n = 3$ and $x = 0$, $N = z_0 y_1 + y_2(y_1 + y_3)$, with $(y_1, y_2, y_3) = \boldsymbol{\mu} + \boldsymbol{L}\boldsymbol{z}$ for a unit lower triangular $\boldsymbol{L}$ and $\boldsymbol{z} = (e_1, e_2, e_3)'$. The quadratic part $y_2(y_1 + y_3)$, with matrix $\boldsymbol{H}$, has the null vector $\boldsymbol{k} = (1, 0, -1)'$, and the projection of the linear term on $\boldsymbol{L}^{-1}\boldsymbol{k}$ is $z_0 + 2\boldsymbol{\mu}'\boldsymbol{H}\boldsymbol{k} = z_0$. The density is therefore finite for $z_0 \neq 0$.

The singularity. In the three exceptional cases $N - xv$ is the product of two jointly normal variables. In case DK with $z_0 = 0$, $\hat{\phi} = y_2(y_1 + y_3)/(y_1^2 + y_2^2)$. In cases DU and SU, with $(a, b, d) = (y_1 - y_0, y_2 - y_0, y_3 - y_0)$,

$$3v\big(\hat{\phi} + \tfrac{1}{2}\big) = (2b - a)d.$$

The vector (a, b, d) has a positive Gaussian density, and $v = 2b^2 > 0$ on the line $a = 2b$, $d = 0$ where both factors vanish, except at $b = 0$. The density of a product of two jointly normal variables with a positive joint density diverges logarithmically at zero, and

integrating it against v near that line gives a logarithmic singularity of the density of $\hat{\phi}$ at $-1/2$; the same argument applies to case DK at 0.

Computation. The first step also gives a representation of the density that the programs use in these smallest samples, $n = 3$ in cases DK, DU and SU and $n = 4$ in cases DT and ST, where the lags residualized on the regressors, $\boldsymbol{h} = (h_1, \dots, h_n)'$ (the centered lags of the first step in cases DU and SU), have two random dimensions. With $\omega = y_n - \boldsymbol{x}'_{n+1}\hat{\boldsymbol{b}}$, where $\boldsymbol{x}'_{n+1}\hat{\boldsymbol{b}}$ is the line fitted to the lags extrapolated to time $n+1$ ($\omega = y_n$ without regressors),

$$N = \sum_{t=1}^{n-1} h_t h_{t+1} + h_n \omega, \qquad v = \sum_{t=1}^{n} h_t^2,$$

and integrating out ω, whose conditional distribution given $\boldsymbol{h}$ is normal with variance at least one,

$$f(x) = \mathbb{E}\left[\frac{v\, p(\omega^* \mid \boldsymbol{h})}{|h_n|}\right], \qquad \omega^* = -\frac{1}{h_n}\left(\sum_{t=1}^{n-1} h_t h_{t+1} - xv\right),$$

a two-dimensional Gaussian integral. The inner integral is taken in $\log|h_n|$, which removes the factor $1/|h_n|$. This avoids the inversion integral, whose integrand decays only like $t_1^{-3/2}$ in these cases, and near a singular point like t_1^{-1} up to t_1 of order one over the distance from it. The programs also use the representation at $n = 4$ in case DK with $z_0 \neq 0$, where the lags have three random dimensions and the inversion integrand decays only like t_1^{-2} while oscillating; the three-dimensional integral is then taken by fixed tanh-sinh and Gauss–Legendre rules, with the logarithmic singularities of the inner integrals at the panel ends.

Proof of Proposition 1

Consider first the cases in which v can approach zero: DU, SU, SK, and DK with $z_0 = 0$. In each,

$$v = \boldsymbol{z}'\boldsymbol{z} = \|\boldsymbol{z}\|^2$$

for a nondegenerate Gaussian vector $\boldsymbol{z} \in \mathbb{R}^d$ obtained from the regressors by a linear change of coordinates; here $\|\cdot\|$ is the Euclidean norm, and $d = \operatorname{rank}(v)$. It is $n-1$ in cases DK and DU, where the fixed start removes one dimension (in case DU the demeaning removes no further one), and in case SU, where the demeaning removes one; it is n in case SK. In case DU, for instance, with $\boldsymbol{w}_- = \boldsymbol{J}_n\boldsymbol{w} = (0, w_1, \dots, w_{n-1})'$ as in the proof of

Theorem 1, $\boldsymbol{z}$ is $\boldsymbol{M}_n\boldsymbol{w}_-$ expressed in an orthonormal basis of the range of $\boldsymbol{M}_n$, which has dimension $n-1$. The remaining Gaussian coordinates are collected in a vector $\boldsymbol{g}$, which is the common level of the regressors in case SU and is empty in the other cases. The last innovation e_n is independent of $(\boldsymbol{z}, \boldsymbol{g})$. Conditionally on $(\boldsymbol{z}, \boldsymbol{g})$ the numerator is affine in e_n,

$$u = \mu_u(\boldsymbol{z}, \boldsymbol{g}) + \ell(\boldsymbol{z})\, e_n, \tag{S.13}$$

where $\mu_u(\boldsymbol{z}, \boldsymbol{g}) = \mathbb{E}[u \mid \boldsymbol{z}, \boldsymbol{g}]$ is the conditional mean and $\ell(\boldsymbol{z})$ is the last (centered) regressor, a nonzero linear form in $\boldsymbol{z}$.

Sufficiency. Let $\boldsymbol{\epsilon} = (e_1, \dots, e_n)'$ be the vector of innovations. By the Cauchy–Schwarz inequality,

$$|u| \le \|\boldsymbol{z}\|\, \|\boldsymbol{\epsilon}\|, \qquad \text{so} \qquad \left|\frac{u}{v}\right|^k \le \frac{\|\boldsymbol{\epsilon}\|^k}{\|\boldsymbol{z}\|^k}.$$

Conditionally on $\boldsymbol{z}$ the moments of $\|\boldsymbol{\epsilon}\|$ grow at most polynomially in $\|\boldsymbol{z}\|$, the Gaussian tails take care of large $\|\boldsymbol{z}\|$, and near the origin the density of $\boldsymbol{z}$ is bounded. In polar coordinates, therefore, $\mathbb{E}|u/v|^k < \infty$ as soon as $\int_0^1 r^{d-1-k}\mathrm{d}r < \infty$, that is, $k < d$.

Necessity. If $X \sim \mathcal{N}(\mu, s^2)$ and $Z \sim \mathcal{N}(0,1)$, then

$$\mathbb{E}|X|^k = s^k\, \mathbb{E}|Z + \mu/s|^k \ge C_k s^k, \qquad C_k = \inf_{a \in \mathbb{R}} \mathbb{E}|Z + a|^k > 0,$$

where C_k depends on k only. By (S.13), conditionally on $(\boldsymbol{z}, \boldsymbol{g})$, u is normal with standard deviation $|\ell(\boldsymbol{z})|$, and $v = \|\boldsymbol{z}\|^2$, hence

$$\mathbb{E}\big[|u/v|^k \,\big|\, \boldsymbol{z}, \boldsymbol{g}\big] \ge C_k\, \frac{|\ell(\boldsymbol{z})|^k}{\|\boldsymbol{z}\|^{2k}}.$$

For small $a > 0$ the angular patch $\{\boldsymbol{z} : |\ell(\boldsymbol{z})| \ge a\|\boldsymbol{z}\|\}$ has positive solid angle, and on it this bound is at least $C_k a^k \|\boldsymbol{z}\|^{-k}$. Since the density of $\boldsymbol{z}$ is bounded below near the origin, for a sufficiently small radius $\varepsilon_0 > 0$ and a constant $C > 0$,

$$\mathbb{E}|u/v|^k \ge C \int_0^{\varepsilon_0} r^{d-1-k}\mathrm{d}r = \infty \qquad \text{when } k \ge d.$$

Adding the constant ϕ does not affect the existence of moments, so $\mathbb{E}|\hat{\phi}|^k < \infty$ if and only if $k < d$. Finally, in case DK with $z_0 \neq 0$, $v \ge z_0^2 > 0$ and $|u/v|$ is bounded by a polynomial in Gaussian variables, so every moment exists. □

Proof of Proposition 3

The identities. From (18) with $m+1=n$,

$$|\boldsymbol{D}_m| = \frac{r_+^n(1-\zeta^n)}{h}, \qquad |\boldsymbol{D}_{m-1}| = \frac{r_+^{n-1}(1-\zeta^{n-1})}{h}.$$

Expanding (10) along its last row and using $q^2 = r_+r_-$ and $r_-\zeta^{n-1} = r_+\zeta^n$,

$$|\boldsymbol{K}| = |\boldsymbol{D}_m| - q^2|\boldsymbol{D}_{m-1}| = \frac{r_+^n}{h}\big[1-\zeta^n - r_-(1-\zeta^{n-1})\big] = \frac{r_+^n}{h}\big[(1-r_-) - \zeta^n(1-r_+)\big],$$

which is (33). Moreover, by (9),

$$e_+e_- = 1-(r_++r_-)+r_+r_- = 1-p+q^2 = 2\mathrm{i}t_2 - t_1^2.$$

Here $e_+ \neq 0$: $r_+ = 1$ would give $1-p+q^2=0$, hence $t_1 = t_2 = 0$, where $r_+ = \phi^2 \neq 1$.

In case DK, (22) and $\eta = -\frac{1}{2}z_0^2(p-1)$ give the exponent

$$\tfrac{1}{2}z_0^2\Big[q^2d_3 + \frac{q^4d_4^2}{\delta_{\mathsf{DK}}} - (p-1)\Big].$$

Multiply the bracket by $\delta_{\mathsf{DK}} = 1-q^2d_3$ and use

$$d_3^2 - d_4^2 = \frac{|\boldsymbol{D}_{m-2}|}{|\boldsymbol{D}_m|}, \qquad q^2|\boldsymbol{D}_{m-2}| = p|\boldsymbol{D}_{m-1}| - |\boldsymbol{D}_m|,$$

the first of which is the identity $|\boldsymbol{D}_{m-1}|^2 - |\boldsymbol{D}_m||\boldsymbol{D}_{m-2}| = q^{2m-2}$ of the Chebyshev recursion. The result is

$$q^2pd_3 - q^2(pd_3-1) - (p-1) = 1-p+q^2 = 2\mathrm{i}t_2 - t_1^2,$$

which is (34).

For (35), the identities

$$q^2d_3 = \frac{q^2|\boldsymbol{D}_{m-1}|}{|\boldsymbol{D}_m|}, \qquad q^2d_4 = \frac{(-q)^{m+1}}{|\boldsymbol{D}_m|}$$

give

$$\Theta = \frac{(-q)^n - |\boldsymbol{K}|}{|\boldsymbol{D}_m|}, \qquad \frac{(-q)^n}{|\boldsymbol{D}_m|} = \frac{(-w)^n h}{1-\zeta^n}.$$

Next, $\Omega = (1-\phi)qd_2 - 1$ with $d_2 = [1 + q(d_3 + d_4)]/(p + 2q)$. After inserting (9), its numerator is

$$(1-\phi)q + (1-\phi)(1+\Theta) - (p+2q) = \mathrm{i}[(1-\phi)t_1 + 2t_2] + (1-\phi)\Theta.$$

In case DU, take the Schur complement of $\boldsymbol{\Delta}$ in $\boldsymbol{K}$ (the last pivot when the core is eliminated first). With $b_1 = \mathrm{i}t_1/n$ (Table 1), $\boldsymbol{b} = b_1\mathbf{1}_m + q\boldsymbol{e}_{m,m}$ and $\boldsymbol{\beta} = \gamma\big((1-\phi)\mathbf{1}_m', 1\big)'$,

$$|\boldsymbol{K}| = |\boldsymbol{\Delta}|(1 - \boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b}) = |\boldsymbol{D}|\rho S,$$
$$\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} = \gamma^2\Big[(1-\phi)^2\mathbf{1}_m'\boldsymbol{\Delta}^{-1}\mathbf{1}_m + \frac{\tilde{\Omega}^2}{S}\Big], \qquad \tilde{\Omega} = (1-\phi)\mathbf{1}_m'\boldsymbol{\Delta}^{-1}\boldsymbol{b} - 1,$$

which is (23). By (20), $\mathbf{1}_m'\boldsymbol{\Delta}^{-1}\boldsymbol{b} = (b_1d_1 + qd_2)/\rho$, so

$$\rho\tilde{\Omega} = (1-\phi)(b_1d_1 + qd_2) - 1 - \kappa d_1 = \Omega + [(1-\phi)b_1 - \kappa]d_1,$$

where

$$(1-\phi)b_1 - \kappa = -\frac{\mathrm{i}[(1-\phi)t_1 + 2t_2]}{n}.$$

Similarly, $\boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b} = b_1^2w_1 + 2b_1qw_2 + q^2w_3$ gives

$$\rho S = \rho(1 - q^2d_3) + \kappa q^2d_2^2 - 2b_1qd_2 - b_1^2d_1.$$

Since

$$\big((1-\phi)t_1 + t_2\big)qd_2 - t_1 = t_1[(1-\phi)qd_2 - 1] + t_2qd_2,$$

we have

$$\kappa q^2d_2^2 - 2b_1qd_2 = qd_2(\kappa qd_2 - 2b_1) = \frac{2\mathrm{i}}{n}\,qd_2\big[((1-\phi)t_1 + t_2)qd_2 - t_1\big] = \frac{2\mathrm{i}}{n}\,qd_2(t_1\Omega + t_2qd_2).$$

With $b_1^2 = -t_1^2/n^2$ this is (36). Finally, by Theorem 1,

$$\varphi = |\boldsymbol{K}|^{-1/2}\exp\big(\tfrac{1}{2}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} + \eta\big), \qquad \eta = -\tfrac{1}{2}n\gamma^2,$$

in case DU, which, together with the two expressions above, gives (37).

The orders. Consider the paths $t_1 = O(|\phi|^{-n})$ and $t_2 = O(\phi^{-2n})$. All orders below are

for fixed ϕ, z_0 and γ as $n \to \infty$, uniformly on the paths. Then

$$p = 1 + \phi^2 + O(|\phi|^{-n}), \qquad q = -\phi + O(|\phi|^{-n}), \qquad r_+ = \phi^2 + O(|\phi|^{-n}),$$
$$r_- = 1 + O(\phi^{-2n}), \qquad w = -\phi^{-1} + O(|\phi|^{-n}),$$
$$\zeta = w^2 = \phi^{-2} + O(|\phi|^{-n}), \qquad h = r_+ - r_- = \phi^2 - 1 + O(|\phi|^{-n}),$$

where the order of r_- follows from $e_+e_- = O(\phi^{-2n})$ and $e_+ = 1 - \phi^2 + O(|\phi|^{-n})$. Hence $\zeta^n = \phi^{-2n}\{1 + O(n|\phi|^{-n})\} = O(\phi^{-2n})$, $(-w)^n = O(|\phi|^{-n})$, and e_- and δ_{DK} are $O(\phi^{-2n})$. Next, $(-w)^n h = O(|\phi|^{-n})$, so $\Theta = O(|\phi|^{-n})$; and $\Omega = O(|\phi|^{-n})$ because $p + 2q = (1 - \phi)^2 + O(|\phi|^{-n})$ is bounded away from zero. From (18),

$$d_3 = \frac{1 - \zeta^{n-1}}{r_+(1 - \zeta^n)} = O(1), \qquad d_4 = O(|\phi|^{-n}),$$

and hence, again because $p + 2q$ is bounded away from zero,

$$d_2 = \frac{1 + q(d_3 + d_4)}{p + 2q} = O(1), \qquad d_1 = \frac{m + 2qd_2}{p + 2q} = O(n).$$

Since $\kappa = O(|\phi|^{-n}/n)$, $\rho = 1 + \kappa d_1 = 1 + O(|\phi|^{-n})$. Each of the three terms of ρS is then $O(\phi^{-2n})$, and each of the two terms of $\rho\tilde{\Omega}$ is $O(|\phi|^{-n})$. Hence ρS and $\rho\tilde{\Omega}$ are at most of these orders, and so are S and $\tilde{\Omega}$, because $\rho = 1 + O(|\phi|^{-n})$. No subtraction of larger terms occurs: e_- is a quotient, $1 - \zeta^n = 1 + O(\phi^{-2n})$ and $e_+ = 1 - \phi^2 + O(|\phi|^{-n})$ involve no cancellation, $e_- - \zeta^n e_+$ combines terms of the same order, and Ω, Θ, ρS, $\rho\tilde{\Omega}$ are sums of terms of the stated order.

The phase. When $|\phi| > 1$, $\mathrm{Re}(p \pm 2q) = (1 \mp \phi)^2 > 0$, so the core has distinct characteristic roots for every real (t_1, t_2). Its root-phase identity (Theorem 2(iii)) is valid even where the programs use a confluent expansion. The quantities δ_{DK} and S are Schur complements of accretive matrices ($\boldsymbol{D}$ in $\boldsymbol{K}$ in case DK, $\boldsymbol{\Delta}$ in $\boldsymbol{K}$ in case DU), hence have positive real parts (Lemma 3(c)), and ρ is off the negative axis by Theorem 2(ii). Hence (29) applies, with $\delta_4 = \delta_{\mathsf{DK}}$ in case DK and $\delta_4 = S$ in case DU. □

Remark: elimination orders and eigenvalues. In case DK, suppose that $\boldsymbol{K}$ is eliminated starting from its last coordinate. The pivots then obey, exactly,

$$\varpi_j = p - \frac{q^2}{\varpi_{j+1}}, \qquad j = n - 1, \ldots, 1, \qquad \varpi_n = 1.$$

At $t_1 = t_2 = 0$ the map $\varpi \mapsto p - q^2/\varpi = 1 + \phi^2 - \phi^2/\varpi$ has the fixed points 1 and ϕ^2, with derivatives ϕ^2 and ϕ^{-2}. Since $\varpi_n = 1$, the exact pivots stay at the unstable fixed point 1: a rounding error of relative size ϵ_{M} in ϖ_{j+1} is multiplied by about ϕ^2 in ϖ_j, hence by about ϕ^{2n} over the recursion. If instead the core is eliminated from its first coordinate, the pivots $\varpi_1^* = p$ and $\varpi_j^* = p - q^2/\varpi_{j-1}^*$, $j = 2, \dots, n-1$, tend to ϕ^2, the stable fixed point, and

$$\prod_{j=1}^{n-1} \varpi_j^* = |\boldsymbol{D}| = O(\phi^{2n}),$$

so the last pivot $|\boldsymbol{K}|/|\boldsymbol{D}| = \delta_{\mathsf{DK}} = O(\phi^{-2n})$ is $1 - q^2 d_3$, formed by cancellation. An evaluation through eigenvalues is affected in the same way when, as in the usual implementation of the method of Imhof (1961), the matrix of $u - rv$ is whitened through the covariance matrix $\boldsymbol{\Sigma}_y$ of $(y_1, \dots, y_n)$. The entries of $\boldsymbol{\Sigma}_y$ are of order ϕ^{2n}, while its smaller eigenvalues are of order one. Once $\epsilon_{\mathrm{M}}\phi^{2n} \gtrsim 1$, the computed $\boldsymbol{\Sigma}_y$ therefore retains only its dominant direction, and its Cholesky factor fails to exist numerically. The eigenvalues λ_j of the whitened matrix are then lost, including the two of order $|\phi|^n$ that dominate $\prod_{j=1}^{n}(1 - 2\mathrm{i}\lambda_j t_1)^{-1/2}$ on the paths. Whitening by the innovations, with the factor $(\boldsymbol{I}_n - \phi \boldsymbol{J}_n)^{-1}$, avoids this loss but costs $O(n^3)$ operations for each value of r.

S7 The regression with a linear trend

This section extends the analysis of the paper to the regression of y_t on a constant, the time index t and y_{t-1}, the setting of the coefficient form of the Dickey–Fuller test with a trend (Dickey and Fuller, 1979; Fuller, 1996), with a fixed initial observation (case DT) or a stationary one (case ST). As in the paper, the first letter of a case label refers to the initial observation (deterministic or stationary); here the T refers to the trend. The structure is that of the paper. The estimation error $\hat{\phi} - \phi$ is again a ratio u/v of quadratic forms in a Gaussian vector, and its joint characteristic function is the inverse square root of the determinant of a complex symmetric matrix $\boldsymbol{K}$ times an exponential. Removing the trend adds a rank-two term to the Toeplitz core $\boldsymbol{D}$ of $\boldsymbol{K}$, and the Woodbury identity reduces $|\boldsymbol{K}|$ and the bilinear forms of $\boldsymbol{K}^{-1}$ to the primitives of Lemma 2 and two more. The three numerical issues of Section 4 have counterparts here. In addition, the loss of accuracy of order $\gamma^2 n^2 \epsilon_{\mathrm{M}}$ that a drift causes near the unit root, as in case DU (Section 4.4), is removed by a reparameterization (Section S7.4.4).

The local-to-unity limit is the Dickey–Fuller distribution of the detrended estimator and its local alternatives with a nonzero initial condition. Its characteristic function at a zero initial condition is in Nabeya and Tanaka (1990, Model 4), and the setting with a nonzero initial value and a local drift and trend is that of Nabeya and Sørensen (1994). We derive a closed form by a change of measure (Theorem S7.2), in which the initial condition enters through a single parameter $\varkappa_\tau$, and treat the issues that its evaluation raises: series near the origin, cancellation on the explosive side, the branch, and the point $r = -c$ at which the inversion integral converges slowly.

The notation is that of the paper, and the symbols introduced below are local to this section. In particular, γ below reduces to the γ of (5) only when $\delta = 0$, and $\varkappa_\tau$ and $\tilde{H}_c$ play the roles of the $\varkappa$ and H_c of Section 6.1.

S7.1 The model, the two cases, and reductions

S7.1.1 The model and the estimator

Consider

$$y_t = \alpha + \delta t + \phi y_{t-1} + \sigma e_t, \qquad t = 1, \dots, n, \tag{S.14}$$

with $e_1, \dots, e_n$ as in (1) and $n \geq 4$, and the least squares estimator $\hat{\phi}$ of ϕ in the regression of y_t on 1, t and y_{t-1} over $t = 1, \dots, n$. Let $\boldsymbol{X} = [\mathbf{1}_n, \boldsymbol{s}]$ with $\boldsymbol{s} = (s_1, \dots, s_n)'$ and $s_t = t - (n+1)/2$, so that $\boldsymbol{X}'\boldsymbol{X} = \mathrm{diag}(n,\, n(n^2-1)/12)$, and let $\boldsymbol{M}_\tau = \boldsymbol{I}_n - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$. With $\boldsymbol{y}_- = (y_0, \dots, y_{n-1})'$ and $\boldsymbol{\epsilon} = (e_1, \dots, e_n)'$, the Frisch–Waugh theorem gives

$$\hat{\phi} - \phi = \frac{u}{v}, \qquad u = \sigma \boldsymbol{y}_-' \boldsymbol{M}_\tau \boldsymbol{\epsilon}, \qquad v = \boldsymbol{y}_-' \boldsymbol{M}_\tau \boldsymbol{y}_-. \tag{S.15}$$

The two cases are:

DT. y_0 fixed and (α, δ) unknown. Let $w_t = (y_t - y_0)/\sigma$, so that $w_0 = 0$ and

$$w_t = \gamma + \gamma_\tau s_t + \phi w_{t-1} + e_t, \qquad \gamma = \frac{\alpha + \delta(n+1)/2 - (1-\phi)y_0}{\sigma}, \qquad \gamma_\tau = \frac{\delta}{\sigma}. \tag{S.16}$$

The estimator is invariant to a common shift and scaling of the data, so $\hat{\phi}$ computed from w equals $\hat{\phi}$ computed from y, and its distribution depends on $(n, \phi, \gamma, \gamma_\tau)$. Any real ϕ is allowed.

ST. $y_t = a_0 + a_1 t + \tilde{x}_t$ and $\tilde{x}_t = \phi \tilde{x}_{t-1} + \sigma e_t$, with $|\phi| < 1$ and $\tilde{x}_0 \sim \mathcal{N}(0, \sigma^2/(1-\phi^2))$

independent of the e_t. This is (S.14) with $\delta = (1-\phi)a_1$ and $\alpha = (1-\phi)a_0 + \phi a_1$. The regression on 1 and t removes $a_0 + a_1 t$ from both y_t and y_{t-1}, so $\hat{\phi}$ is the estimator computed from $\tilde{x}$, and its distribution depends on (n, ϕ) only.

S7.1.2 The effective start

Proposition S7.1 (Effective start)**.** *In case* DT*, the distribution of $\hat{\phi}$ depends on (γ, γ_τ) only through*

$$\xi = (1-\phi)\gamma - \Big[(1-\phi)\frac{n+1}{2} + \phi\Big]\gamma_\tau. \tag{S.17}$$

For $\phi \neq 1$ it is the distribution of $\hat{\phi}$ in the regression on 1, t and x_{t-1} of the driftless autoregression $x_t = \phi x_{t-1} + e_t$ started at the fixed value $x_0 = z_*$, the effective start,

$$z_* = -\frac{\gamma - \gamma_\tau(n+1)/2}{1-\phi} + \frac{\phi\gamma_\tau}{(1-\phi)^2} = -\frac{\xi}{(1-\phi)^2}. \tag{S.18}$$

In terms of (S.14), $z_* = (y_0 - m_0)/\sigma$, where

$$m_t = a_0 + a_1 t, \qquad a_1 = \frac{\delta}{1-\phi}, \qquad a_0 = \frac{\alpha}{1-\phi} - \frac{\phi\delta}{(1-\phi)^2},$$

is the trend that solves $m_t = \alpha + \delta t + \phi m_{t-1}$; so z_* does not depend on n. For $\phi = 1$, $\xi = -\gamma_\tau$ and the distribution does not depend on γ. Case ST *is case* DT *with $\gamma_\tau = 0$ and $\gamma = -(1-\phi)\tilde{x}_0/\sigma \sim \mathcal{N}(0, \sigma_\gamma^2)$, $\sigma_\gamma^2 = (1-\phi)/(1+\phi)$, independent of $\boldsymbol{\epsilon}$.*

Proof. For $\phi \neq 1$ let $\tilde{m}_t = \tilde{a} + \tilde{b}s_t$ with $\tilde{b} = \gamma_\tau/(1-\phi)$ and $\tilde{a} = \gamma/(1-\phi) - \phi\gamma_\tau/(1-\phi)^2$. Since $s_t - s_{t-1} = 1$, $\tilde{m}_t = \gamma + \gamma_\tau s_t + \phi\tilde{m}_{t-1}$, so $x_t = w_t - \tilde{m}_t$ satisfies $x_t = \phi x_{t-1} + e_t$ with $x_0 = -\tilde{m}_0 = -\tilde{a} + \tilde{b}(n+1)/2 = z_*$. The regression on 1 and t removes $\tilde{m}_t$ from w_t and $\tilde{m}_{t-1}$ from w_{t-1}, so $\hat{\phi}$ computed from w equals $\hat{\phi}$ computed from x. Since $z_* = -\xi/(1-\phi)^2$, the distribution depends on ξ only. The same argument applied to (S.14) with m_t gives $z_* = (y_0 - m_0)/\sigma$. For $\phi = 1$,

$$w_t = \gamma t + \gamma_\tau \sum_{j=1}^{t} s_j + \sum_{j=1}^{t} e_j;$$

the term γt is removed by the regression, and $\gamma_\tau \sum_{j=1}^{t} s_j = \gamma_\tau t(t-n)/2$, whose linear part is removed by the regression and whose quadratic coefficient is $\gamma_\tau/2 = -\xi/2$. Case ST follows from the reduction to $\tilde{x}$ and $w_t = (\tilde{x}_t - \tilde{x}_0)/\sigma = -(1-\phi)\tilde{x}_0/\sigma + \phi w_{t-1} + e_t$. □

The proposition is used throughout this section: to reduce ST to DT (Section S7.3.3), to remove a cancellation near the unit root (Section S7.4.4), to identify the limits (Section S7.5), and to interpret the exact critical values (Section S7.6.2).

S7.1.3 Inversion formulas and existence of moments

The inversion formulas (6) and (7) apply to (u, v) of (S.15) without change. So does the moment representation (8): $v > 0$ with probability one, and the joint moment generating function of (u, v) is finite in a neighborhood of the origin, so Lemma 1 of Meng (2005) applies as in Section 2. The density formula requires the rank condition stated after (7). In cases DT and ST it fails only at $n = 4$, at $\hat{\phi} = -2/3$, for every ϕ, and in case DT for every (γ, γ_τ), where the quadratic part of $u - rv$ again has rank two. There are no other exceptions. Write $x = \phi + r$ and let h_t be the lag y_{t-1} after projection on 1 and t. The elimination of the coordinates h_n and ω in Section S6 shows that the rank of the quadratic part of $u - rv$ is two plus the rank of

$$Q_x(\boldsymbol{h}) = \sum_{t=1}^{n-2} h_t h_{t+1} - x \sum_{t=1}^{n-1} h_t^2$$

on $h_n = 0$, that is, on the subspace $\sum_{t=1}^{n-1} h_t = 0$, $\sum_{t=1}^{n-1} t h_t = 0$, of dimension $n - 3$. For $n = 4$ it is spanned by $(1, -2, 1)$, whose ratio $\sum_{t=1}^{n-2} h_t h_{t+1} / \sum_{t=1}^{n-1} h_t^2$ is $-2/3$. For $n \geq 5$ the vectors $(1, -2, 1, 0, \ldots, 0)$ and $(1, -1, -1, 1, 0, \ldots, 0)$ lie in the subspace and give the ratios $-2/3$ and $-1/4$, so Q_x vanishes identically for no x, and the rank is at least three. To see the singularity, subtract from the series the line through y_0 and y_1, and let a, b and d be the residuals at $t = 2$, 3 and 4. Projection on the constant and the trend gives

$$30v\big(\hat{\phi} + \tfrac{2}{3}\big) = (4a - 3b)(2a + 2b - 3d).$$

The vector (a, b, d) has a positive Gaussian density, and at $(a, b, d) = (9, 12, 14)$ both factors vanish while $v = 27/2$; as in Section S6, the density of $\hat{\phi}$ has a logarithmic singularity at $-2/3$, and the programs return $+\infty$.

Proposition S7.2 (Existence of moments)**.** *In cases* DT *and* ST*, for real $k > 0$, $\mathbb{E}|\hat{\phi}|^k < \infty$ if and only if $k < n - 2$, for every (γ, γ_τ).*

Sketch. In case DT the vector $\boldsymbol{y}_-$ in the coordinates of w has the fixed first entry $w_0 = 0$

and $n-1$ entries with a nondegenerate normal distribution. The matrix $\boldsymbol{M}_\tau$ has rank $n-2$, and its null space $\operatorname{span}\{\mathbf{1}_n, \boldsymbol{s}\}$ meets the coordinate subspace $\{\boldsymbol{x} : x_1 = 0\}$ in the line spanned by $\boldsymbol{s} - s_1\mathbf{1}_n$. So v has rank $n-2$ in nondegenerate coordinates, and the mean of $\boldsymbol{M}_\tau \boldsymbol{y}_-$ lies in the range of the random part. In case ST all n entries of $\boldsymbol{y}_-$ are random and $\boldsymbol{M}_\tau$ has rank $n-2$. In both cases v can be arbitrarily close to zero, with a density of order $v^{(n-2)/2-1}$ near the origin, and the numerator vanishes there only to first order in the residualized lags: conditional on the lags, the coefficient of the last innovation in u is the last residualized lag, a nonzero linear form in them (the last coordinate vector is not in the span of the constant and the trend), whose modulus exceeds a fixed multiple of their norm on a cone of positive solid angle. The argument of the proof of Proposition 1 (Section S6) then gives the threshold $n-2$, the rank of v. □

So the variance exists for $n \geq 5$ and the kurtosis for $n \geq 7$; the programs compute moments of order up to $n-3$.

S7.2 The joint characteristic function

Set $\sigma = 1$, let $m = n-1$, and let p, q and the $m \times m$ matrix $\boldsymbol{D}$ be those of (9). Let $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_m)'$ with $\ell_j = j - n/2$ (the centered index in the core), $\boldsymbol{Z} = [\mathbf{1}_m, \boldsymbol{\ell}]$, $G_1 = 1/n$, $G_2 = 12/(n(n^2-1))$, $k_0 = -(1+\phi)/2$, and

$$\boldsymbol{j}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{k}_1 = \begin{bmatrix} 1-\phi \\ 0 \end{bmatrix}, \quad \boldsymbol{j}_2 = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}, \quad \boldsymbol{k}_2 = \begin{bmatrix} k_0 \\ 1-\phi \end{bmatrix}.$$

Theorem S7.1 (Joint characteristic function, case DT)**.** *The joint characteristic function of (u, v) in (S.15) is*

$$\varphi(t_1, t_2) = |\boldsymbol{K}|^{-1/2} \exp\left(\tfrac{1}{2}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} + \eta\right),$$

the square root being the branch continuous in (t_1, t_2) with $\varphi(0,0) = 1$, where

$$\boldsymbol{K} = \begin{bmatrix} \boldsymbol{D} + \boldsymbol{Z}\boldsymbol{C}\boldsymbol{Z}' & \boldsymbol{b} \\ \boldsymbol{b}' & 1 \end{bmatrix}, \qquad \boldsymbol{b} = q\boldsymbol{e}_{m,m} + \boldsymbol{Z}\boldsymbol{b}_v, \tag{S.19}$$

$$\boldsymbol{C} = \sum_{i=1}^{2} G_i\big[\mathrm{i}t_1(\boldsymbol{j}_i\boldsymbol{k}_i' + \boldsymbol{k}_i\boldsymbol{j}_i') + 2\mathrm{i}t_2\boldsymbol{j}_i\boldsymbol{j}_i'\big], \qquad \boldsymbol{b}_v = \mathrm{i}t_1\Big(G_1\boldsymbol{j}_1 + \frac{m}{2}G_2\boldsymbol{j}_2\Big),$$
$$\boldsymbol{\beta} = \begin{bmatrix}\boldsymbol{Z}\boldsymbol{b}_c \\ b_n\end{bmatrix}, \qquad \boldsymbol{b}_c = \begin{bmatrix}(1-\phi)\gamma + k_0\gamma_\tau \\ (1-\phi)\gamma_\tau\end{bmatrix},$$
$$b_n = \gamma + \frac{m}{2}\gamma_\tau, \qquad \eta = -\frac{1}{2}\Big[n\gamma^2 + \frac{n(n^2-1)}{12}\gamma_\tau^2\Big].$$

The real part $\mathrm{Re}\,\boldsymbol{K} = \boldsymbol{F}'\boldsymbol{F}$*, with* $\boldsymbol{F} = \boldsymbol{I}_n - \phi\boldsymbol{J}_n$ *and* $\boldsymbol{J}_n$ *the shift matrix of Theorem 1, does not depend on* (t_1, t_2) *and is positive definite for every real* ϕ*.*

Proof. Let $\boldsymbol{w} = (w_1, \ldots, w_n)'$ and $\boldsymbol{\mu} = \gamma\mathbf{1}_n + \gamma_\tau\boldsymbol{s}$, so that $\boldsymbol{\epsilon} = \boldsymbol{F}\boldsymbol{w} - \boldsymbol{\mu}$ and $\boldsymbol{y}_- = \boldsymbol{J}_n\boldsymbol{w}$ in the coordinates of w. Since $\boldsymbol{M}_\tau\boldsymbol{\mu} = \mathbf{0}$, $u = \boldsymbol{w}'\boldsymbol{J}_n'\boldsymbol{M}_\tau\boldsymbol{F}\boldsymbol{w}$ and $v = \boldsymbol{w}'\boldsymbol{J}_n'\boldsymbol{M}_\tau\boldsymbol{J}_n\boldsymbol{w}$, and

$$\mathrm{i}t_1u + \mathrm{i}t_2v - \tfrac{1}{2}(\boldsymbol{F}\boldsymbol{w} - \boldsymbol{\mu})'(\boldsymbol{F}\boldsymbol{w} - \boldsymbol{\mu}) = -\tfrac{1}{2}\boldsymbol{w}'\boldsymbol{K}\boldsymbol{w} + \boldsymbol{w}'\boldsymbol{F}'\boldsymbol{\mu} - \tfrac{1}{2}\boldsymbol{\mu}'\boldsymbol{\mu},$$

with

$$\boldsymbol{K} = \boldsymbol{F}'\boldsymbol{F} - \mathrm{i}t_1(\boldsymbol{J}_n'\boldsymbol{M}_\tau\boldsymbol{F} + \boldsymbol{F}'\boldsymbol{M}_\tau\boldsymbol{J}_n) - 2\mathrm{i}t_2\boldsymbol{J}_n'\boldsymbol{M}_\tau\boldsymbol{J}_n.$$

Since $|\boldsymbol{F}| = 1$, Gaussian integration gives the formula with $\boldsymbol{\beta} = \boldsymbol{F}'\boldsymbol{\mu}$ and $\eta = -\frac{1}{2}\boldsymbol{\mu}'\boldsymbol{\mu}$, both sides being analytic in (t_1, t_2), since $\mathrm{Re}\,\boldsymbol{K} = \boldsymbol{F}'\boldsymbol{F}$ is positive definite. Write $\boldsymbol{M}_\tau = \boldsymbol{I}_n - G_1\mathbf{1}_n\mathbf{1}_n' - G_2\boldsymbol{s}\boldsymbol{s}'$. The terms with $\boldsymbol{I}_n$ give the matrix (10) of case DK with $z_0 = 0$ (core $\boldsymbol{D}$, border $q\boldsymbol{e}_{m,m}$, corner 1). Let $\boldsymbol{x}_1 = \mathbf{1}_n$ and $\boldsymbol{x}_2 = \boldsymbol{s}$. For $i = 1, 2$, the vector $\boldsymbol{J}_n'\boldsymbol{x}_i$, which moves the entries of $\boldsymbol{x}_i$ up by one place and ends with a zero, has core part $\boldsymbol{Z}\boldsymbol{j}_i$ and last entry zero. The vector $\boldsymbol{F}'\boldsymbol{x}_i = \boldsymbol{x}_i - \phi\boldsymbol{J}_n'\boldsymbol{x}_i$ has core part $\boldsymbol{Z}\boldsymbol{k}_i$ and last entry 1 for $i = 1$ and $m/2$ for $i = 2$; for $i = 2$ this uses $s_{j+1} = \ell_j + 1/2$ and $s_j - \phi s_{j+1} = (1-\phi)\ell_j + k_0$. Hence the correction

$$\sum_{i=1}^{2} G_i\big[\mathrm{i}t_1(\boldsymbol{J}_n'\boldsymbol{x}_i\boldsymbol{x}_i'\boldsymbol{F} + \boldsymbol{F}'\boldsymbol{x}_i\boldsymbol{x}_i'\boldsymbol{J}_n) + 2\mathrm{i}t_2\boldsymbol{J}_n'\boldsymbol{x}_i\boldsymbol{x}_i'\boldsymbol{J}_n\big],$$

is $\boldsymbol{Z}\boldsymbol{C}\boldsymbol{Z}'$ in the core, $\boldsymbol{Z}\boldsymbol{b}_v$ in the border and zero in the corner. The same computation gives $\boldsymbol{F}'\boldsymbol{\mu}$, and $\boldsymbol{\mu}'\boldsymbol{\mu} = n\gamma^2 + \gamma_\tau^2\boldsymbol{s}'\boldsymbol{s}$. □

S7.3 $O(1)$ evaluation

S7.3.1 Seven primitives

To the primitives $|\boldsymbol{D}|$ and d_1 to d_4 of Lemma 2 the trend adds

$$d_\ell = \boldsymbol{e}'_{m,m}\boldsymbol{D}^{-1}\boldsymbol{\ell}, \qquad d_{\ell\ell} = \boldsymbol{\ell}'\boldsymbol{D}^{-1}\boldsymbol{\ell}. \tag{S.20}$$

Lemma S7.1. *The matrix $\boldsymbol{D}$ is persymmetric and $\boldsymbol{\ell}$ is antisymmetric under reversal of the coordinates, so $\mathbf{1}'_m\boldsymbol{D}^{-1}\boldsymbol{\ell} = 0$, $\boldsymbol{e}'_{m,m}\boldsymbol{D}^{-1}\mathbf{1}_m = d_2$ and $\boldsymbol{e}'_{1,m}\boldsymbol{D}^{-1}\boldsymbol{\ell} = -d_\ell$. Moreover*

$$\boldsymbol{D}\boldsymbol{\ell} = (p+2q)\boldsymbol{\ell} + \frac{n}{2}q(\boldsymbol{e}_{1,m} - \boldsymbol{e}_{m,m}), \tag{S.21}$$

and hence, for $p+2q \neq 0$ (at $p+2q=0$ the continuous values of (S.27) are used),

$$d_\ell = \frac{(m-1)/2 + (n/2)q(d_3 - d_4)}{p+2q}, \qquad d_{\ell\ell} = \frac{m(m^2-1)/12 + nqd_\ell}{p+2q}. \tag{S.22}$$

Proof. The interior rows of $\boldsymbol{D}\boldsymbol{\ell}$ are $p\ell_j + q(\ell_{j-1} + \ell_{j+1}) = (p+2q)\ell_j$; the first and last rows lack $q\ell_0 = -qn/2$ and $q\ell_{m+1} = qn/2$. Multiplying (S.21) by $\boldsymbol{e}'_{m,m}\boldsymbol{D}^{-1}$ and by $\boldsymbol{\ell}'\boldsymbol{D}^{-1}$, and using $\ell_m = (m-1)/2$, $\boldsymbol{\ell}'\boldsymbol{D}^{-1}(\boldsymbol{e}_{1,m} - \boldsymbol{e}_{m,m}) = -2d_\ell$ and $\boldsymbol{\ell}'\boldsymbol{\ell} = m(m^2-1)/12$, gives (S.22). □

S7.3.2 The assembly

Let $\boldsymbol{W} = \boldsymbol{Z}'\boldsymbol{D}^{-1}\boldsymbol{Z} = \operatorname{diag}(d_1, d_{\ell\ell})$ (Lemma S7.1), $\boldsymbol{M}_C = \boldsymbol{I}_2 + \boldsymbol{C}\boldsymbol{W}$ and $\boldsymbol{\Delta} = \boldsymbol{D} + \boldsymbol{Z}\boldsymbol{C}\boldsymbol{Z}'$, which takes the place of $\boldsymbol{D} + \kappa\mathbf{1}_m\mathbf{1}'_m$ in (15), and let κ_1 and κ_2 be the eigenvalues of $\boldsymbol{C}\boldsymbol{W}$. The Woodbury identity

$$\boldsymbol{\Delta}^{-1} = \boldsymbol{D}^{-1} - \boldsymbol{D}^{-1}\boldsymbol{Z}\boldsymbol{M}_C^{-1}\boldsymbol{C}\boldsymbol{Z}'\boldsymbol{D}^{-1}$$

gives, for $\boldsymbol{x} = \boldsymbol{Z}\boldsymbol{a} + a_e\boldsymbol{e}_{m,m}$ and $\boldsymbol{z} = \boldsymbol{Z}\boldsymbol{b}_z + b_e\boldsymbol{e}_{m,m}$,

$$\boldsymbol{x}'\boldsymbol{\Delta}^{-1}\boldsymbol{z} = \boldsymbol{x}'\boldsymbol{D}^{-1}\boldsymbol{z} - \big(\boldsymbol{W}\boldsymbol{a} + a_e\boldsymbol{h}_0\big)'\boldsymbol{M}_C^{-1}\boldsymbol{C}\big(\boldsymbol{W}\boldsymbol{b}_z + b_e\boldsymbol{h}_0\big), \qquad \boldsymbol{h}_0 = \begin{bmatrix} d_2 \\ d_\ell \end{bmatrix}, \tag{S.23}$$

where

$$\boldsymbol{x}'\boldsymbol{D}^{-1}\boldsymbol{z} = \boldsymbol{a}'\boldsymbol{W}\boldsymbol{b}_z + (a_e\boldsymbol{b}_z + b_e\boldsymbol{a})'\boldsymbol{h}_0 + a_e b_e d_3.$$

Taking the Schur complement of $\boldsymbol{\Delta}$ in $\boldsymbol{K}$,

$$|\boldsymbol{K}| = |\boldsymbol{D}|\,(1+\kappa_1)(1+\kappa_2)\,S, \qquad S = 1 - \boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b}, \tag{S.24}$$

$$\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} = \boldsymbol{\beta}_c'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c + \frac{(b_n - \boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c)^2}{S}, \qquad \boldsymbol{\beta}_c = \boldsymbol{Z}\boldsymbol{b}_c, \tag{S.25}$$

and every term is an explicit expression in the seven primitives, q and (t_1, t_2), so the cost does not depend on n. The eigenvalues of $\boldsymbol{C}\boldsymbol{W}$ are the roots of $\kappa^2 - \operatorname{tr}(\boldsymbol{C}\boldsymbol{W})\kappa + |\boldsymbol{C}\boldsymbol{W}|$, with

$$\operatorname{tr}(\boldsymbol{C}\boldsymbol{W}) = C_{11}d_1 + C_{22}d_{\ell\ell}, \qquad |\boldsymbol{C}\boldsymbol{W}| = (C_{11}C_{22} - C_{12}^2)d_1 d_{\ell\ell};$$

only their values (the constant coefficients of the jets) are needed, for the phase (Section S7.4.2).

S7.3.3 Stationary start

By Proposition S7.1, case ST is case DT with $\gamma_\tau = 0$ and $\gamma \sim \mathcal{N}(0, \sigma_\gamma^2)$. At $\gamma_\tau = 0$ the exponent of Theorem S7.1 is $\gamma^2 C_\gamma$, with C_γ the value of $\frac{1}{2}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} + \eta$ at $(\gamma, \gamma_\tau) = (1, 0)$, so

$$\varphi_{\mathsf{ST}}(t_1, t_2) = |\boldsymbol{K}|^{-1/2}\big(1 - 2\sigma_\gamma^2 C_\gamma\big)^{-1/2}, \qquad \sigma_\gamma^2 = \frac{1-\phi}{1+\phi}. \tag{S.26}$$

At the arguments $(t_1, t_2) = (0, \mathrm{i}s_2)$, $s_2 \geq 0$, of the moment formula, $\boldsymbol{K} = \boldsymbol{F}'\boldsymbol{F} + 2s_2\boldsymbol{J}_n'\boldsymbol{M}_\tau\boldsymbol{J}_n$ is real and at least $\boldsymbol{F}'\boldsymbol{F}$, so $1 - 2\sigma_\gamma^2 C_\gamma \geq 1$. For real (t_1, t_2), $\operatorname{Re}(1 - 2\sigma_\gamma^2 C_\gamma) \geq 1$, because $\operatorname{Re}(\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta}) \leq \boldsymbol{\beta}'(\operatorname{Re}\,\boldsymbol{K})^{-1}\boldsymbol{\beta}$ for accretive $\boldsymbol{K}$ and real $\boldsymbol{\beta}$, and $\boldsymbol{\beta}'(\boldsymbol{F}'\boldsymbol{F})^{-1}\boldsymbol{\beta} = \boldsymbol{\mu}'\boldsymbol{\mu}$. So the principal square root is the continuous one.

S7.3.4 Derivatives

Every scalar in the assembly is carried as a jet in one variable, exactly as in Section 3.3: the density needs the first derivative in t_2 along the inversion paths, and the moments need the derivatives of any order in s_1 along the paths $(t_1, t_2) = (-\mathrm{i}s_1, \mathrm{i}(s_2 + s_1\tilde{c}))$. The primitives (S.20) are computed together with the others, in the same regimes.

S7.4 Numerical issues in finite samples

S7.4.1 Confluence near $\phi = \pm 1$

With $x = p/(2q)$ and U_k the Chebyshev polynomial of the second kind, $|\boldsymbol{D}| = q^m U_m(x)$, and the primitives d_1 to d_4 are expressed through U_m and U_{m-1}, with Taylor expansions at the confluent point on the bands $|\tau| < 5/2$ (Section 4.1). The new primitives (S.22) divide by $p + 2q$ as well, and their numerators vanish with it. In the confluent regime at $x = -1$ they are evaluated as

$$d_\ell = \frac{P_\ell}{4qU_m}, \qquad d_{\ell\ell} = \frac{R_\ell}{8qU_m}, \tag{S.27}$$

where P_ℓ and R_ℓ are the polynomials defined by the identities

$$(m-1)U_m + (m+1)\big(U_{m-1} + (-1)^m\big) = (x+1)P_\ell, \qquad 4\boldsymbol{\ell}'\boldsymbol{\ell}\, U_m + (m+1)P_\ell = (x+1)R_\ell,$$

the divisions by $x + 1$ being carried out on the Taylor coefficients at $x = -1$ (an index shift), so that no cancellation survives. At $x = +1$ (near $\phi = -1$) and in the generic regime, (S.22) is used, with $p + 2q$ supplied in closed form by (26).

S7.4.2 The branch

Proposition S7.3 (Continuous phase)**.** *Along any path of real (t_1, t_2) starting at the origin, the continuous argument of $|\boldsymbol{K}|$ is*

$$\Phi(t_1, t_2) + \operatorname{Arg}(1 + \kappa_1) + \operatorname{Arg}(1 + \kappa_2) + \operatorname{Arg} S,$$

where Φ is the continuous phase of $|\boldsymbol{D}|$ of Theorem 2 and the three other terms are principal arguments.

Proof. For $\lambda \in [0, 1]$ replace $\boldsymbol{M}_\tau$ by $\boldsymbol{I}_n - \lambda \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ in the proof of Theorem S7.1. The resulting matrix $\boldsymbol{K}_\lambda$ has $\operatorname{Re} \boldsymbol{K}_\lambda = \boldsymbol{F}'\boldsymbol{F}$, so it is accretive, and so is its core $\boldsymbol{D} + \lambda \boldsymbol{Z}\boldsymbol{C}\boldsymbol{Z}'$. An accretive matrix is nonsingular (Lemma 3(a)), so $|\boldsymbol{I}_2 + \lambda \boldsymbol{C}\boldsymbol{W}| = (1+\lambda\kappa_1)(1+\lambda\kappa_2) \neq 0$ for $\lambda \in [0, 1]$; that is, $1 + \kappa_j$ is not on the closed negative real axis. The scalar S is the Schur complement of the core in the accretive $\boldsymbol{K}$, so $\operatorname{Re} S > 0$ (Lemma 3(c)). A product of factors none of which reaches the closed negative real axis has the sum of their principal arguments as a continuous argument (the unordered pair $\{\kappa_1, \kappa_2\}$ is continuous,

and the sum is symmetric), and at the origin the factors $1+\kappa_j$ equal one and $|\boldsymbol{D}|$ and S are positive, so all the arguments vanish. □

S7.4.3 Cancellation in the explosive region

For $|\phi|>1$, $|\boldsymbol{D}|$ is of order ϕ^{2n} while $|\boldsymbol{K}|=O(1)$, so (S.24) multiplies a huge number by a small one, S, which is itself a difference of $O(1)$ quantities, and loses about $2n\log_{10}|\phi|$ digits. Proposition 3 rearranges cases DK and DU; we extend the rearrangement. In the notation of that proposition, the quantities δ_{DK}, Θ and Ω of (33) and (35) are small and free of cancellation. Define $\nu_1=qd_2$, $\nu_2=qd_\ell$, $\boldsymbol{\nu}=(\nu_1,\nu_2)'$, $a_n=n(1-\phi)+1+\phi$, and

$$\Upsilon \equiv k_0\nu_1+(1-\phi)\nu_2-\frac{m}{2}=\frac{\frac{\mathrm{i}}{2}\big(a_nt_1+2mt_2\big)-\frac{1}{2}a_n\Theta-n(1-\phi)\delta_{\mathsf{DK}}}{p+2q},$$

which is $O(n|\phi|^{-n})$, because $a_n=n(1-\phi)+O(1)$ and $p+2q$ stays bounded away from zero; at $t_1=t_2=0$, for instance, $\Theta\sim(\phi^2-1)\phi^{-n}$ and $\delta_{\mathsf{DK}}=O(\phi^{-2n})$, so that $\Upsilon\sim n(\phi+1)\phi^{-n}/2$. Then, by the definition of Υ and by (S.22),

$$\nu_2=\frac{\Upsilon-k_0\nu_1+m/2}{1-\phi},\qquad d_{\ell\ell}=\frac{n\nu_2+m(m^2-1)/12}{p+2q}.$$

Proposition S7.4 (Cancellation-free forms, case DT)**.** *With* $\boldsymbol{s}_C=\boldsymbol{C\nu}-2\boldsymbol{b}_v$,

$$\begin{aligned}S&=\delta_{\mathsf{DK}}+\boldsymbol{\nu}'\boldsymbol{s}_C-\boldsymbol{\nu}'\boldsymbol{C}\boldsymbol{W}\boldsymbol{M}_C^{-1}\boldsymbol{s}_C-\boldsymbol{b}_v'\boldsymbol{W}\boldsymbol{M}_C^{-1}\boldsymbol{b}_v,\\ \boldsymbol{\nu}'\boldsymbol{s}_C&=\mathrm{i}t_1\big[2G_1\nu_1\Omega+G_2\Upsilon(\nu_1+2\nu_2)\big]+2\mathrm{i}t_2\big[G_1\nu_1^2+G_2(\nu_1/2+\nu_2)^2\big].\end{aligned}$$

These formulas remove the exponential cancellation from the evaluation of S: every term on the right-hand sides is a product of small factors or an $O(1)$ quantity formed without subtraction of terms of larger exponential order.

Moreover, with $\boldsymbol{\nu}=q\boldsymbol{Z}'\boldsymbol{D}^{-1}\boldsymbol{e}_{m,m}=(\nu_1,\nu_2)'$, *let* $\bar{\boldsymbol{\nu}}=(\nu_1,-\nu_2)'=q\boldsymbol{Z}'\boldsymbol{D}^{-1}\boldsymbol{e}_{1,m}$ *(Lemma S7.1),* $\delta_{\mathsf{DK}}=1-q^2d_3$ *and*

$$\begin{aligned}\boldsymbol{a}&=G_1\begin{bmatrix}\mathrm{i}t_1(1-2\phi)+2\mathrm{i}t_2\\0\end{bmatrix}+\frac{m}{2}G_2\begin{bmatrix}\mathrm{i}t_1(\phi+\frac{1}{2})-\mathrm{i}t_2\\\mathrm{i}t_1(2\phi-1)-2\mathrm{i}t_2\end{bmatrix},\\ c_{00}&=\Big(G_1+\frac{m^2}{4}G_2\Big)(2\mathrm{i}t_2-2\mathrm{i}\phi t_1),\qquad c_{0n}=\mathrm{i}t_1\Big(G_1-\frac{m^2}{4}G_2\Big).\end{aligned}$$

Then the exponent of Theorem S7.1 is

$$\tfrac{1}{2}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta}+\eta=-\frac{z_*^2}{2}\,\frac{\delta_4}{S}, \tag{S.28}$$

with z_* the effective start (S.18), S as above,

$$\begin{aligned}
\delta_4 &= t_1^2-2\mathrm{i}t_2+\delta_{\mathsf{DK}}\Xi_{11}+\delta_1\Xi_{22}+2q^2d_4\Xi_{12}+\Xi_{11}\Xi_{22}-\Xi_{12}^2,\\
\delta_1 &= q^2(1-d_3)+t_1^2-2\mathrm{i}t_2,\\
\Xi_{11} &= c_{00}+\bar{\boldsymbol{\nu}}'\boldsymbol{M}_C^{-1}\boldsymbol{C}\bar{\boldsymbol{\nu}}-2\bar{\boldsymbol{\nu}}'\boldsymbol{M}_C^{-1}\boldsymbol{a}-\boldsymbol{a}'\boldsymbol{W}\boldsymbol{M}_C^{-1}\boldsymbol{a},\\
\Xi_{12} &= c_{0n}+\bar{\boldsymbol{\nu}}'\boldsymbol{M}_C^{-1}(\boldsymbol{C}\boldsymbol{\nu}-\boldsymbol{b}_v)-\boldsymbol{a}'\boldsymbol{M}_C'^{-1}(\boldsymbol{\nu}+\boldsymbol{W}\boldsymbol{b}_v),
\end{aligned}$$

and $\Xi_{22}=S-\delta_{\mathsf{DK}}$, the sum of the three explicit terms in the formula for S above.

Sketch. On the scale $t_1=O(|\phi|^{-n})$, $t_2=O(\phi^{-2n})$ on which the characteristic function varies in the explosive region, $\boldsymbol{C}$, $\boldsymbol{b}_v$ and $\boldsymbol{s}_C$ are $O(|\phi|^{-n})$ and $S=O(|\phi|^{-2n})$. Expanding $S=1-\boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b}$ with $\boldsymbol{b}=q\boldsymbol{e}_{m,m}+\boldsymbol{Z}\boldsymbol{b}_v$ by (S.23) gives $\delta_{\mathsf{DK}}=1-q^2d_3$, the two Woodbury corrections, which are products of two small factors, and $\boldsymbol{\nu}'\boldsymbol{s}_C$, a sum of terms of order $|\phi|^{-n}$ whose leading parts cancel. Rewriting it with $\nu_2=(\Upsilon-k_0\nu_1+m/2)/(1-\phi)$ and $(1-\phi)\nu_1=\Omega+1$ removes those parts identically and leaves the stated combination, in which $\Omega=O(|\phi|^{-n})$ and $\Upsilon=O(n|\phi|^{-n})$. In the term containing Υ, $G_2=O(n^{-3})$ and $\nu_1+2\nu_2=O(n)$, so the extra factor n does not affect the order of S. Υ itself follows from (S.21) and the row-sum identity $\boldsymbol{D}\mathbf{1}_m=(p+2q)\mathbf{1}_m-q(\boldsymbol{e}_{1,m}+\boldsymbol{e}_{m,m})$, which give $(p+2q)d_\ell$ and $(p+2q)d_2$ in terms of δ_{DK} and Θ. The algebra was done and checked with SymPy, and numerically against a dense evaluation in high-precision arithmetic (Section S7.6.1).

For the exponent: in the driftless autoregression from $x_0=z_*$, add x_0 as the first coordinate of the Gaussian integral, with a flat density. The matrix of $(x_0,x_1,\dots,x_n)$ then has the form (15) with the start as the first border coordinate. Its coupling to the core is $q\boldsymbol{e}_{1,m}+\boldsymbol{Z}\boldsymbol{a}$, its coupling to x_n is c_{0n}, and its diagonal entry is $q^2+t_1^2-2\mathrm{i}t_2+c_{00}$. Here $\boldsymbol{a}$, c_{00} and c_{0n} come from the projection on $(1,t)$, since the start is the lag at $t=1$ and enters the innovation $x_1-\phi x_0$. Integrating out $x_1,\dots,x_n$ at fixed $x_0=z_*$ gives $|\boldsymbol{K}|^{-1/2}\exp(-\frac{1}{2}z_*^2\Sigma_0)$, with Σ_0 the Schur complement of the start, which vanishes at $t_1=t_2=0$. By Lemma 1, $\Sigma_0=\delta_4/S$, where δ_4 is the determinant of the 2×2 Schur complement of $\boldsymbol{\Delta}$ in the bordered matrix.

Without the projection (case DK) the border is δ_1, $-q^2d_4$, δ_{DK}, and its determinant

equals $t_1^2 - 2it_2$ exactly. This is the identity $(1-d_3)(1-q^2d_3) - q^2d_4^2 = (t_1^2 - 2it_2)d_3$ of the closed forms (18), equivalent to the form $-\frac{1}{2}z_0^2(t_1^2 - 2it_2)/\delta_{\mathsf{DK}}$ of the exponent in case DK. The projection changes the border by $\boldsymbol{\Xi}$, and the Woodbury identity (S.23) gives its entries in the stated form, with $\boldsymbol{M}_C^{-1}\boldsymbol{C}\boldsymbol{W} + \boldsymbol{M}_C^{-1} = \boldsymbol{I}_2$ used to collect the terms. Every entry of $\boldsymbol{\Xi}$ is proportional to (t_1, t_2), so no term of δ_4 cancels a term of larger order. On the scale of the inversion, Ξ_{11}, Ξ_{12} and $\boldsymbol{a}$ are of order $|\phi|^{-n}$, and Ξ_{22} and δ_4 are of order ϕ^{-2n}. The terms of Ξ_{11} and Ξ_{12} are of the size of the entries themselves, so the form is free of cancellation also within them. The identities were checked against a dense evaluation in 60-digit arithmetic to 50 digits. □

The exponent $\frac{1}{2}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} + \eta$ of Theorem S7.1 cannot be formed as written in the explosive region: it is a difference of terms of order $n\gamma^2 + n^3\gamma_\tau^2/12$ that cancel at $t_1 = t_2 = 0$. By Proposition S7.1 it is that of the driftless autoregression started at the effective start z_*, and with the start as a further coordinate of the Gaussian integral, with a flat density, it is $-\frac{1}{2}z_*^2$ times the Schur complement of the start. Bordering the core by the start and by the last observation, as in Lemma 1, turns this Schur complement into a ratio of two border quantities in which nothing cancels.

The earlier assembly, $\frac{1}{2}[E_0 - \boldsymbol{b}_c'\boldsymbol{W}\boldsymbol{C}\boldsymbol{W}\boldsymbol{M}_C^{-1}\boldsymbol{b}_c + F_\beta^2/S]$ with $E_0 = \boldsymbol{b}_c'\boldsymbol{W}\boldsymbol{b}_c - n\gamma^2 - n(n^2-1)\gamma_\tau^2/12$ and $F_\beta = b_n - \boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c$, removed the exponential cancellation but not the polynomial one. At the origin with $\gamma = 0$ and $\gamma_\tau \neq 0$, for instance, $S = \delta_{\mathsf{DK}}$ and $F_\beta = -\gamma_\tau\Upsilon$, and F_β^2/S, of order n^2, cancels E_0 exactly. It lost about $\epsilon_{\mathrm{M}}(\gamma_\tau n)^2$: at $\phi = 1.2$, $\gamma = 0.4$ and $\gamma_\tau = 1$, its value at $t_1 = t_2 = 0$ was 4×10^{-3} at $n = 5 \times 10^6$ and -64 at $n = 5 \times 10^8$, instead of zero. With (S.28) the value at the origin is zero to rounding at every n. At $n = 5 \times 10^6$ and 5×10^8 the scaled distribution function approaches the limit at the effective start like n^{-1}, with n times the largest difference over four points equal to 4.85 at both sample sizes; the earlier form was off by 1.6×10^{-3} and 0.41 there.

The phase is assembled as in Proposition S7.3, with Φ from (31), and, as in cases DK and DU, the distribution of $\hat{\phi} - \phi$ can be evaluated for ϕ^{2n} up to about 10^{300}, the range in which ϕ^{-2n} is representable, and that of $|\phi|^n(\hat{\phi} - \phi)/(\phi^2 - 1)$ for every n (Section 4.3). In the scaled arguments $|\phi|^n t_1$ and $\phi^{2n}t_2$ the quantities $\boldsymbol{C}$, $\boldsymbol{b}_v$, Ω and Υ carry the factor $|\phi|^{-n}$ and S the factor ϕ^{-2n}, and the programs carry them divided by these factors. At $\phi = 1.05$ with effective start $z_* = 2$, n times the largest difference between the computed distribution function and the limit at the effective start, over a few fixed x, is 27.18, 27.09 and 27.08 at $n = 1600$, 1.6×10^4 and 1.6×10^5, the last two far beyond $\phi^{2n} = 10^{300}$: the

$O(n^{-1})$ approach of Proposition S7.7.

S7.4.4 The drift near the unit root

The exponent of Theorem S7.1 is $O(1)$, but it is assembled from $\eta = -\frac{1}{2}[n\gamma^2 + \gamma_\tau^2 n(n^2-1)/12]$ and $\frac{1}{2}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta}$, each of the size of η. Near the unit root these cancel. At $\phi = 1$ the drift γ is removed by the regression and has no effect at all (Proposition S7.1). Yet with $\gamma = 10$, the characteristic function computed from (S.25) differs from the one at $\gamma = 0$ by 2.7×10^{-9} at $n = 1000$, 3.9×10^{-7} at $n = 5000$ and 2.4×10^{-6} at $n = 20000$ (maximum over five points of the inversion region). The error is of the order $\gamma^2 n^2 \epsilon_{\mathrm{M}}$, as for the drift in case DU (Section 4.4). Proposition S7.1 removes the problem. Write $\xi = a_\gamma\gamma + a_\tau\gamma_\tau$ in (S.17), with $a_\gamma = 1-\phi$ and $a_\tau = -[(1-\phi)(n+1)/2 + \phi]$. Since only ξ matters, the programs replace (γ, γ_τ) by the pair with the same ξ that minimizes $n\gamma^2 + n^3\gamma_\tau^2/12$, approximately $\boldsymbol{\mu}'\boldsymbol{\mu} = n\gamma^2 + n(n^2-1)\gamma_\tau^2/12$,

$$\gamma^\circ = \frac{\lambda a_\gamma}{n}, \qquad \gamma_\tau^\circ = \frac{12\lambda a_\tau}{n^3}, \qquad \lambda = \frac{\xi}{a_\gamma^2/n + 12a_\tau^2/n^3}. \tag{S.29}$$

At $\phi = 1$, $\gamma^\circ = 0$ and the result is exactly the one with $\gamma = 0$; in general the cancellation is reduced to nearly the intrinsic one, set by the smallest value of $\boldsymbol{\mu}'\boldsymbol{\mu}$ compatible with ξ (the pair (S.29) minimizes the proxy $n\gamma^2 + n^3\gamma_\tau^2/12$, not $\boldsymbol{\mu}'\boldsymbol{\mu}$ itself). The explosive forms do not need the reparameterization: they use the effective start directly (Proposition S7.4).

S7.5 Limiting distributions

S7.5.1 The local-to-unity limit

Let $\phi = 1 + c/n$ with c fixed, and let

$$\tilde{H}_c(r) = \int_0^r H_c(s)\,\mathrm{d}s = \frac{e^{cr} - 1 - cr}{c^2},$$

with H_c of Section 6.1 ($\tilde{H}_c(r) = r^2/2$ at $c = 0$). For a function Z on $[0,1]$ let $\hat{Z}$ be the residual of its L^2 regression on 1 and r, $\hat{Z}(r) = Z(r) - \hat{a} - \hat{b}r$, where

$$\begin{bmatrix}\hat{a}\\ \hat{b}\end{bmatrix} = \boldsymbol{G}^{-1}\begin{bmatrix}\int_0^1 Z(r)\,\mathrm{d}r\\ \int_0^1 rZ(r)\,\mathrm{d}r\end{bmatrix}, \qquad \boldsymbol{G} = \begin{bmatrix}1 & 1/2\\ 1/2 & 1/3\end{bmatrix}, \qquad \boldsymbol{G}^{-1} = \begin{bmatrix}4 & -6\\ -6 & 12\end{bmatrix}.$$

Proposition S7.5 (Local-to-unity limit)**.** *In case* DT *let*

$$\varkappa_{\tau,n} = -n^{3/2}\xi = \sqrt{n}\left[c\big(\gamma - \gamma_\tau(n+1)/2\big) + n\phi\gamma_\tau\right],$$

with ξ *of (S.17), and suppose* $\varkappa_{\tau,n} \to \varkappa_\tau$*. Then* $n(\hat\phi - \phi) \Rightarrow A_\tau/B_\tau$*, where*

$$A_\tau = \int_0^1 \hat X_\tau(r)\,\mathrm{d}W(r), \qquad B_\tau = \int_0^1 \hat X_\tau(r)^2\,\mathrm{d}r, \qquad X_\tau = J_c + \varkappa_\tau \tilde H_c. \tag{S.30}$$

Here $\int_0^1 \hat X_\tau\,\mathrm{d}W$ *stands for* $\int_0^1 X_\tau\,\mathrm{d}W - \hat a W(1) - \hat b\int_0^1 r\,\mathrm{d}W(r)$*, where* $\hat a + \hat b r$ *is the least squares fit of a line to the whole path of* X_τ*; this defines the integral although* $\hat a$ *and* $\hat b$ *are anticipative. In case* ST *(*$c < 0$*) the limit is (S.30) with* $\varkappa_\tau \sim \mathcal{N}(0, -c^3/2)$ *independent of* W*.*

Sketch. For $c \neq 0$, by Proposition S7.1 $\hat\phi$ is the detrended estimator of $x_t = \phi^t z_* + \sum_{j=1}^t \phi^{t-j} e_j$ with $z_* = -\xi/(1-\phi)^2 = -n^2\xi/c^2$. So $x_{\lfloor nr\rfloor}/\sqrt{n} \Rightarrow (\varkappa_\tau/c^2)e^{cr} + J_c(r)$, and $(\varkappa_\tau/c^2)e^{cr} = \varkappa_\tau \tilde H_c(r)$ up to a linear function of r, which the regression removes. For $c = 0$, $\varkappa_{\tau,n} = n^{3/2}\gamma_\tau$, and the quadratic term $\gamma_\tau t(t-n)/2$ in the proof of Proposition S7.1 gives $n^{-1/2}\gamma_\tau\lfloor nr\rfloor^2/2 \to \varkappa_\tau r^2/2$ up to a linear function. The numerator and denominator of $n(\hat\phi - \phi)$ are

$$\frac{1}{n}\sum_{t=1}^n \hat x_{t-1} e_t \qquad \text{and} \qquad \frac{1}{n^2}\sum_{t=1}^n \hat x_{t-1}^2,$$

with $\hat x_{t-1}$ the detrended lagged values, and the usual arguments (Phillips, 1987; Chan and Wei, 1987) give (S.30). In case ST, $z_* = \tilde x_0/\sigma \sim \mathcal{N}(0, 1/(1-\phi^2))$, so $z_*/\sqrt{n} \Rightarrow \mathcal{N}(0, -1/(2c))$ and $\varkappa_\tau = c^2 \lim z_*/\sqrt{n} \sim \mathcal{N}(0, -c^3/2)$. □

At $\varkappa_\tau = 0$ and $c = 0$ the limit is the coefficient form of the Dickey–Fuller distribution with a trend. Since $\tilde H_c'(r) = c\tilde H_c(r) + r$, $X_\tau(0) = 0$ and

$$\mathrm{d}X_\tau(r) = \{cX_\tau(r) + \varkappa_\tau r\}\,\mathrm{d}r + \mathrm{d}W(r),$$

and since $\hat X_\tau$ is orthogonal to 1 and r in $L^2[0,1]$,

$$\int_0^1 \hat X_\tau(r)\,\mathrm{d}W(r) = \int_0^1 \hat X_\tau(r)\,\mathrm{d}X_\tau(r) - c\int_0^1 \hat X_\tau(r)^2\,\mathrm{d}r,$$

so that

$$A_\tau + cB_\tau = \int_0^1 \hat{X}_\tau(r)\,\mathrm{d}X_\tau(r). \tag{S.31}$$

S7.5.2 The characteristic function in closed form

For real (t_1, t_2) let ω, θ^2 and τ_0 be as in Proposition 4, and let

$$\varsigma = 2\mathrm{i}(t_2 - ct_1) = c^2 + \theta^2, \qquad \vartheta = (c^2 - \varsigma)^{1/2} = (-\theta^2)^{1/2}, \qquad E = e^{-\vartheta},$$

with the principal root, so that $\operatorname{Re}\vartheta \geq 0$ and $\vartheta^2 - \omega^2 = \tau_0$. Define

$$\begin{aligned}
h_1 &= (\vartheta - 2) + E(\vartheta + 2),\\
h_2 &= -(\vartheta^2 - 6\vartheta + 12) + E(\vartheta^2 + 6\vartheta + 12),\\
B_0(\vartheta) &= c^4(\vartheta^3 - 8\vartheta^2 + 24\vartheta - 24) - 8c^3(\vartheta^3 - 7\vartheta^2 + 18\vartheta - 15)\\
&\quad + 8c^2(4\vartheta^3 - 24\vartheta^2 + 51\vartheta - 36) - 72c(\vartheta - 1)(\vartheta - 2)^2 + 72\vartheta(\vartheta - 2)^2,\\
B_1(\vartheta) &= (c - 3)(c^2 - 4c + 6)\vartheta^2 - 6(c^3 - 5c^2 + 12c - 12),
\end{aligned}$$

and

$$\begin{aligned}
\mathcal{D} = h_1h_2\big[c(c-2)(c^2 - 6c + 12)\,\mathrm{i}t_1 - 4(c^2 - 3c + 3)\tau_0\big]\\
+ (\vartheta - c)B_0(\vartheta) + 8\varsigma B_1(\vartheta)E - (\vartheta + c)B_0(-\vartheta)E^2.
\end{aligned} \tag{S.32}$$

Theorem S7.2 (Limiting characteristic functions)**.** *In case* ST*,*

$$\mathbb{E}\big[e^{\mathrm{i}t_1A_\tau + \mathrm{i}t_2B_\tau}\big] = e^{-(\omega+\vartheta)/2}\Big(\frac{\mathcal{D}}{2\vartheta^8}\Big)^{-1/2}\exp\Big(\frac{\varkappa_\tau^2\tau_0h_1h_2}{2\mathcal{D}}\Big), \tag{S.33}$$

and in case ST *($c < 0$),*

$$\mathbb{E}\big[e^{\mathrm{i}t_1A_\tau + \mathrm{i}t_2B_\tau}\big] = e^{-(\omega+\vartheta)/2}\Big(\frac{\mathcal{D}}{2\vartheta^8}\Big)^{-1/2}\Big(1 + \frac{c^3\tau_0h_1h_2}{2\mathcal{D}}\Big)^{-1/2}, \tag{S.34}$$

the square roots being taken continuously in (t_1, t_2) from their positive values at the origin, where both right-hand sides equal one (there $\varsigma = \tau_0 = 0$, $\omega = c$ and $\vartheta = |c|$, and $B_0(-c) = -c^7$ gives $\mathcal{D}/(2\vartheta^8) = e^{-(c+|c|)}$). The functions $e^\vartheta\mathcal{D}/\vartheta^8$ and $e^\vartheta h_1h_2/\vartheta^8$ are entire functions of ϑ^2 (removable singularities filled in by continuity), so (S.33) and (S.34) do not depend on the choice of the root ϑ.

Proof. Let $Y = (X_\tau(1), I_0, I_1)'$, with $I_0 = \int_0^1 X_\tau(r)\,\mathrm{d}r$ and $I_1 = \int_0^1 rX_\tau(r)\,\mathrm{d}r$. The exponent is

$$\mathrm{i}t_1 A_\tau + \mathrm{i}t_2 B_\tau = \mathrm{i}t_1(A_\tau + cB_\tau) + \mathrm{i}(t_2 - ct_1)B_\tau.$$

By (S.31) and Itô's formula,

$$\int_0^1 \hat{X}_\tau(r)\,\mathrm{d}X_\tau(r) = \tfrac{1}{2}\big(X_\tau(1)^2 - 1\big) - \hat{a}X_\tau(1) - \hat{b}\big(X_\tau(1) - I_0\big),$$
$$\int_0^1 \hat{X}_\tau(r)^2\,\mathrm{d}r = \int_0^1 X_\tau(r)^2\,\mathrm{d}r - (I_0, I_1)\boldsymbol{G}^{-1}(I_0, I_1)'.$$

Under the original measure X_τ satisfies the equation displayed before (S.31). For real μ let Q_μ be the measure under which $\mathrm{d}X_\tau(r) = \mu X_\tau(r)\,\mathrm{d}r + \mathrm{d}W(r)$, $X_\tau(0) = 0$. By Girsanov's theorem, with $\int_0^1 X_\tau(r)\,\mathrm{d}X_\tau(r) = (X_\tau(1)^2 - 1)/2$ and $\int_0^1 r\,\mathrm{d}X_\tau(r) = X_\tau(1) - I_0$,

$$\frac{\mathrm{d}P}{\mathrm{d}Q_\mu} = \exp\Big\{\frac{c-\mu}{2}\big(X_\tau(1)^2 - 1\big) + \varkappa_\tau\big(X_\tau(1) - I_0\big) - \frac{c^2 - \mu^2}{2}\int_0^1 X_\tau(r)^2\,\mathrm{d}r - c\varkappa_\tau I_1 - \frac{\varkappa_\tau^2}{6}\Big\}.$$

Choose $\mu^2 = c^2 - \varsigma$, so that the terms in $\int_0^1 X_\tau(r)^2\,\mathrm{d}r$ cancel, and take $\mu = -\vartheta$. Then

$$\mathbb{E}\big[e^{\mathrm{i}t_1 A_\tau + \mathrm{i}t_2 B_\tau}\big] = e^{-(\omega+\vartheta)/2 - \varkappa_\tau^2/6}\,\mathbb{E}_{Q_\mu}\big[\exp(Y'\boldsymbol{A}Y + \varkappa_\tau \boldsymbol{b}_0'Y)\big], \qquad \boldsymbol{b}_0 = (1, -1, -c)',$$

$$\boldsymbol{A} = \begin{bmatrix} (\omega+\vartheta)/2 & \mathrm{i}t_1 & -3\mathrm{i}t_1 \\ \mathrm{i}t_1 & -6\mathrm{i}t_1 - 2\varsigma & 6\mathrm{i}t_1 + 3\varsigma \\ -3\mathrm{i}t_1 & 6\mathrm{i}t_1 + 3\varsigma & -6\varsigma \end{bmatrix},$$

and under Q_μ, Y is normal with mean zero and covariance $\boldsymbol{\Sigma}(\mu)$ with entries, writing $\varepsilon = e^\mu$,

$$\Sigma_{11} = \frac{\varepsilon^2 - 1}{2\mu}, \quad \Sigma_{12} = \frac{(\varepsilon-1)^2}{2\mu^2}, \quad \Sigma_{13} = \frac{\varepsilon^2+1}{2\mu^2} - \frac{\varepsilon^2-1}{2\mu^3}, \quad \Sigma_{22} = \frac{1}{\mu^2} + \frac{\varepsilon^2 - 4\varepsilon + 3}{2\mu^3},$$
$$\Sigma_{23} = \frac{1}{2\mu^2} + \frac{(\varepsilon-1)^2}{2\mu^3} - \frac{(\varepsilon-1)^2}{2\mu^4}, \quad \Sigma_{33} = \frac{1}{3\mu^2} + \frac{\varepsilon^2+1}{2\mu^3} - \frac{\varepsilon^2}{\mu^4} + \frac{\varepsilon^2-1}{2\mu^5}.$$

When $\mathrm{i}t_1$ and $\mathrm{i}t_2$ are real (moment generating function arguments) and μ is real, the change of measure is legitimate, and

$$\mathbb{E}_{Q_\mu}\big[\exp(Y'\boldsymbol{A}Y + \varkappa_\tau \boldsymbol{b}_0'Y)\big] = |\boldsymbol{I}_3 - 2\boldsymbol{\Sigma}\boldsymbol{A}|^{-1/2}\exp\big(\tfrac{1}{2}\varkappa_\tau^2 \boldsymbol{b}_0'\boldsymbol{\Sigma}(\boldsymbol{I}_3 - 2\boldsymbol{A}\boldsymbol{\Sigma})^{-1}\boldsymbol{b}_0\big);$$

both sides of the resulting identity are analytic in (t_1, t_2), so it extends to real (t_1, t_2) along continuous paths. Symbolic simplification (SymPy) gives

$$|\boldsymbol{I}_3 - 2\boldsymbol{\Sigma}\boldsymbol{A}| = \frac{\mathcal{D}}{2\vartheta^8}, \qquad -\frac{1}{6} + \frac{1}{2}\boldsymbol{b}_0'\boldsymbol{\Sigma}(\boldsymbol{I}_3 - 2\boldsymbol{A}\boldsymbol{\Sigma})^{-1}\boldsymbol{b}_0 = \frac{\tau_0 h_1 h_2}{2\mathcal{D}};$$

the dependence on ω is through $\omega^2 - \vartheta^2 = -\tau_0$ and $\omega - c = \mathrm{i}t_1$ only, which gives (S.32). For case ST, with $\Lambda_\tau = \tau_0 h_1 h_2/(2\mathcal{D})$ and $\operatorname{Var} \varkappa_\tau = -c^3/2$,

$$\mathbb{E}\big[e^{\varkappa_\tau^2 \Lambda_\tau}\big] = \big(1 - 2\Lambda_\tau \operatorname{Var} \varkappa_\tau\big)^{-1/2} = \big(1 + c^3\Lambda_\tau\big)^{-1/2}.$$

Finally,

$$e^{\vartheta/2} h_1 = 2\vartheta \cosh\frac{\vartheta}{2} - 4\sinh\frac{\vartheta}{2}, \qquad e^{\vartheta/2} h_2 = 12\vartheta \cosh\frac{\vartheta}{2} - 2(\vartheta^2 + 12)\sinh\frac{\vartheta}{2}$$

are odd entire functions of ϑ vanishing to orders three and five. Let $\mathcal{D}_E$ be the sum of the last three terms of (S.32) and $f(\vartheta) = (\vartheta - c)B_0(\vartheta)e^{\vartheta}$; then

$$e^{\vartheta}\mathcal{D}_E = f(\vartheta) + f(-\vartheta) + 8\varsigma B_1(\vartheta)$$

is an even entire function vanishing to order eight. □

At $\varkappa_\tau = 0$, (S.33) agrees with the Fredholm determinant of Nabeya and Tanaka (1990, Model 4) to 3×10^{-12} (relative, on rays covering the support, for $-30 \le c \le 5$). For $\varkappa_\tau \neq 0$ and in case ST it agrees with the limit of the finite-sample characteristic function (four-point Richardson extrapolation from $n = 2^{12}, \ldots, 2^{15}$) to 10^{-11}.

S7.5.3 Evaluation

The closed form is evaluated as follows; all quantities are jets.

(i) *Near the origin,* $|\vartheta^2| < 16$. With $z = \vartheta^2$, the entire function $\Psi(z) = e^{\vartheta}\mathcal{D}/\vartheta^8$ is

$$\Psi(z) = \chi_1(z)\chi_2(z)\big[c(c-2)(c^2 - 6c + 12)\mathrm{i}t_1 - 4(c^2 - 3c + 3)\tau_0\big] + \mathcal{R}(z),$$

where $\chi_1 = e^{\vartheta/2}h_1/\vartheta^3$, $\chi_2 = e^{\vartheta/2}h_2/\vartheta^5$ and $\mathcal{R} = e^{\vartheta}\mathcal{D}_E/\vartheta^8$ have the power series

$$\chi_1(z) = \sum_{j=1}^{\infty} \frac{4j\, z^{j-1}}{4^j(2j+1)!}, \qquad \chi_2(z) = \sum_{j=2}^{\infty} \frac{z^{j-2}}{4^j}\Big[\frac{12}{(2j)!} - \frac{4}{(2j-1)!} - \frac{12}{(2j+1)!}\Big],$$

$$\mathcal{R}(z) = \sum_{k=4}^{\infty} z^{k-4}\, 2 \sum_{j=0}^{4} \frac{\upsilon_j}{(2k-j)!},$$

with $(\vartheta - c)B_0(\vartheta) = \sum_{j=0}^{4} \upsilon_j \vartheta^j$,

$$\upsilon_0 = 24c^2(c-2)(c^2-3c+6), \qquad \upsilon_1 = -\upsilon_0,$$
$$\upsilon_2 = 8(c^5 - 4c^4 + 6c^3 + 6c^2 - 36c + 36), \qquad \upsilon_3 = -c^5 + 24c^3 - 120c^2 + 288c - 288,$$
$$\upsilon_4 = c^4 - 8c^3 + 32c^2 - 72c + 72.$$

Then $\log \psi_0 = -\omega/2 - \frac{1}{2}\log(\Psi/2)$, with ψ_0 the characteristic function (S.33) at $\varkappa_\tau = 0$, and the exponent of (S.33) is $\varkappa_\tau^2 \tau_0 \chi_1 \chi_2/(2\Psi)$. Forty terms give full accuracy on $|z| < 16$. The closed forms in E lose about two digits in the higher derivatives near $|z| = 4$, which is why the threshold is 16 rather than the 4 of Section 6.1.

(ii) *Elsewhere*, (S.32) is evaluated with $\vartheta - c$ and $\vartheta + c$ formed without cancellation: for $c \geq 0$, $\vartheta - c = -\varsigma/(\vartheta + c)$, and for $c < 0$, $\vartheta + c = -\varsigma/(\vartheta - c)$. Both $B_0(\vartheta)$ and $B_0(-\vartheta)$ are expanded about $\vartheta = |c|$, where $\vartheta - |c|$ vanishes:

$$\begin{aligned} B_0(c+y) &= c(c-2)^2(c^2-6c+12)^2 \\ &\quad + y(3c^6 - 40c^5 + 232c^4 - 744c^3 + 1344c^2 - 1152c + 288) \\ &\quad + y^2(3c^5 - 32c^4 + 152c^3 - 408c^2 + 576c - 288) + y^3\upsilon_4, \\ B_0(-c+y) &= -c^7 + y(3c^6 - 8c^5 + 8c^4 + 24c^3 - 96c^2 + 288) \\ &\quad + y^2(-3c^5 + 16c^4 - 40c^3 + 24c^2 + 144c - 288) + y^3\upsilon_4, \end{aligned}$$

with $y = \vartheta - c$ in $B_0(\vartheta) = B_0(c+y)$ and $y = c - \vartheta$ in $B_0(-\vartheta) = B_0(-c+y)$ when $c \geq 0$, and $y = \vartheta + c$ in $B_0(\vartheta) = B_0(-c+y)$ and $y = -(\vartheta + c)$ in $B_0(-\vartheta) = B_0(c+y)$ when $c < 0$. On the explosive side ($c > 0$) the characteristic function varies on the scale $t_1 = O(e^{-c})$ (up to powers of c) and $t_2 = O(e^{-2c})$, where $\vartheta - c$, ς and $\mathrm{i}t_1$ are $O(e^{-c})$ while τ_0, E^2 and $\mathcal{D}$ are $O(e^{-2c})$, so that the first two terms of (S.32) cancel to leading order. With $K_c = c(c-2)(c^2-6c+12)$, the identities $\mathrm{i}t_1 = (\vartheta - c) - \tau_0/(\vartheta + \omega)$

(from $\vartheta^2 - \omega^2 = \tau_0$),

$$h_1 h_2 = -(\vartheta^3 - 8\vartheta^2 + 24\vartheta - 24) + 8(\vartheta^2 - 6)E + (\vartheta^3 + 8\vartheta^2 + 24\vartheta + 24)E^2$$

and

$$B_0(\vartheta) - K_c(\vartheta^3 - 8\vartheta^2 + 24\vartheta - 24) = 8(\vartheta - c)\{(c-3)\vartheta - 3(c-2)\}^2$$

give, for $c > 0$,

$$\begin{aligned}\mathcal{D} &= 8(\vartheta - c)^2\{(c-3)\vartheta - 3(c-2)\}^2 \\ &\quad + K_c(\vartheta - c)\big[8(\vartheta^2 - 6)E + (\vartheta^3 + 8\vartheta^2 + 24\vartheta + 24)E^2\big] \\ &\quad - \tau_0 h_1 h_2\Big[\frac{K_c}{\vartheta + \omega} + 4(c^2 - 3c + 3)\Big] + 8\varsigma B_1(\vartheta)E - (\vartheta + c)B_0(-\vartheta)E^2,\end{aligned}$$

with $\mathrm{Re}(\vartheta + \omega) \geq c$, in which every term is $O(e^{-2c})$ and no larger terms are subtracted; for instance, $\mathcal{D} = 2c^8 e^{-2c}$ at the origin comes from the last term alone. This is the counterpart of the explosive rearrangement in finite samples; the result is accurate to a few units of rounding for c up to 300 or so.

(iii) *The phase* is that of Proposition S7.6 below.

S7.5.4 The branch of the limit

Let $\psi_{\mathsf{DK}} = e^{-\omega/2}(\mathcal{C} - \omega\mathcal{S})^{-1/2}$ be the limiting characteristic function (43) of case DK with $\gamma_0 = 0$, whose continuous phase is given in Section S1, and let

$$\begin{aligned}g_{11} &= \frac{h_1}{\vartheta^3(1+E)} = \frac{1}{\vartheta^2}\Big(1 - \frac{2}{\vartheta}\tanh\frac{\vartheta}{2}\Big), \\ g_{\ell\ell} &= -\frac{h_2}{12\vartheta^4(1-E)} = \frac{1}{\vartheta^2}\Big(\frac{1}{12} - \frac{1}{2\vartheta}\coth\frac{\vartheta}{2} + \frac{1}{\vartheta^2}\Big).\end{aligned}$$

These are

$$\int_0^1\int_0^1 G_\vartheta(r,s)\,\mathrm{d}r\,\mathrm{d}s \qquad \text{and} \qquad \int_0^1\int_0^1 \big(r - \tfrac{1}{2}\big)G_\vartheta(r,s)\big(s - \tfrac{1}{2}\big)\,\mathrm{d}r\,\mathrm{d}s$$

for the Green function G_ϑ of $-y'' + \vartheta^2 y$ with $y(0) = y(1) = 0$. Let $\bar\kappa_1$ and $\bar\kappa_2$ be the eigenvalues of

$$\tilde{\boldsymbol{C}}\tilde{\boldsymbol{W}} = \begin{bmatrix} \varsigma & -12\mathrm{i}t_1 \\ -12\mathrm{i}t_1 & 12\varsigma \end{bmatrix} \begin{bmatrix} g_{11} & 0 \\ 0 & g_{\ell\ell} \end{bmatrix}.$$

Proposition S7.6 (Continuous phase of the limit)**.** *Let ψ_0 be (S.33) at $\varkappa_\tau = 0$, and let*

$$\rho_\tau = \frac{\psi_{\mathsf{DK}}^2}{\psi_0^2(1+\bar\kappa_1)(1+\bar\kappa_2)}$$

(formed from the values of ψ_{DK}^2, ψ_0^2 and $1+\bar\kappa_j$, so that no branch is involved). Along every continuous path of real (t_1, t_2) from the origin,

$$\operatorname{Arg_c}\psi_0 = \operatorname{Arg_c}\psi_{\mathsf{DK}} - \tfrac{1}{2}\big[\operatorname{Arg}(1+\bar\kappa_1) + \operatorname{Arg}(1+\bar\kappa_2) + \operatorname{Arg}\rho_\tau\big],$$

where $\operatorname{Arg_c}$ is the continuous argument of Lemma S1.3 and the three terms in brackets are principal arguments. In case ST the mixing factor of (S.34) has positive real part, so its principal root is the continuous one.

Sketch. At finite n, $|\boldsymbol{K}_{\mathsf{DT}}| = |\boldsymbol{D}|(1+\kappa_1^{(n)})(1+\kappa_2^{(n)})S$ (Section S7.3.2) and $|\boldsymbol{K}_{\mathsf{DK}}| = |\boldsymbol{D}|\delta_{\mathsf{DK}}$ (Proposition 2), the two matrices sharing the core $\boldsymbol{D}$. The factors $1+\kappa_j^{(n)}$ avoid the closed negative real axis (Proposition S7.3), and S and δ_{DK}, Schur complements of accretive matrices, have positive real parts, so $\operatorname{Arg}(S/\delta_{\mathsf{DK}}) = \operatorname{Arg} S - \operatorname{Arg}\delta_{\mathsf{DK}}$. With the transform arguments scaled as $(t_1/n, t_2/n^2)$,

$$n^3 C_{11} \to \varsigma, \qquad n^4 C_{12} \to -12\mathrm{i}t_1, \qquad n^5 C_{22} \to 12\varsigma, \qquad \frac{d_1}{n^3} \to g_{11}, \qquad \frac{d_{\ell\ell}}{n^5} \to g_{\ell\ell};$$

the core $\boldsymbol{D}$ is n^{-2} times a discretization of $-y'' + \vartheta^2 y$ with the Dirichlet conditions that the dominant boundary terms impose. So $\kappa_j^{(n)} \to \bar\kappa_j$ (the eigenvalues are invariant under the rescaling of $\boldsymbol{\ell}$ by $1/n$), and $S/\delta_{\mathsf{DK}} \to \rho_\tau$. The identity holds at finite n, and it passes to the limit because no factor reaches the closed negative real axis in the limit. For $1+\bar\kappa_j$: the Dirichlet operator $L = -\partial_r^2 + \vartheta^2$ has $\operatorname{Re}\vartheta^2 = c^2$, so for $f = 1$ or $f = r - \frac{1}{2}$ and $g = L^{-1}f$, integration by parts gives

$$\operatorname{Re}\langle f, g\rangle = \int_0^1 \big(|g'(r)|^2 + c^2|g(r)|^2\big)\,\mathrm{d}r > 0,$$

so both diagonal entries of $\tilde{\boldsymbol{W}}$ have positive real parts. For $0 \le \lambda \le 1$ the matrix

$\tilde{\boldsymbol{W}}^{-1} + \lambda\tilde{\boldsymbol{C}}$ is then accretive, $\tilde{\boldsymbol{C}}$ being symmetric with purely imaginary entries, so $|\boldsymbol{I}_2 + \lambda\tilde{\boldsymbol{C}}\tilde{\boldsymbol{W}}| = (1+\lambda\bar{\kappa}_1)(1+\lambda\bar{\kappa}_2) \neq 0$, and $1+\bar{\kappa}_j$ is not on the closed negative real axis. For ρ_τ: S and δ_{DK} are Schur complements $\boldsymbol{h}^*\boldsymbol{K}\boldsymbol{h}$ with elimination vectors whose last coordinate is one, and $\operatorname{Re}\boldsymbol{K} = \boldsymbol{F}'\boldsymbol{F}$ in both models, so the telescoping bound in the proof of Lemma S1.4 gives

$$\operatorname{Re}(nS),\ \operatorname{Re}(n\delta_{\mathsf{DK}}) \ \geq\ n\Big(\sum_{k=0}^{n-1}\phi^{2k}\Big)^{-1} \to \frac{2c}{e^{2c}-1} > 0$$

(one at $c=0$). Both scaled complements have finite limits. First, $n\delta_{\mathsf{DK}} \to \theta\cot\theta - \omega$, which is finite because $\theta^2 + c^2 = 2\mathrm{i}(t_2 - ct_1)$ is purely imaginary, so that $\sin\theta \neq 0$ for $\theta \neq 0$. Second, $S/\delta_{\mathsf{DK}} \to \rho_\tau$, which is finite because $1+\bar{\kappa}_j \neq 0$ and $\psi_0 \neq 0$: at $\varkappa_\tau = 0$, $t_1A_\tau + t_2B_\tau$ is a quadratic functional of a zero-mean Gaussian process with square-summable eigenvalues λ_j, whose characteristic function has modulus $\prod_{j=1}^{\infty}(1+4\lambda_j^2)^{-1/4} > 0$ at unit argument. So ρ_τ is the ratio of two numbers with positive real parts, and is not on the closed negative real axis. For case ST, let Λ_τ be as in the proof of Theorem S7.2. As in the proof of Lemma S1.2, $|\psi_0|e^{\varkappa_\tau^2 \operatorname{Re}\Lambda_\tau} \leq 1$ for every real $\varkappa_\tau$ gives $\operatorname{Re}\Lambda_\tau \leq 0$, so $\operatorname{Re}(1+c^3\Lambda_\tau) \geq 1$ for $c<0$. □

The phase of ψ_{DK} is computed as in Section S1; then ρ_τ is formed from the values, and the constant term of $\log\psi_0$ receives the phase of the proposition. Against the phase of the Nabeya–Tanaka determinant continued numerically along the paths, the result agrees to 5×10^{-12} on rays covering the support for c from -30 to 5 (at $c=10$ the reference itself loses accuracy, to 6×10^{-8}).

S7.5.5 The point $r=-c$

Along the ray $t_2 = ct_1$ the characteristic function of $A_\tau + cB_\tau$ decays only like $|t_1|^{-1}$, so the inversion integral (6) at and near $r=-c$ converges slowly. Near $r=-c$ the programs complete it by the asymptotic tail expansion of Section 6.1; at $r=-c$ itself they use the representation below. By (S.31) and the proof of Theorem S7.2,

$$A_\tau + cB_\tau = Y'\boldsymbol{\Lambda}Y - \tfrac{1}{2}, \qquad \boldsymbol{\Lambda} = \begin{bmatrix} 1/2 & 1 & -3 \\ 1 & -6 & 6 \\ -3 & 6 & 0 \end{bmatrix}, \tag{S.35}$$

with $Y = (X_\tau(1), I_0, I_1)'$ normal, with mean $\varkappa_\tau \boldsymbol{m}$, where

$$\boldsymbol{m} = \Big(\tilde{H}_c(1),\ \int_0^1 \tilde{H}_c(r)\,\mathrm{d}r,\ \int_0^1 r\tilde{H}_c(r)\,\mathrm{d}r \Big)',$$

and covariance $\boldsymbol{\Sigma}(c)$, the matrix of the proof of Theorem S7.2 at $\mu = c$; in case ST the mean is zero and the covariance $\boldsymbol{\Sigma}(c) - \frac{c^3}{2}\boldsymbol{m}\boldsymbol{m}'$. The matrix $\boldsymbol{\Lambda}$ has rank two, with null vector $(6, 3, 2)'$, so $A_\tau + cB_\tau$ is a weighted difference of two noncentral χ^2_1 variables minus $\frac{1}{2}$, as in case DU (Section S2), and $\mathbb{P}[A_\tau/B_\tau < -c] = \mathbb{P}[Y'\boldsymbol{\Lambda}Y < \frac{1}{2}]$ is the distribution function of an indefinite quadratic form of rank two in three normal variables. For $c > 2$ the entries of Y are dominated by e^c and nearly collinear. The coordinates

$$\tilde{Y} = \Big(X_\tau(1),\ I_0 - \frac{X_\tau(1)}{c},\ I_1 - \Big(\frac{1}{c} - \frac{1}{c^2}\Big) X_\tau(1) \Big)',$$

whose last two kernels are $-1/c$ and $1/c^2 - s/c$ (they do not grow like $e^{c(1-s)}$), make the covariance well conditioned, with means $-1/(2c)$ and $1/(2c^2) - 1/(3c)$ per unit $\varkappa_\tau$. Given $(\tilde{Y}_2, \tilde{Y}_3)$, $\tilde{Y}_1$ is normal and

$$\tilde{Y}'\tilde{\boldsymbol{\Lambda}}\tilde{Y} - \tfrac{1}{2} = \tilde{\Lambda}_{11}\tilde{Y}_1^2 + 2l_1\tilde{Y}_1 + l_0, \qquad \tilde{\Lambda}_{11} = \frac{(c-2)(c^2 - 6c + 12)}{2c^3},$$

is a quadratic in it, where $\tilde{\boldsymbol{\Lambda}}$ is $\boldsymbol{\Lambda}$ in the new coordinates, $\tilde{\Lambda}_{11}$ is positive for $c > 2$ and vanishes at $c = 2$, and l_1 and l_0 are linear and quadratic in $(\tilde{Y}_2, \tilde{Y}_3)$; its discriminant $l_1^2 - \tilde{\Lambda}_{11} l_0$ does not involve the conditional mean. The conditional probability is a difference of two normal distribution functions, and the remaining two-dimensional normal integral is done by nested adaptive quadrature over $[-12, 12]^2$, with breakpoints where the discriminant, a quadratic in the inner variable, vanishes. For $c \le 2$, Y itself is used, with $\boldsymbol{\Sigma}(c)$ and $\boldsymbol{m}$ by 64-point Gauss–Legendre quadrature of the kernels ($|c| \le 20$) or in closed form. At $c = 0$ and $\varkappa_\tau = 0$ the two weights are $\pm 1/(4\sqrt{15})$, and

$$\mathbb{P}[A_\tau < 0] = 1 - \frac{1}{\pi}\int_{\sqrt{15}}^{\infty} K_0(x)\,\mathrm{d}x = 0.99628533,$$

with K_0 the modified Bessel function of the second kind of order zero; this is the value the programs return.

S7.5.6 Moments of the limit

The limit A_τ/B_τ has moments of all orders: B_τ is a positive definite quadratic form of infinite rank, so $\mathbb{E}[B_\tau^{-k}] < \infty$ for every k (cf. Nabeya, 1999, and Section S1.5). They are computed from (8) with the limiting moment generating function, exactly as in finite samples. At $c = \varkappa_\tau = 0$, $\mathbb{E}[A_\tau/B_\tau] = -10.2455$, and Table S14 gives the standard deviation, skewness and excess kurtosis.

S7.5.7 The explosive limit

Proposition S7.7 (Explosive limit)**.** *Let $|\phi| > 1$ be fixed, (α, δ, y_0) fixed in (S.14), and $z_* = (y_0 - m_0)/\sigma$ the effective start of Proposition S7.1. Then $|\phi|^n(\hat\phi - \phi)/(\phi^2 - 1)$ has the limit (47) of case* DK *with $z_0 = z_*$.*

Sketch. By Proposition S7.1, $\hat\phi$ is the detrended estimator of $x_t = \phi^t z_t$, with $z_t = z_* + \sum_{j=1}^t \phi^{-j} e_j \to z_\infty$. With $\boldsymbol{x}_- = (x_0, \ldots, x_{n-1})'$, the regression on 1 and t changes $\sum_{t=1}^n x_{t-1}^2$ by $\boldsymbol{x}_-'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{x}_- = O_p(\phi^{2n}/n)$, a relative $O_p(1/n)$, and $\sum_{t=1}^n x_{t-1}e_t$ by $O_p(|\phi|^n n^{-1/2})$, which is $o_p(1)$ after the numerator is scaled by $|\phi|^{-n}$. The scaled denominator has a positive limit almost surely, so Slutsky's theorem applies. The limit is therefore that of case DK with start z_*, as for case DU (Wang and Yu, 2015). □

Table S16 shows the approach to the limit: as in case DU it is like $1/n$, much slower than the $|\phi|^{-n}$ of case DK. One numerical point calls for attention: at large n the point $\phi + x(\phi^2 - 1)/|\phi|^n$ is not representable, so the distribution function is evaluated at the centered argument $\hat\phi - \phi$.

S7.6 Numerical results

All entries were computed with the MATLAB programs, with the default tolerances; the high-precision references use the Python library `mpmath`.

S7.6.1 Accuracy

Table S11 compares the $O(1)$ evaluation of $\varphi(t_1, -rt_1)$ with a dense evaluation through the eigenvalues of the quadratic form of $u - rv$ in independent standard normal coordinates (whose product form gives the continuous branch), over $t_1 \in [10^{-5}, 10^3]$ (2000 logarithmically spaced points) and four values of r, at $n = 60$ and 200. The differences are of

Table S11: Accuracy of the $O(1)$ evaluation of the characteristic function, cases DT and ST

Case	ϕ	parameters	$n = 60$	$n = 200$
DT	0.5	$\gamma = \gamma_\tau = 0$	8.3×10^{-15}	2.6×10^{-14}
DT	0.9	$(\gamma, \gamma_\tau) = (1, 0.05)$	2.2×10^{-14}	3.1×10^{-13}
DT	1	$\gamma = \gamma_\tau = 0$	3.4×10^{-14}	1.8×10^{-13}
DT	1	$(\gamma, \gamma_\tau) = (0.5, 0.02)$	1.2×10^{-13}	3.9×10^{-12}
DT	-1	$\gamma = \gamma_\tau = 0$	1.6×10^{-14}	6.6×10^{-14}
ST	0.95	–	1.7×10^{-14}	2.9×10^{-14}
ST	0.999	–	2.0×10^{-13}	2.5×10^{-13}
DT	$-1.3^\dagger$	$(\gamma, \gamma_\tau) = (0.5, 0.1)$	4.0×10^{-14}	$(n = 80)$
DT	$1.05^\dagger$	$(\gamma, \gamma_\tau) = (0, 0.2)$	4.9×10^{-11}	$(n = 200)$
DT	$1.2^\dagger$	$(\gamma, \gamma_\tau) = (0, 0)$	1.4×10^{-14}	$(n = 60)$
DT	$1.2^\dagger$	$(\gamma, \gamma_\tau) = (0.7, 0.3)$	4.4×10^{-12}	$(n = 120)$
DT	$1.5^\dagger$	$(\gamma, \gamma_\tau) = (0.4, -0.2)$	1.3×10^{-13}	$(n = 50)$
DT	$1.5^\dagger$	$(\gamma, \gamma_\tau) = (1, 0)$	1.4×10^{-14}	$(n = 120)$

Notes: Maximum absolute difference between the $O(1)$ evaluation of $\varphi(t_1, -rt_1)$ (Sections S7.3 and S7.4) and the dense eigenvalue evaluation, over $t_1 \in [10^{-5}, 10^3]$ and $r \in \{-15, -2.5, -0.25, 2.5\}/n$. †Explosive region: maximum relative error of φ^2 against the high-precision dense reference at the sample size shown, over $t_1 = z/\operatorname{sd}(u - rv)$, $z \in \{0.01, \ldots, 10\}$, and three values of r.

order 10^{-14} to 10^{-13}; the largest, 3.9×10^{-12}, is at the unit root with a drift and $n = 200$, as in case DU (Table 2). In the explosive region, where the dense evaluation in double precision is itself inaccurate, the reference is a dense evaluation in $(2n \log_{10} |\phi| + 60)$-digit arithmetic. The relative error of φ^2 (the dense determinant fixes the root only up to sign) is 1.4×10^{-14} at $\phi = 1.5$, $n = 120$ ($|\phi|^{2n} \approx 10^{42}$), 4.4×10^{-12} at $\phi = 1.2$, $n = 120$, and 4.9×10^{-11} at $\phi = 1.05$, $n = 200$ ($|\phi|^{2n} \approx 3 \times 10^8$), where the generic assembly is accurate only to 10^{-2} (with the drift $\gamma_\tau = 0.2$, whose exponent loses further digits; without a drift the loss is about $2n \log_{10} |\phi|$ digits). For the limit, the closed form of Theorem S7.2 agrees with a direct evaluation of the 3×3 determinant and bilinear form of its proof in 80- to 300-digit arithmetic to 5×10^{-14} in the logarithm of (S.33) for c from -50 to 300, on the scale on which the characteristic function varies, and $\varkappa_\tau \in \{0, 1.3, -4\}$. Its jets of orders one to six agree with high-precision Taylor coefficients to 2×10^{-14} (order one) and 6×10^{-11} (order six), and its phase with the continued phase of the Nabeya–Tanaka determinant as stated in Section S7.5.4.

S7.6.2 Exact critical values at the unit root

Table S12 gives the exact quantiles of $n(\hat{\phi} - 1)$ at $\phi = 1$ in case DT; by Proposition S7.1 they hold for every value of the drift γ (and of the initial value), with $\gamma_\tau = 0$. The sample sizes are those of the panel $\hat{\rho}_\tau$ of Table 10.A.1 of Fuller (1996), with $n = 1000$ added; that panel was constructed by D. A. Dickey by Monte Carlo, with standard errors that Fuller reports as mostly below 0.10 in the left half of the table and below 0.02 in the right half. As for his panel $\hat{\rho}_\mu$ (Section S4), his entry at sample size n estimates $n/(n-1)$ times the exact quantile from $n - 1$ regression observations; the exact quantile from n observations in Panel A does not match it (at $n = 25$ and $p = 0.01$, -21.97 against his -22.5). Panel B shows that, so interpreted, his table agrees with the exact distribution to within the accuracy one expects from its standard errors. The differences are at most 0.14 in the left tail (at $n = 25$ and 50, $p = 0.01$: -22.5 and -25.8 against -22.64 and -25.67), at most 0.02 at the median, and at most 0.04 in the right tail. His limiting row agrees with ours to within 0.06 (-29.4, -25.0, -21.7 and -18.3 against -29.359, -25.058, -21.711 and -18.245 in the left tail); the first edition of his table, reproduced in Table B.5 of Hamilton (1994), has -29.5, -25.1, -21.8 and -18.3.

S7.6.3 Moments

Table S13 gives the bias, standard deviation, skewness and excess kurtosis of $\hat{\phi}$ in cases DT ($\gamma = \gamma_\tau = 0$) and ST. The trend increases the bias further: at $\phi = 0.5$, $n = 100$, it is -0.0411, against -0.0253 with an intercept only and -0.0098 without (Table S5); the approximations $-(1 + 3\phi)/n$ and $-2\phi/n$ of Marriott and Pope (1954) (the first also in Kendall (1954)) give -0.025 and -0.010. At $\phi = 0$ the bias in case ST is $-2/n$ to the digits shown. The stationary and the fixed start differ little, much less than in the cases without a trend, because the trend absorbs the effect of the initial observation; even at $\phi = 0.99$ and $n = 400$ the bias is -0.0201 against -0.0205. Table S14 shows the slow convergence of the moments of $n(\hat{\phi} - 1)$ at the unit root to those of the limit: the excess kurtosis at $n = 1000$ is still 2.7% below its limit 2.3725.

S7.6.4 Local-to-unity quantiles

Table S15 gives the 5%, 50% and 95% quantiles of the limit A_τ/B_τ for c from -20 to 5, in case DT with $\varkappa_\tau = 0$ and $\varkappa_\tau = 10$ and, for $c < 0$, in case ST, together with the exact

Table S12: Exact quantiles of $n(\hat{\phi}-1)$, case DT, $\phi=1$, $\gamma_\tau=0$

Panel A: exact quantiles									
n	0.01	0.025	0.05	0.10	0.50	0.90	0.95	0.975	0.99
25	−21.973	−19.474	−17.385	−15.076	−8.201	−3.517	−2.425	−1.517	−0.491
50	−25.229	−21.973	−19.344	−16.530	−8.629	−3.643	−2.555	−1.667	−0.687
100	−27.168	−23.433	−20.471	−17.351	−8.859	−3.705	−2.615	−1.735	−0.773
250	−28.450	−24.386	−21.201	−17.878	−9.004	−3.742	−2.650	−1.774	−0.822
500	−28.899	−24.718	−21.453	−18.060	−9.053	−3.755	−2.661	−1.787	−0.838
1000	−29.127	−24.887	−21.582	−18.152	−9.078	−3.761	−2.667	−1.793	−0.846
∞	−29.359	−25.058	−21.711	−18.245	−9.103	−3.767	−2.673	−1.800	−0.854

Panel B: Fuller (1996, Table 10.A.1) against the exact distribution

n		0.01	0.025	0.05	0.10	0.50	0.90	0.95	0.975	0.99
25	Fuller	−22.5	−20.0	−17.9	−15.6	−8.49	−3.65	−2.51	−1.53	−0.46
	exact	−22.64	−20.09	−17.96	−15.59	−8.51	−3.65	−2.51	−1.57	−0.49
50	Fuller	−25.8	−22.4	−19.7	−16.8	−8.80	−3.71	−2.60	−1.67	−0.67
	exact	−25.67	−22.36	−19.69	−16.83	−8.80	−3.71	−2.60	−1.70	−0.70
100	Fuller	−27.4	−23.7	−20.6	−17.5	−8.96	−3.74	−2.63	−1.74	−0.76
	exact	−27.42	−23.65	−20.67	−17.52	−8.95	−3.74	−2.64	−1.75	−0.78
250	Fuller	−28.5	−24.4	−21.3	−17.9	−9.05	−3.76	−2.65	−1.79	−0.83
	exact	−28.56	−24.48	−21.28	−17.95	−9.04	−3.76	−2.66	−1.78	−0.83
500	Fuller	−28.9	−24.7	−21.5	−18.1	−9.08	−3.76	−2.66	−1.80	−0.86
	exact	−28.96	−24.77	−21.50	−18.10	−9.07	−3.76	−2.67	−1.79	−0.84
∞	Fuller	−29.4	−25.0	−21.7	−18.3	−9.11	−3.77	−2.67	−1.81	−0.88
	exact	−29.36	−25.06	−21.71	−18.25	−9.10	−3.77	−2.67	−1.80	−0.85

Notes: Panel A: x_p with $\mathbb{P}[n(\hat{\phi}-1)<x_p]=p$ for the regression of y_t on 1, t and y_{t-1} over $t=1,\dots,n$; the last row is the limit (Theorem S7.2 with $c=\varkappa_\tau=0$). Panel B: "Fuller" is the panel $\hat{\rho}_\tau$ of Fuller (1996, Table 10.A.1); "exact" is $n/(n-1)$ times the exact quantile at sample size $n-1$, the quantity that Fuller's entry at sample size n estimates, and, in the last row, the limit.

Table S13: Bias, standard deviation, skewness and excess kurtosis of $\hat{\phi}$, cases DT and ST

		$n=25$				$n=100$				$n=400$			
Case	ϕ	bias	sd	skew	kurt	bias	sd	skew	kurt	bias	sd	skew	kurt
DT	0	−0.0830	0.1951	0.06	−0.18	−0.0202	0.0995	0.01	−0.06	−0.0050	0.0499	0.00	−0.01
DT	0.5	−0.1719	0.1957	−0.38	0.01	−0.0411	0.0906	−0.30	0.09	−0.0101	0.0438	−0.17	0.03
DT	0.9	−0.2875	0.1846	−0.71	0.62	−0.0667	0.0639	−0.86	1.08	−0.0149	0.0247	−0.57	0.51
DT	0.99	−0.3438	0.1846	−0.73	0.69	−0.0906	0.0562	−1.07	1.77	−0.0205	0.0155	−1.11	1.96
DT	1	−0.3533	0.1846	−0.73	0.69	−0.0985	0.0561	−1.07	1.79	−0.0254	0.0148	−1.17	2.21
ST	0	−0.0800	0.1921	0.05	−0.18	−0.0200	0.0990	0.01	−0.06	−0.0050	0.0499	0.00	−0.01
ST	0.5	−0.1658	0.1927	−0.38	0.03	−0.0406	0.0900	−0.30	0.09	−0.0100	0.0438	−0.17	0.03
ST	0.9	−0.2832	0.1845	−0.71	0.63	−0.0649	0.0631	−0.86	1.10	−0.0147	0.0245	−0.57	0.51
ST	0.99	−0.3438	0.1846	−0.73	0.69	−0.0903	0.0562	−1.07	1.78	−0.0201	0.0154	−1.12	1.99

Notes: Exact values; "kurt" is the excess kurtosis. Case DT with $\gamma=\gamma_\tau=0$.

quantiles of $n(\hat{\phi}-\phi)$ at $n=1000$ in case ST for comparison. The initial condition matters most near the unit root: $\varkappa_\tau=10$ moves the median from −9.10 to −2.90 at $c=0$, and

Table S14: Moments of $n(\hat{\phi}-1)$ at the unit root, case DT, and of the limit

n	mean	sd	skewness	excess kurtosis
25	−8.8328	4.6151	−0.7278	0.6928
50	−9.4867	5.2382	−0.9484	1.3363
100	−9.8515	5.6109	−1.0706	1.7922
250	−10.0842	5.8576	−1.1491	2.1231
1000	−10.2047	5.9881	−1.1900	2.3078
10000	−10.2414	6.0282	−1.2025	2.3660
∞	−10.2455	6.0327	−1.2039	2.3725

Notes: Case DT with $\gamma_\tau = 0$ (any γ). The last row is the limit A_τ/B_τ with $c = \varkappa_\tau = 0$.

Table S15: Quantiles of the local-to-unity limits of $n(\hat{\phi}-\phi)$, cases DT and ST

	DT, $\varkappa_\tau = 0$			DT, $\varkappa_\tau = 10$			ST			ST, $n = 1000$		
c	5%	50%	95%	5%	50%	95%	5%	50%	95%	5%	50%	95%
−20	−21.972	−5.701	4.633	−21.965	−5.698	4.633	−21.681	−5.592	4.641	−21.360	−5.512	4.653
−10	−20.583	−6.250	1.946	−20.504	−6.214	1.958	−20.217	−6.079	2.000	−20.005	−6.029	2.010
−5	−20.259	−6.950	0.127	−19.746	−6.656	0.291	−19.956	−6.769	0.229	−19.793	−6.733	0.236
−2	−20.576	−7.815	−1.269	−17.993	−6.043	−0.115	−20.470	−7.733	−1.201	−20.332	−7.705	−1.195
0	−21.711	−9.103	−2.673	−10.418	−2.901	0.588	–	–	–	–	–	–
2	−21.621	−8.563	−1.525	−2.162	−0.484	0.973	–	–	–	–	–	–
5	−4.178	−0.018	0.419	−1.251	−0.010	0.255	–	–	–	–	–	–

Notes: Quantiles of A_τ/B_τ in (S.30); last block: exact quantiles of $n(\hat{\phi}-\phi)$ with $\phi = 1 + c/n$, $n = 1000$, case ST.

Table S16: The explosive region with a trend: $\mathbb{P}[|\phi|^n(\hat{\phi}-\phi)/(\phi^2-1) < x]$, $\phi = 1.05$, $z_* = 2$

	DT				DU			
n	$x=-1$	$x=0$	$x=0.5$	$x=2$	$x=-1$	$x=0$	$x=0.5$	$x=2$
100	0.4124	0.5403	0.6152	0.8055	0.3934	0.6883	0.7964	0.9263
200	0.2481	0.4270	0.5508	0.8015	0.2662	0.5619	0.7037	0.8759
400	0.2306	0.4531	0.6050	0.8421	0.2397	0.5281	0.6857	0.8745
800	0.2251	0.4749	0.6385	0.8603	0.2291	0.5136	0.6788	0.8753
1600	0.2223	0.4871	0.6555	0.8686	0.2242	0.5067	0.6756	0.8758
3200	0.2209	0.4934	0.6640	0.8725	0.2218	0.5033	0.6740	0.8761
∞	0.2195	0.5000	0.6725	0.8763	0.2195	0.5000	0.6725	0.8763

Notes: Case DT with $\gamma = (\phi-1)z_*$ and $\gamma_\tau = 0$; case DU with $\gamma = (\phi-1)z_*$. The rows $n = \infty$ are the limit of Proposition S7.7.

has almost no effect at $c = -20$, where the stationary distribution of $\varkappa_\tau$ has standard deviation 63. At $n = 1000$ the exact quantiles in case ST are within 0.33 of the limit.

S7.6.5 The explosive region: approach to the limit

Table S16 gives $\mathbb{P}[|\phi|^n(\hat{\phi}-\phi)/(\phi^2-1) < x]$ at $\phi = 1.05$ with effective start $z_* = 2$, in case DT (with $\gamma = (\phi-1)z_*$ and $\gamma_\tau = 0$; any (γ, γ_τ) with the same ξ gives the same values) and, for comparison, in case DU with the same effective start, together with the limit of Proposition S7.7. Both approach the limit like $1/n$. With a trend the absolute difference between the finite-sample and the limiting distribution functions is larger except at $x = -1$: at $n = 3200$ it is 0.0066 against 0.0033 at $x = 0$, and 0.0085 against 0.0015 at $x = 0.5$.

S7.6.6 Cost

Table S17 shows that the cost does not grow with n, much as in the cases without a trend (Table 3). At every sample size from 10^2 to 10^6, a value of the distribution function costs two and a half to four milliseconds, a density value under a hundredth of a second, a quantile about five hundredths of a second and four moments about three hundredths.

Table S17: Cost against the sample size, cases DT and ST (seconds)

n	cdf	pdf	quantile	four moments	cdf, ST
10^2	0.0026	0.0054	0.0447	0.0310	0.0046
10^3	0.0030	0.0056	0.0397	0.0277	0.0045
10^4	0.0042	0.0062	0.0464	0.0349	0.0069
10^5	0.0037	0.0088	0.0539	0.0387	0.0108
10^6	0.0042	0.0096	0.0575	0.0326	0.0041

Notes: Case DT, $\phi = 1$, $\gamma = \gamma_\tau = 0$: one value of $\mathbb{P}[n(\hat{\phi}-1) < -20]$, the density at the same point, the 5% quantile and $\mathbb{E}[(\hat{\phi}-1)^k]$, $k = 1, \ldots, 4$. Last column: $\mathbb{P}[\hat{\phi} < 0.95]$ in case ST with $\phi = 0.95$. Averages of five runs after a warm-up call, on the laptop of Table 3.

S7.7 Programs

The trend cases are part of the MATLAB suite, which also runs in GNU Octave, and every finite-sample and limit routine accepts them. In case DT the parameter of the finite-sample routines is (γ, γ_τ) and that of the limit routines is $\varkappa_\tau$; the explosive limit is evaluated at the effective start z_* of (S.18). The validation suite checks the trend cases, a driver program reproduces the tables of this section, and a Python script distributed

with the programs computes the high-precision references. The user manual describes the routines.

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